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(54) **VEHICLE WITH THERMAL MANAGEMENT SYSTEM**

(71) Applicant: **Oshkosh Corporation**, Oshkosh, WI (US)

(72) Inventors: **Jon Zeamer**, Oshkosh, WI (US); **Greg Steffens**, Oshkosh, WI (US); **Jeff Verhagen**, Oshkosh, WI (US); **Chris Goodman**, Oshkosh, WI (US); **Jon Branson**, Oshkosh, WI (US); **McCall Groen**, Oshkosh, WI (US); **Jon Pandolfini**, Oshkosh, WI (US); **Ryan Wolf**, Oshkosh, WI (US); **Sam Mueller**, Oshkosh, WI (US); **Kevin Pennau**, Oshkosh, WI (US); **Jeremy Andringa**, Oshkosh, WI (US); **Colin Wheeler**, Oshkosh, WI (US); **David Palmer**, Oshkosh, WI (US); **David Cockrell**, Oshkosh, WI (US); **Travis Zuleger**, Oshkosh, WI (US)

(73) Assignee: **Oshkosh Corporation**, Oshkosh, WI (US)

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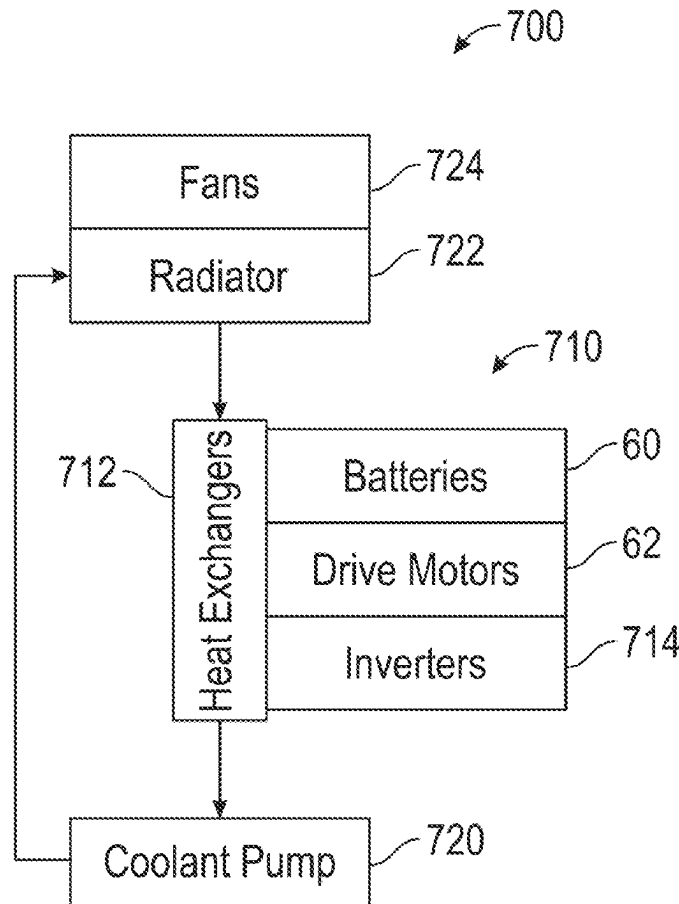
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CPC ..... **B60L 58/26** (2019.02); **B60K 11/08** (2013.01); **B60L 50/64** (2019.02); **B60Y 2200/144** (2013.01)

(57) **ABSTRACT**

A vehicle includes a chassis, a cab coupled to the chassis, a body coupled to the chassis, a cowl coupled to at least one of the body or the cab and extending above the cab, and a heat exchanger coupled to the cab and positioned at least partially within the cowl such that the heat exchanger extends above the cab.



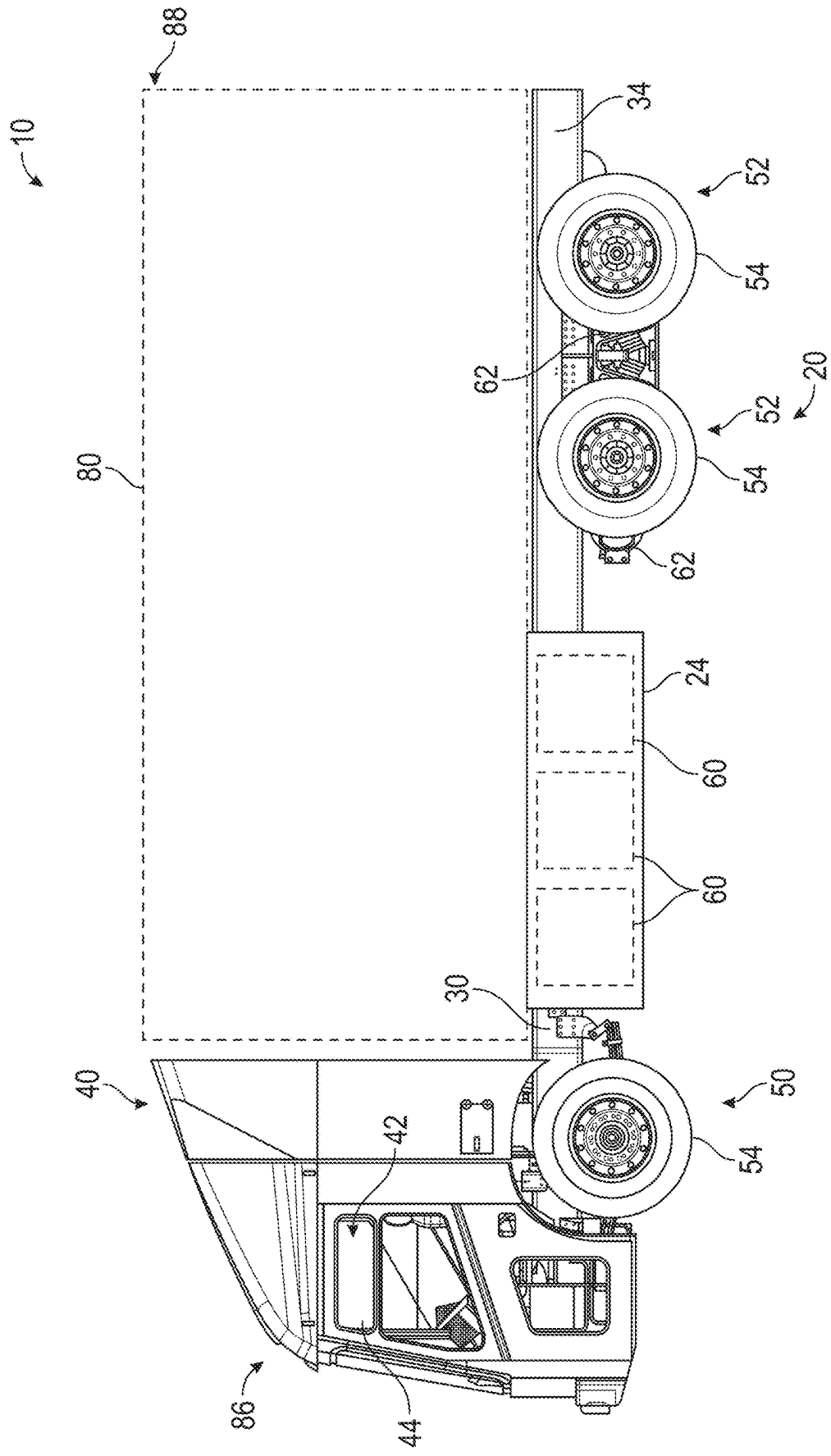
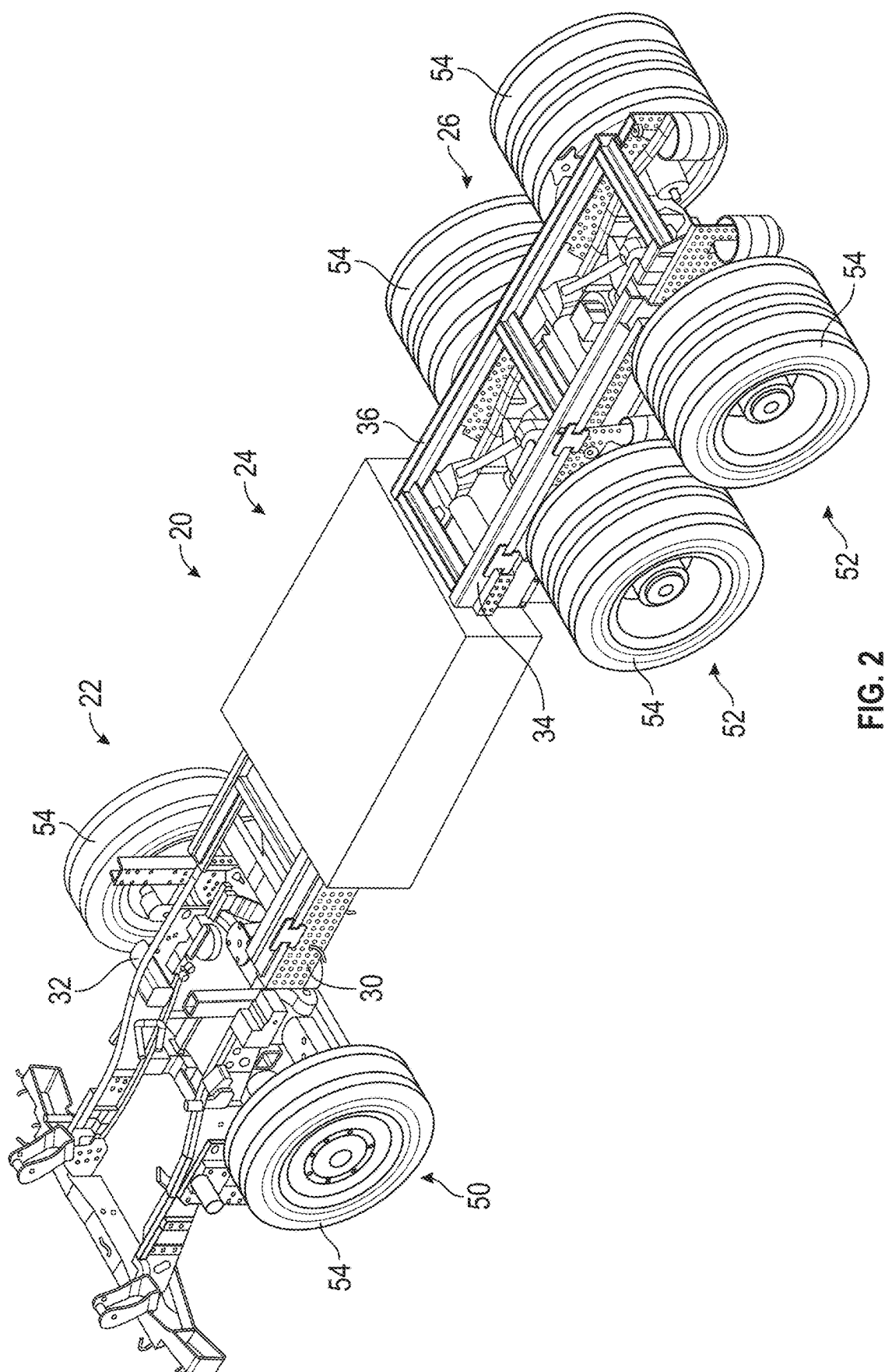
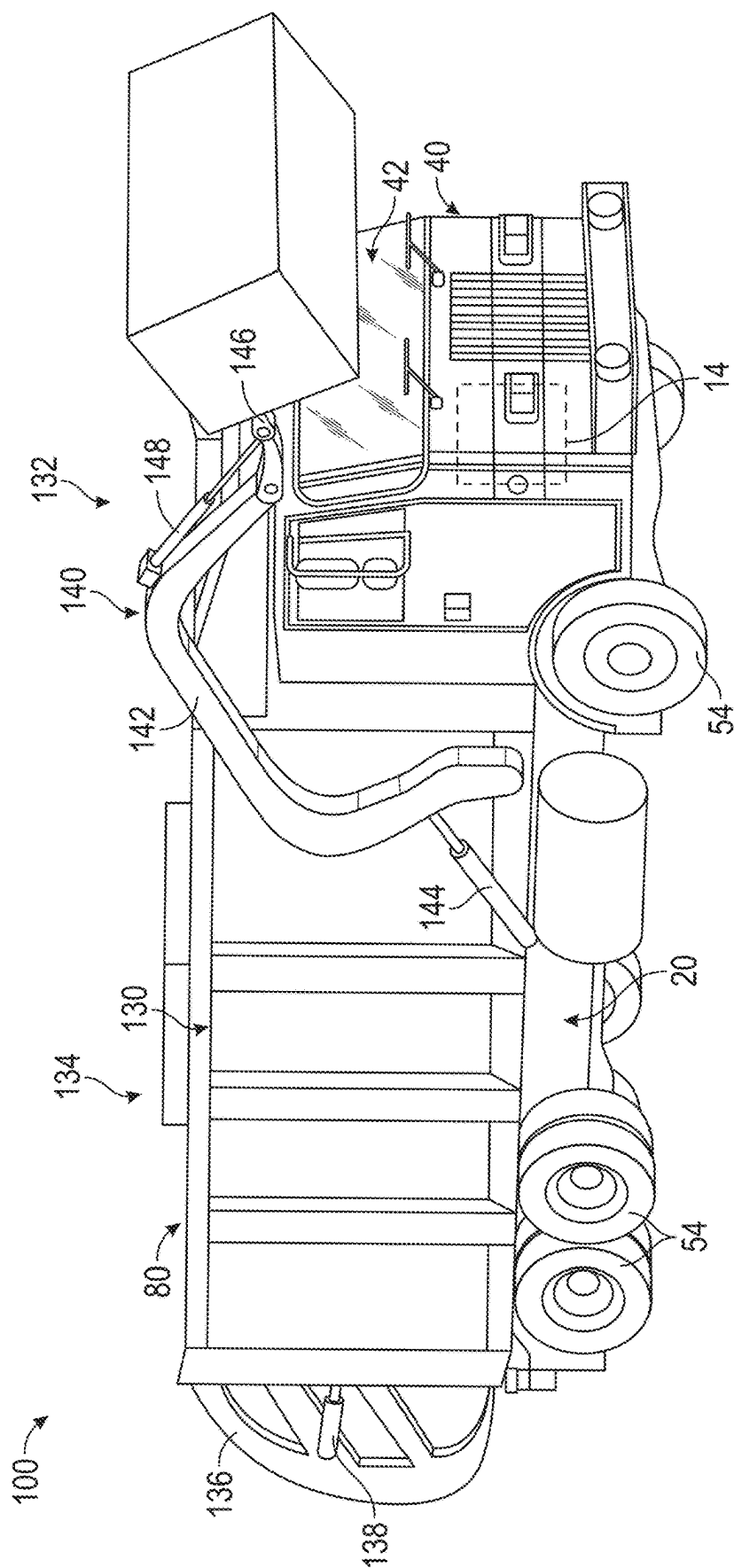


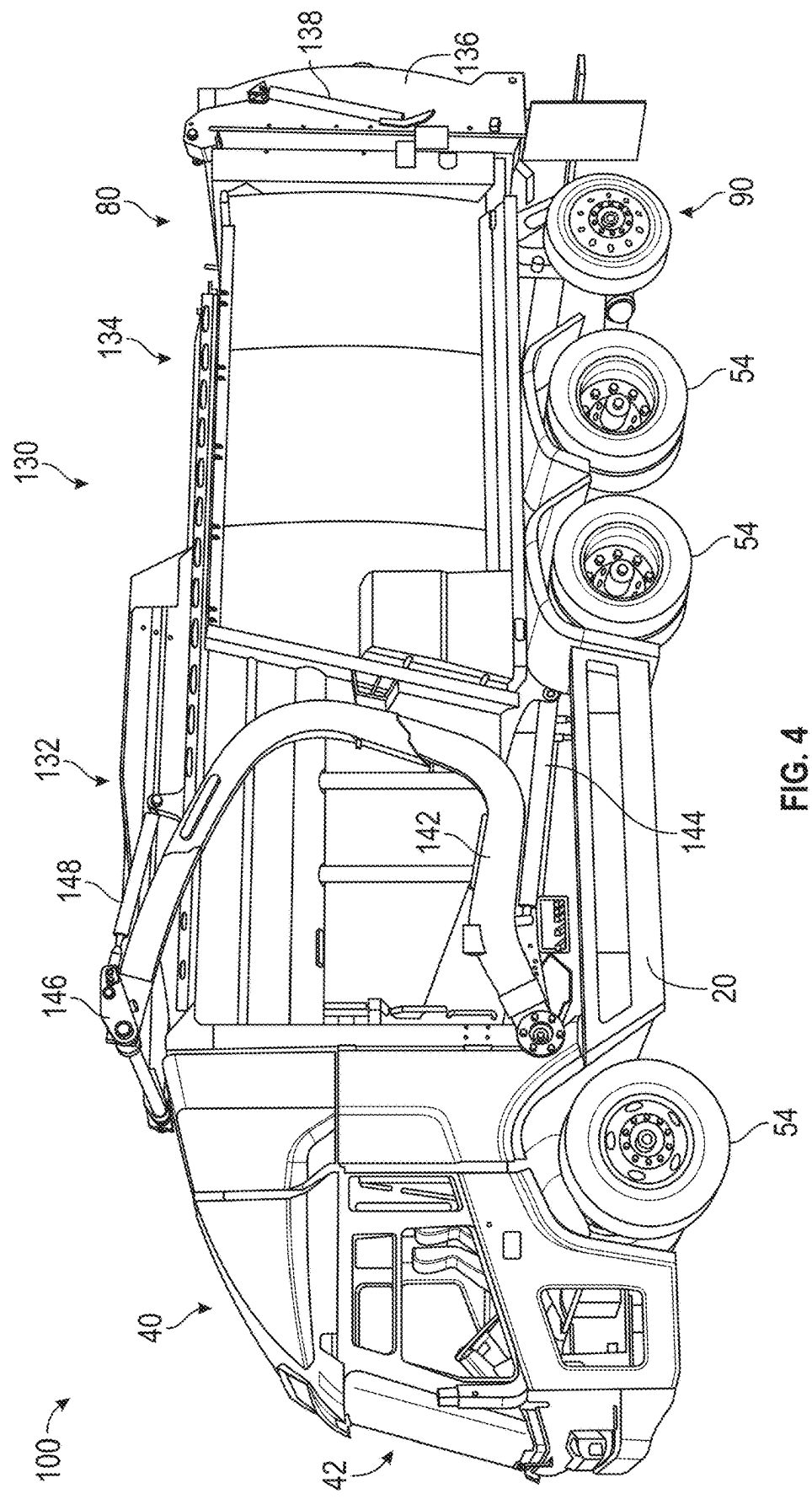
FIG. 1

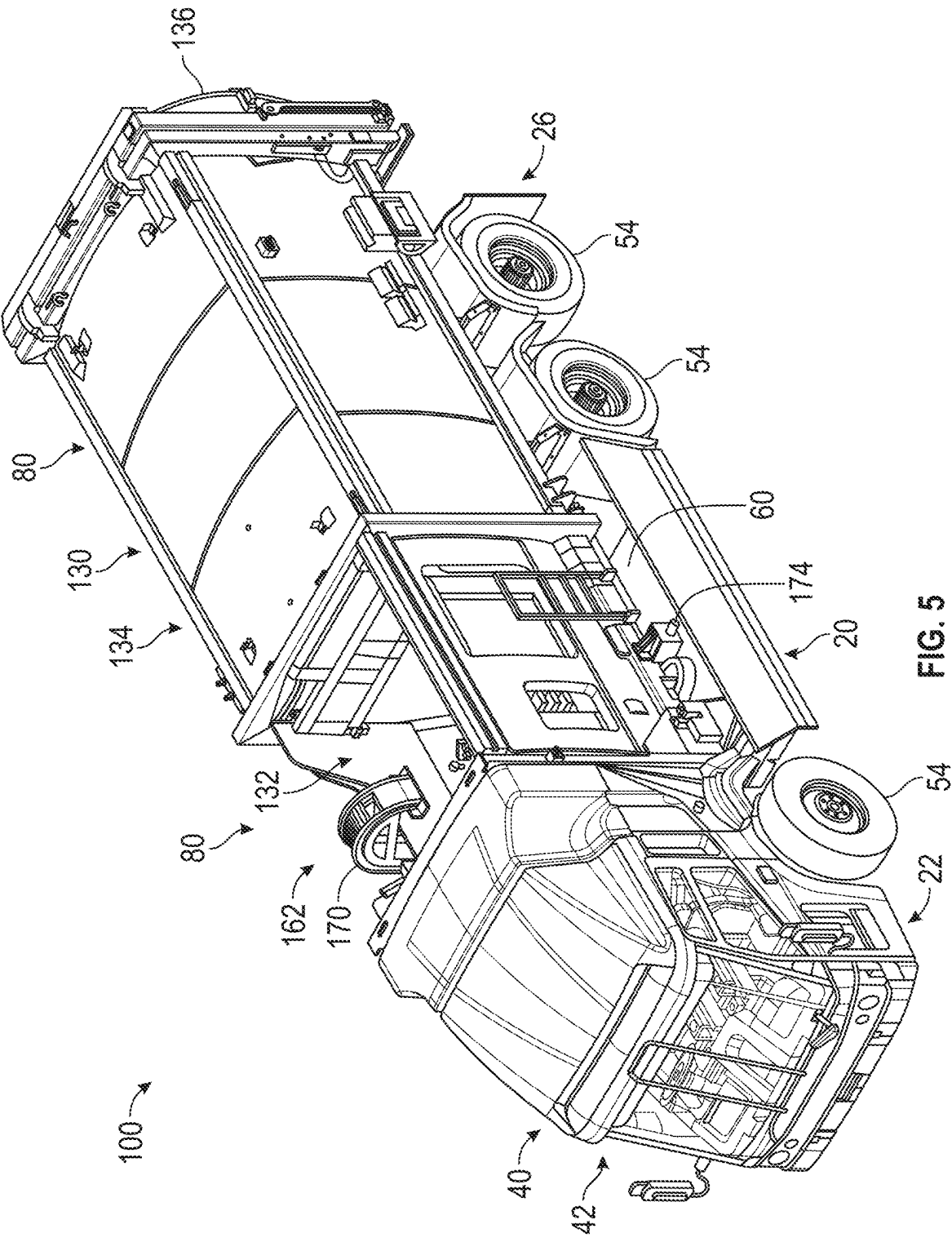


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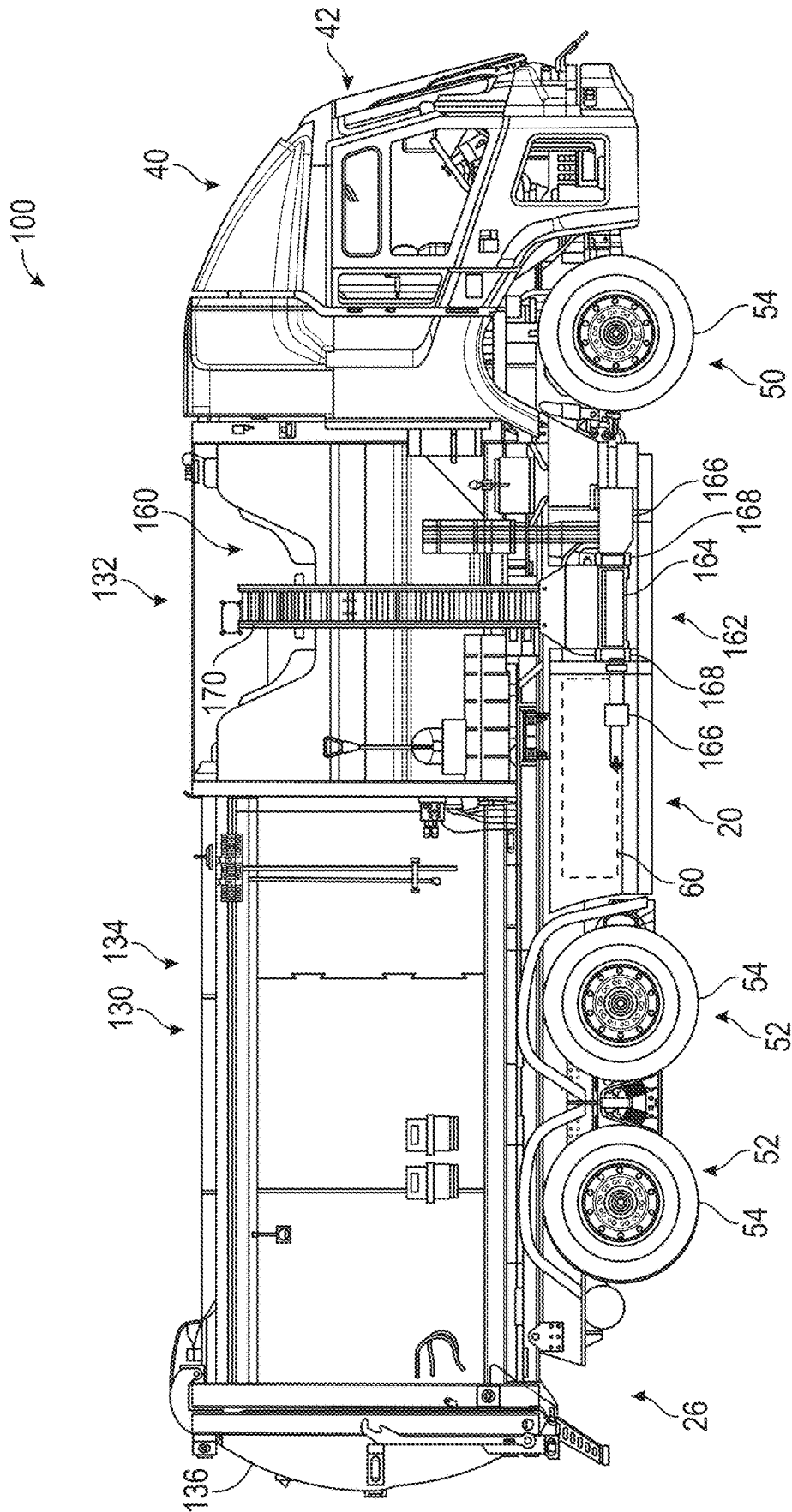


FIG. 6

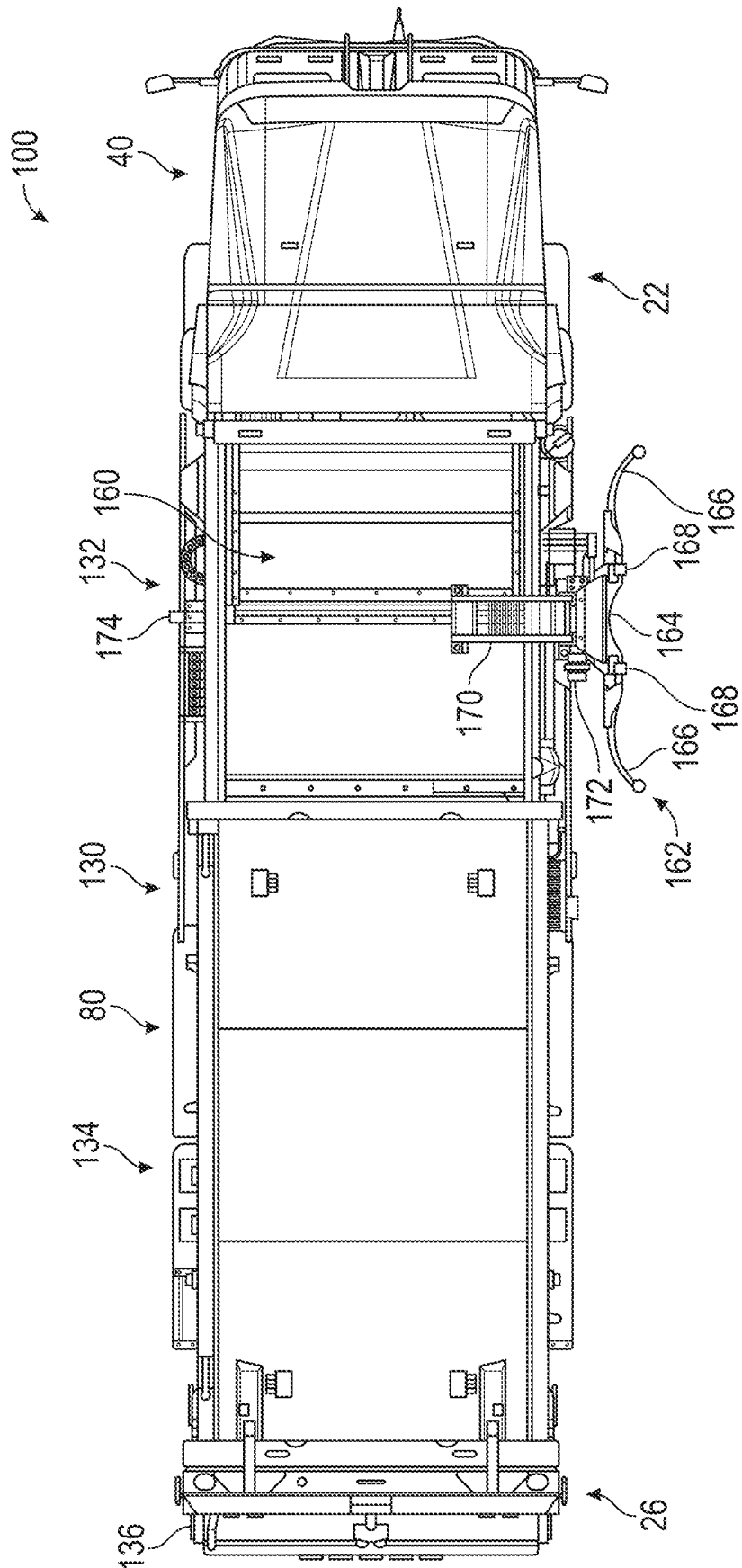


FIG. 7



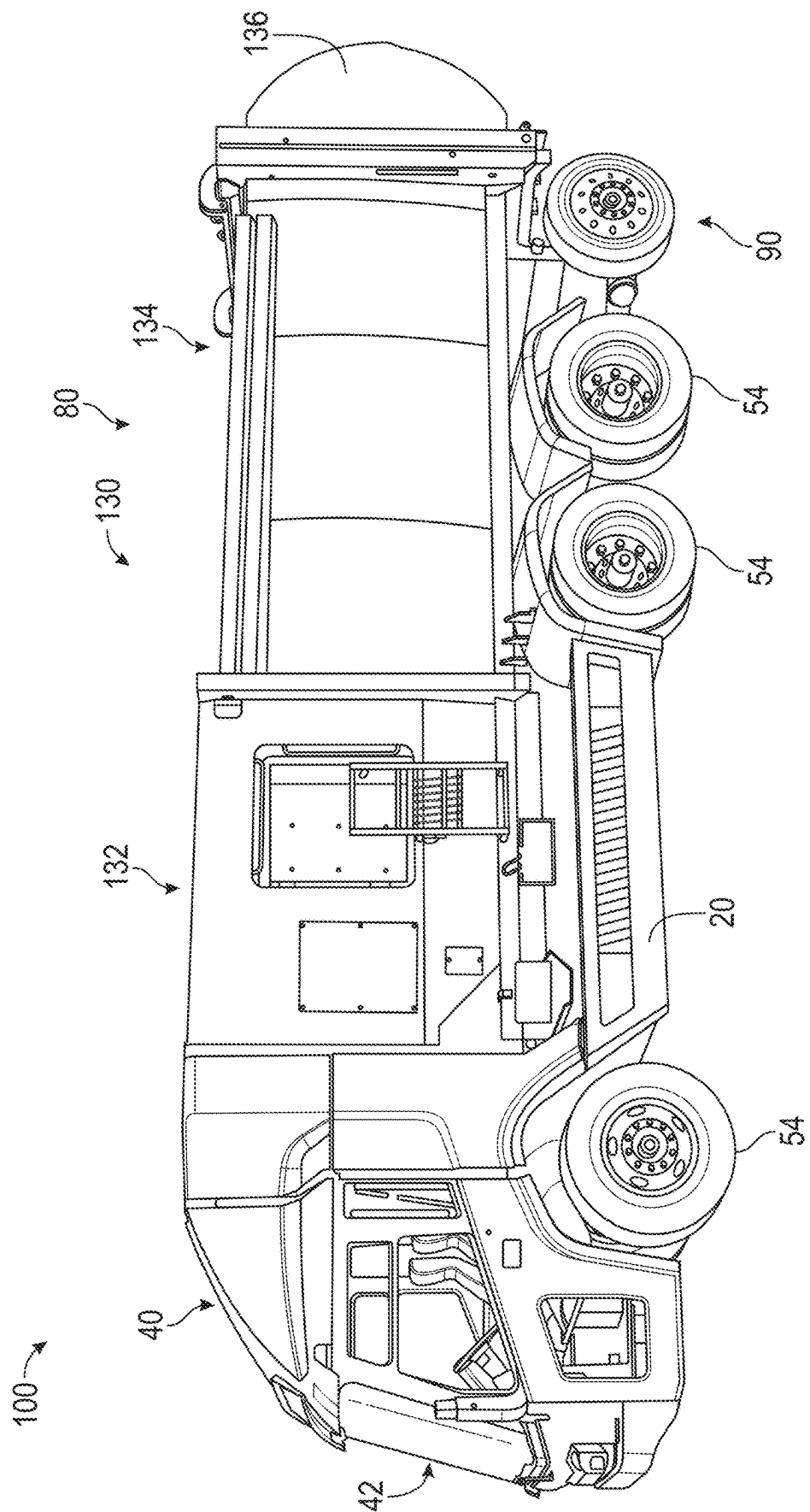


FIG. 8

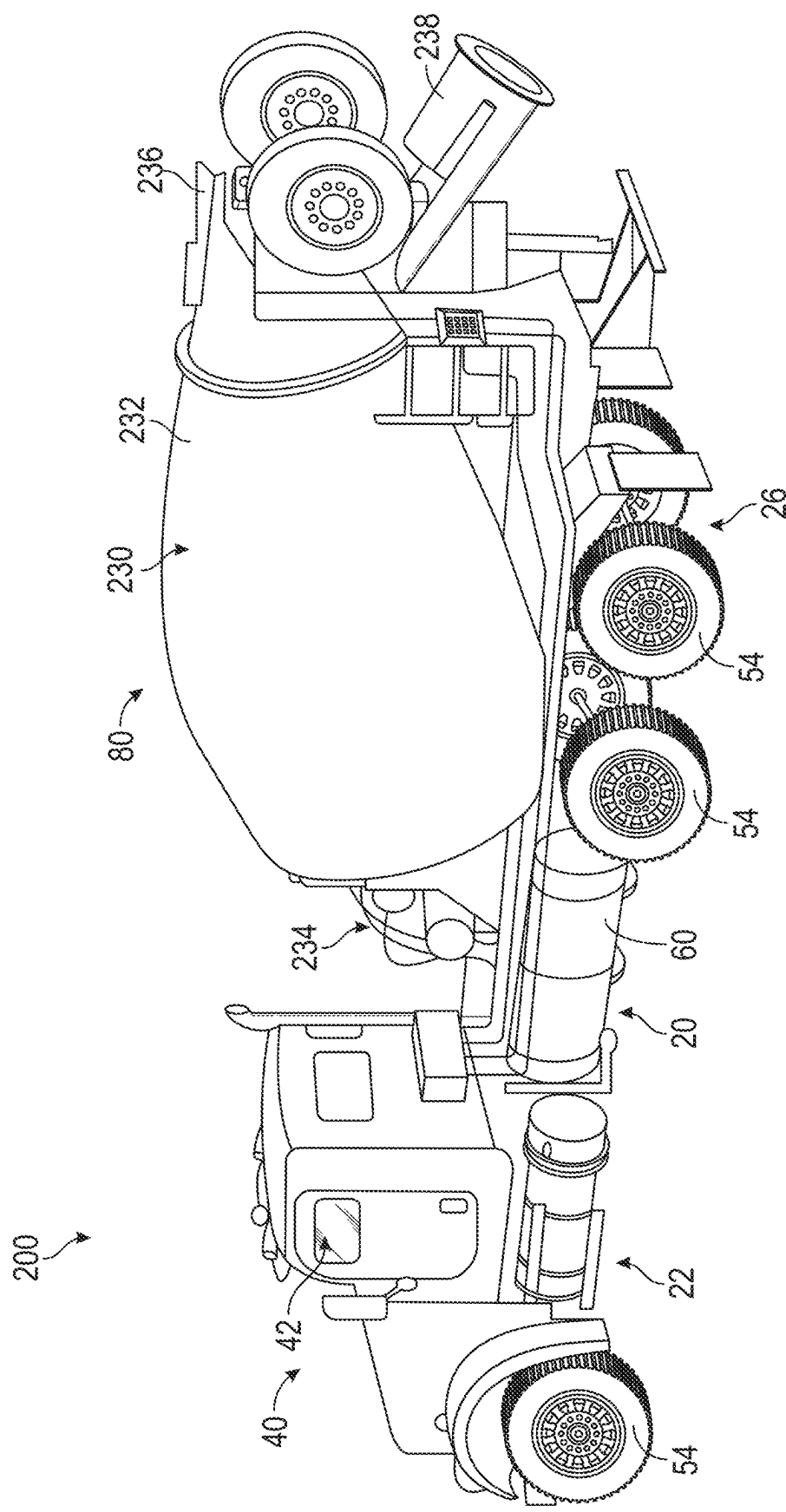


FIG. 9

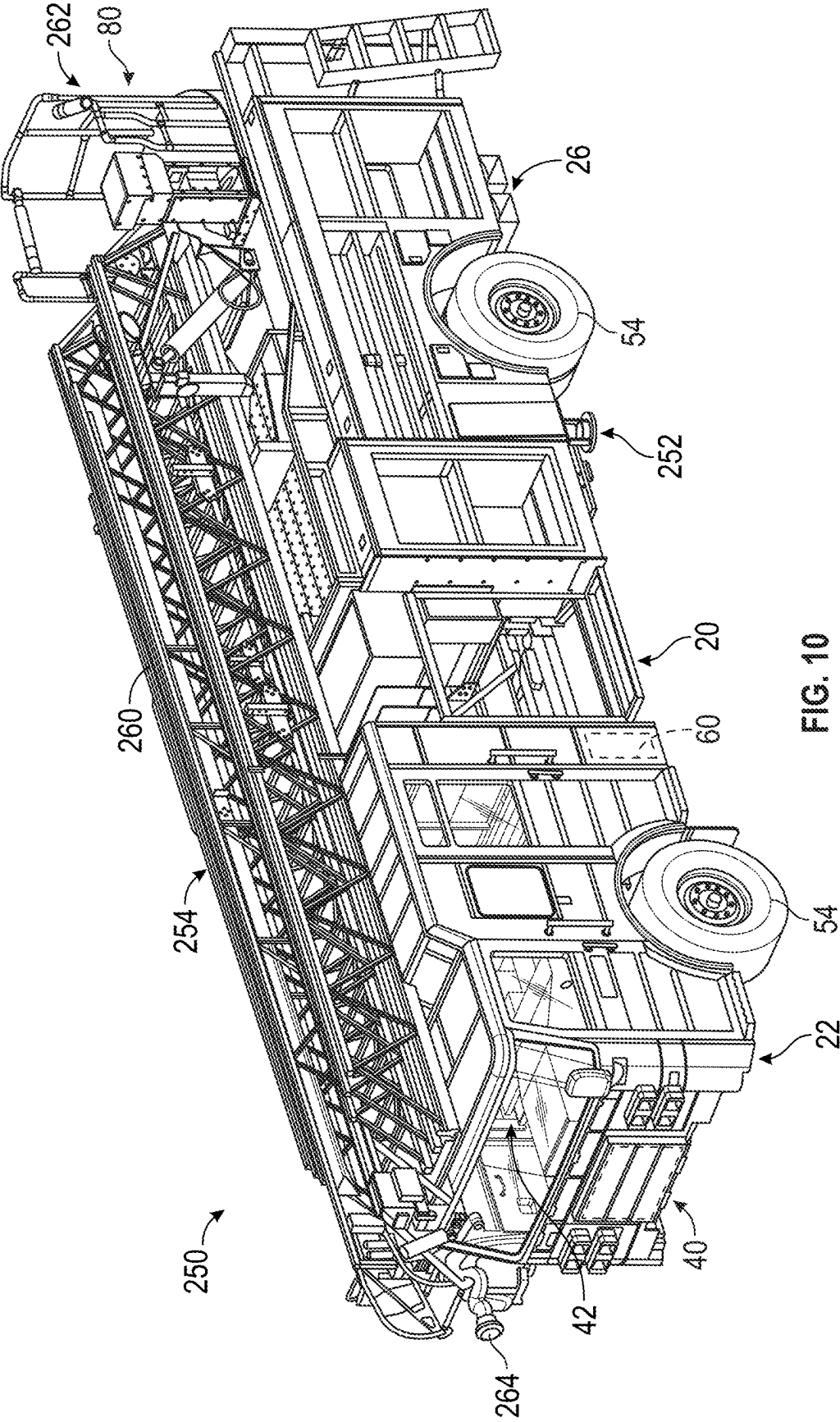


FIG. 10

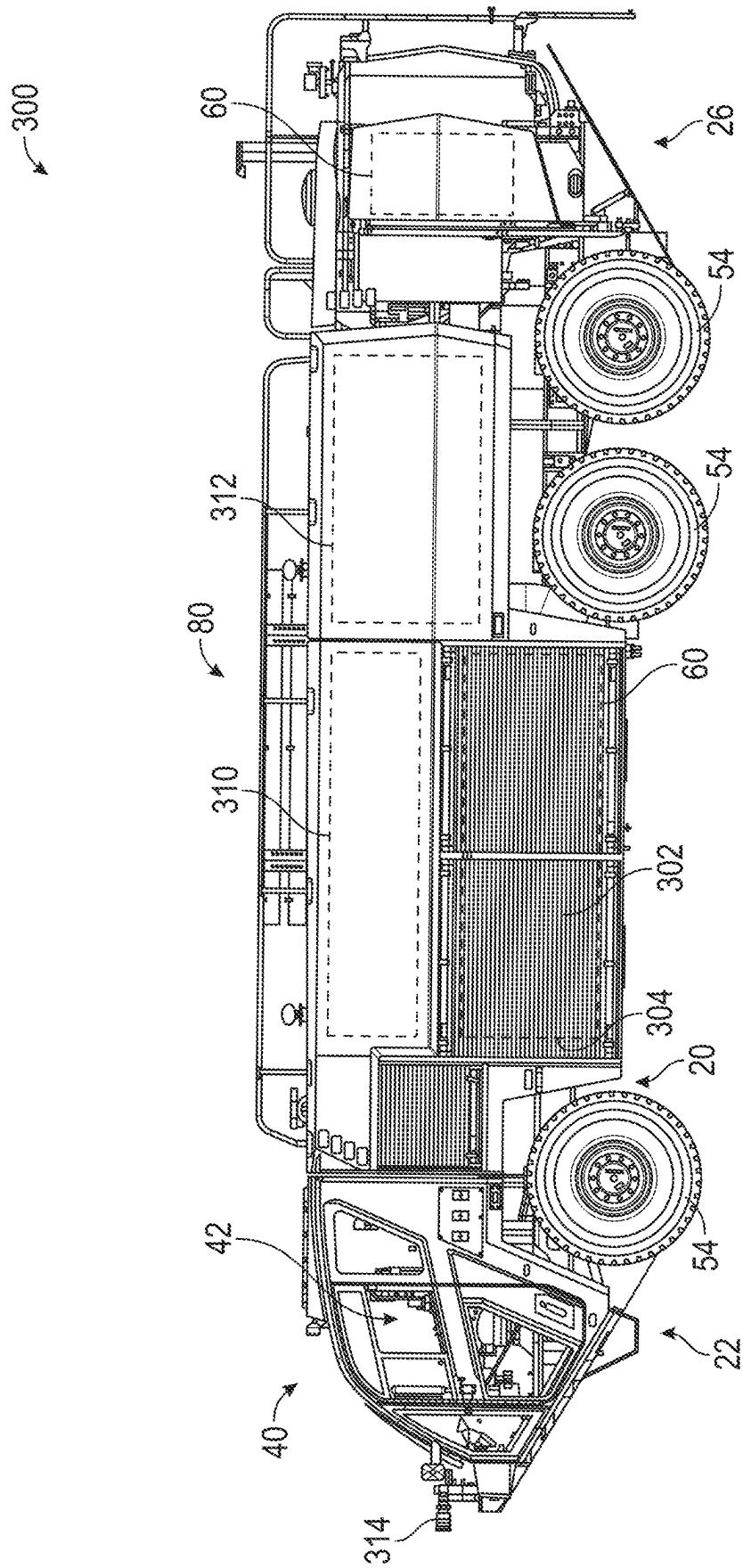


FIG. 11

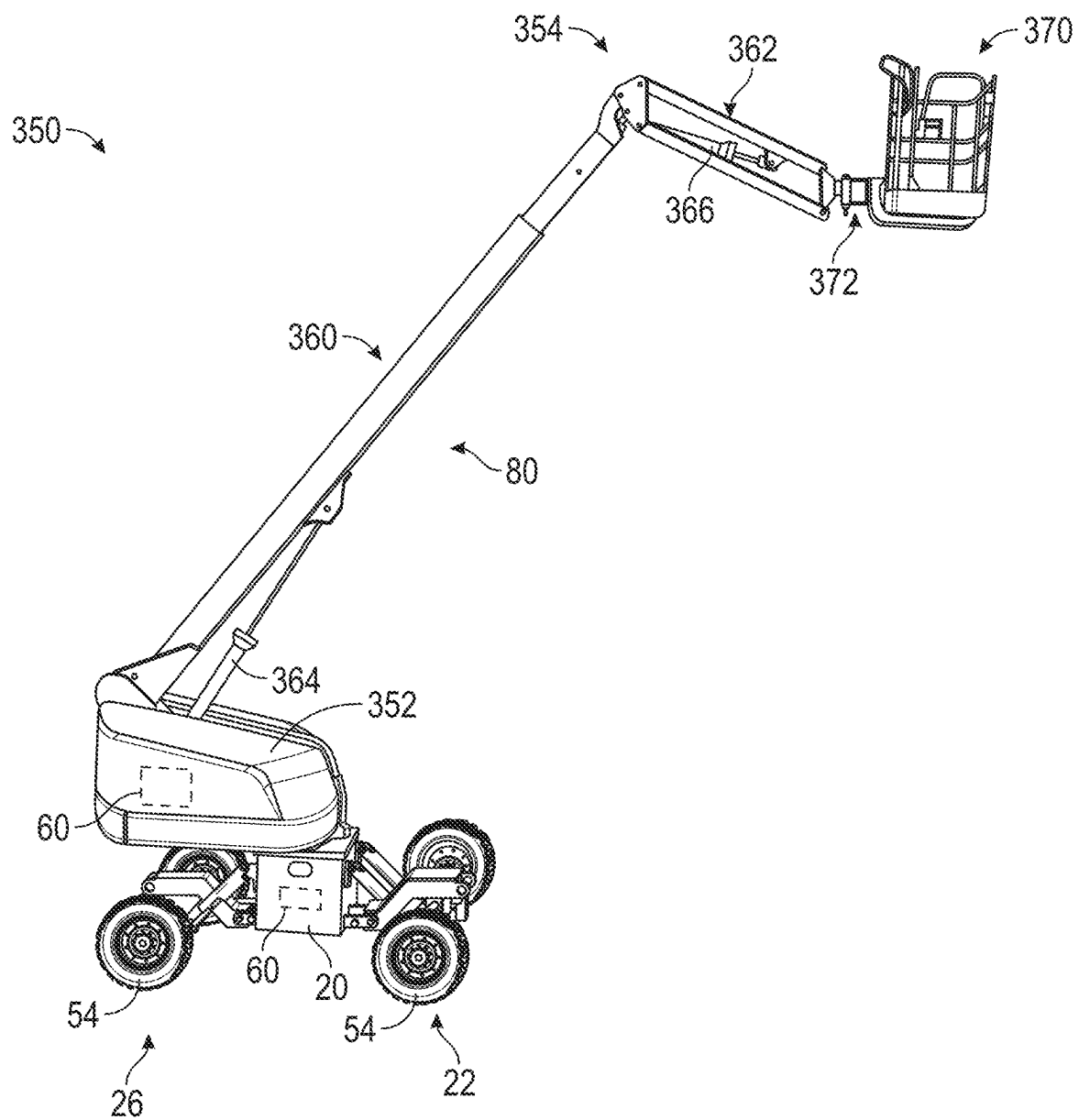


FIG. 12

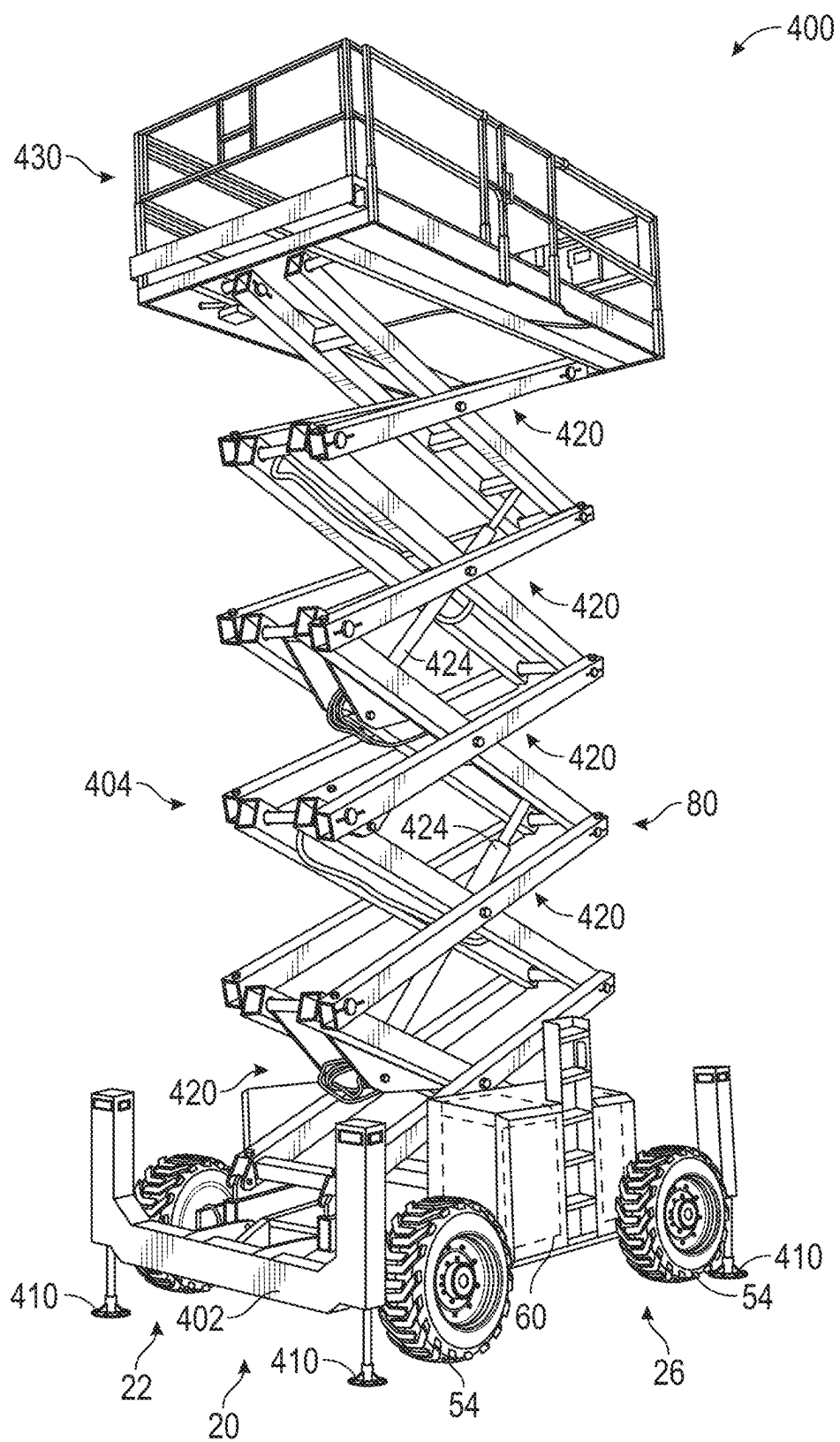


FIG. 13

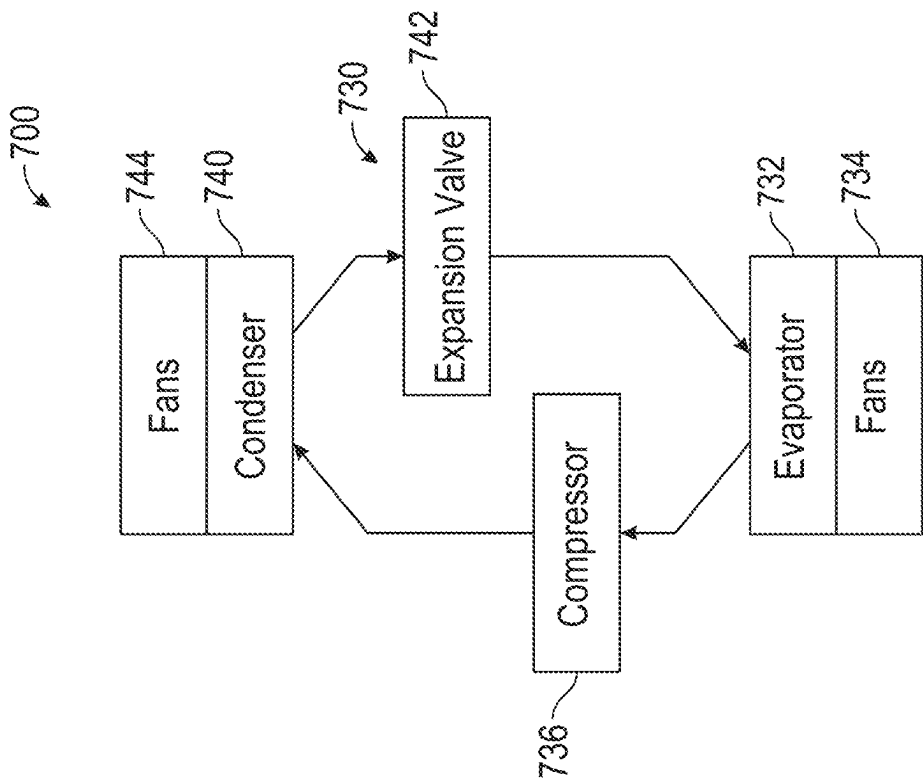


FIG. 15

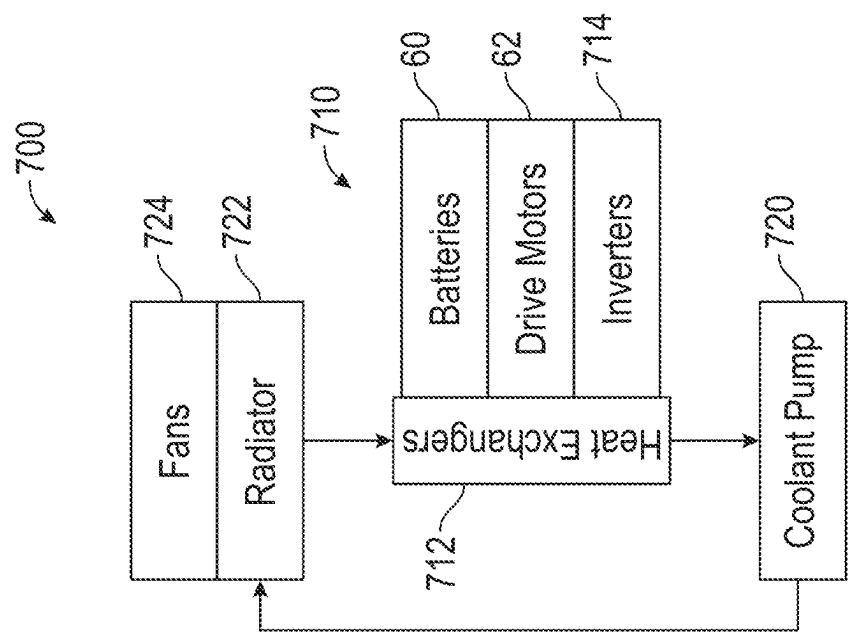


FIG. 14

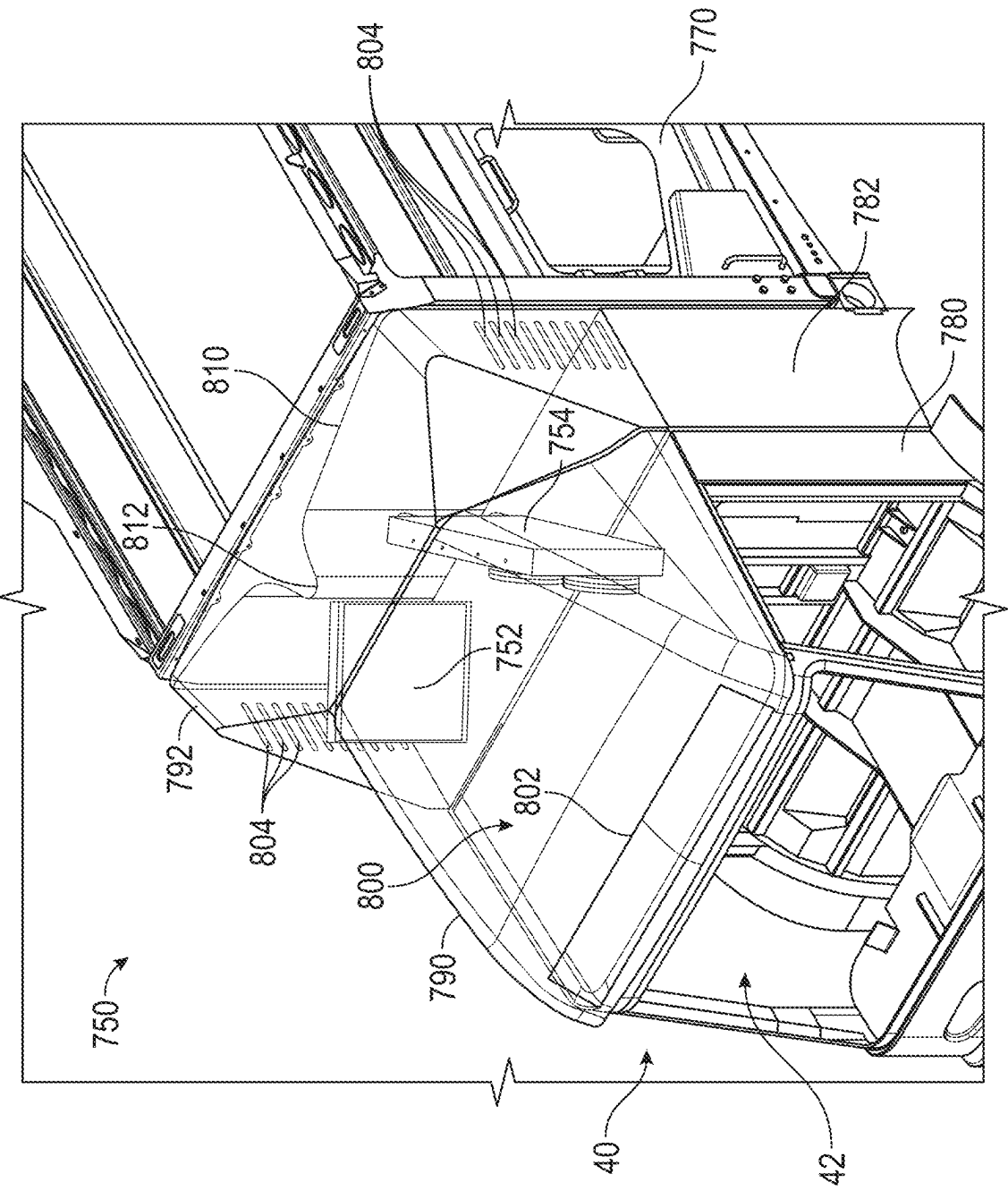


FIG. 16



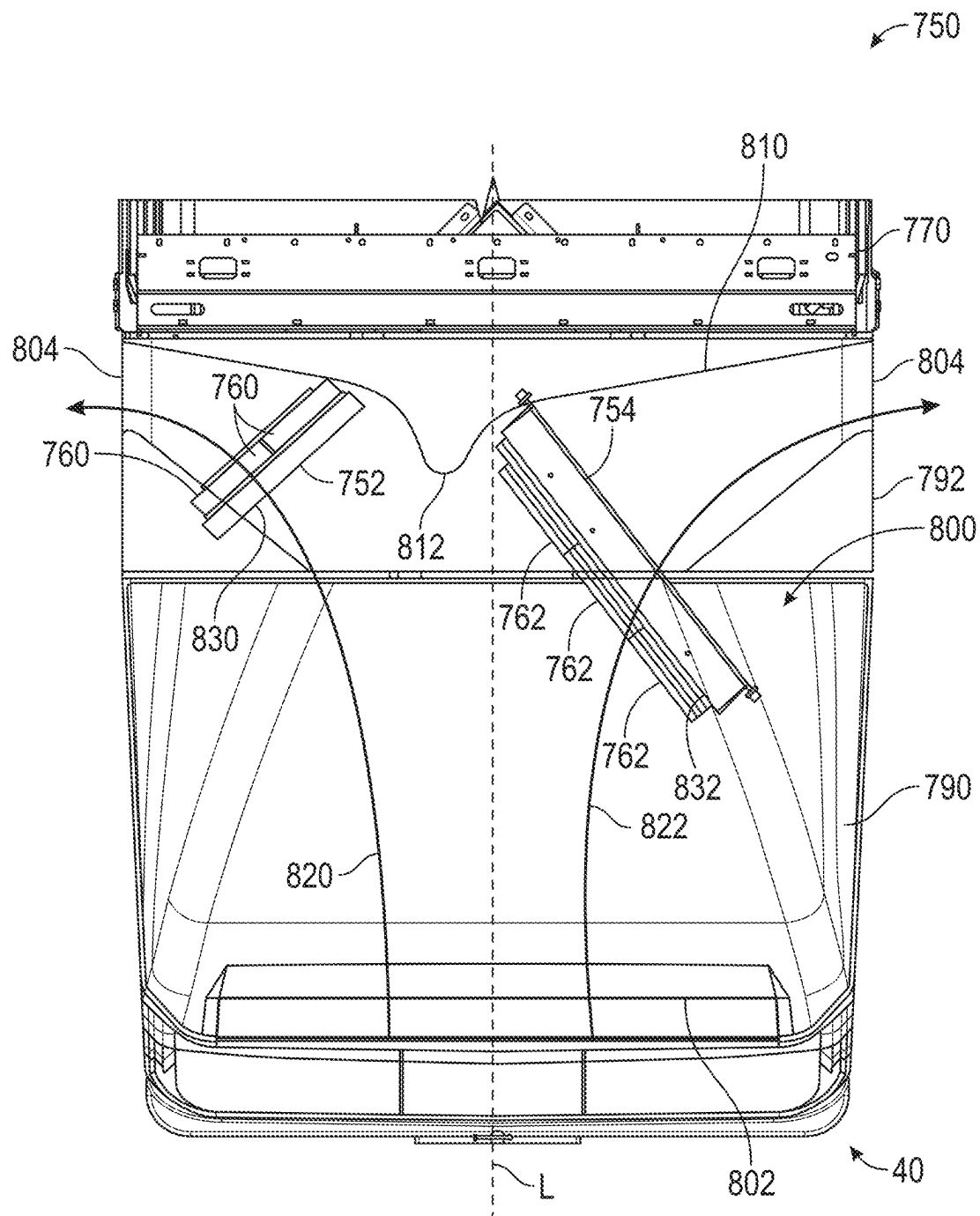
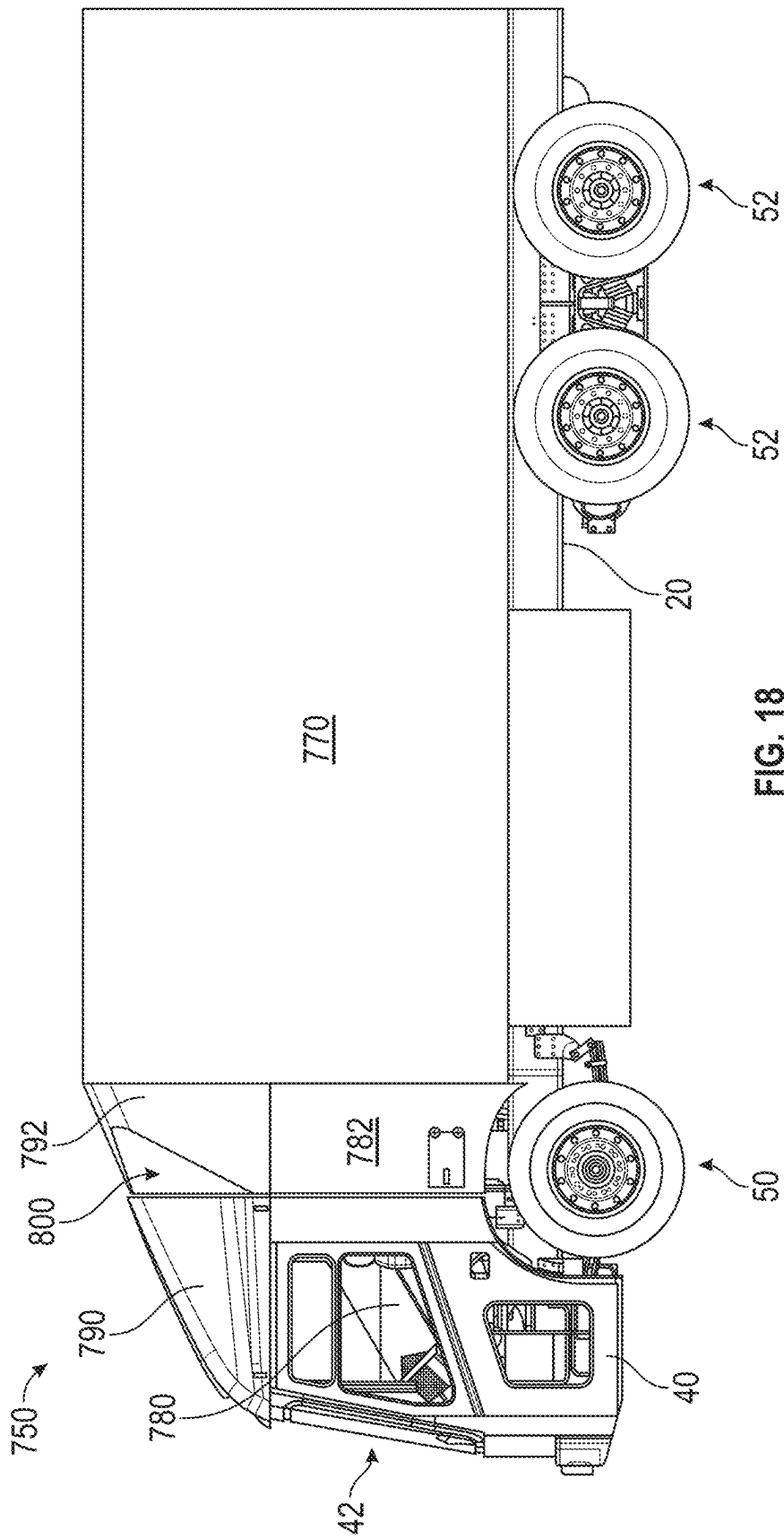


FIG. 17



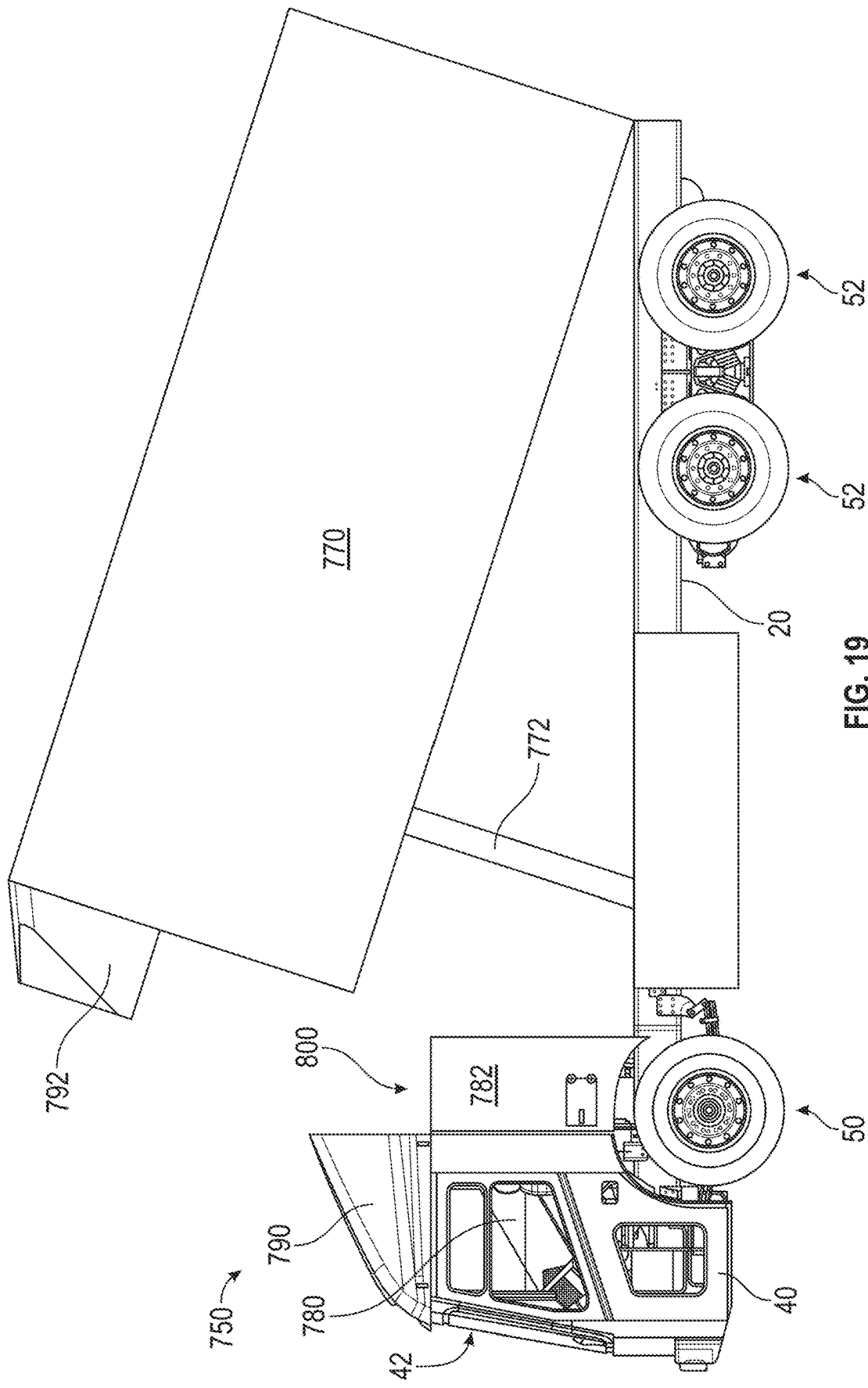


FIG. 19

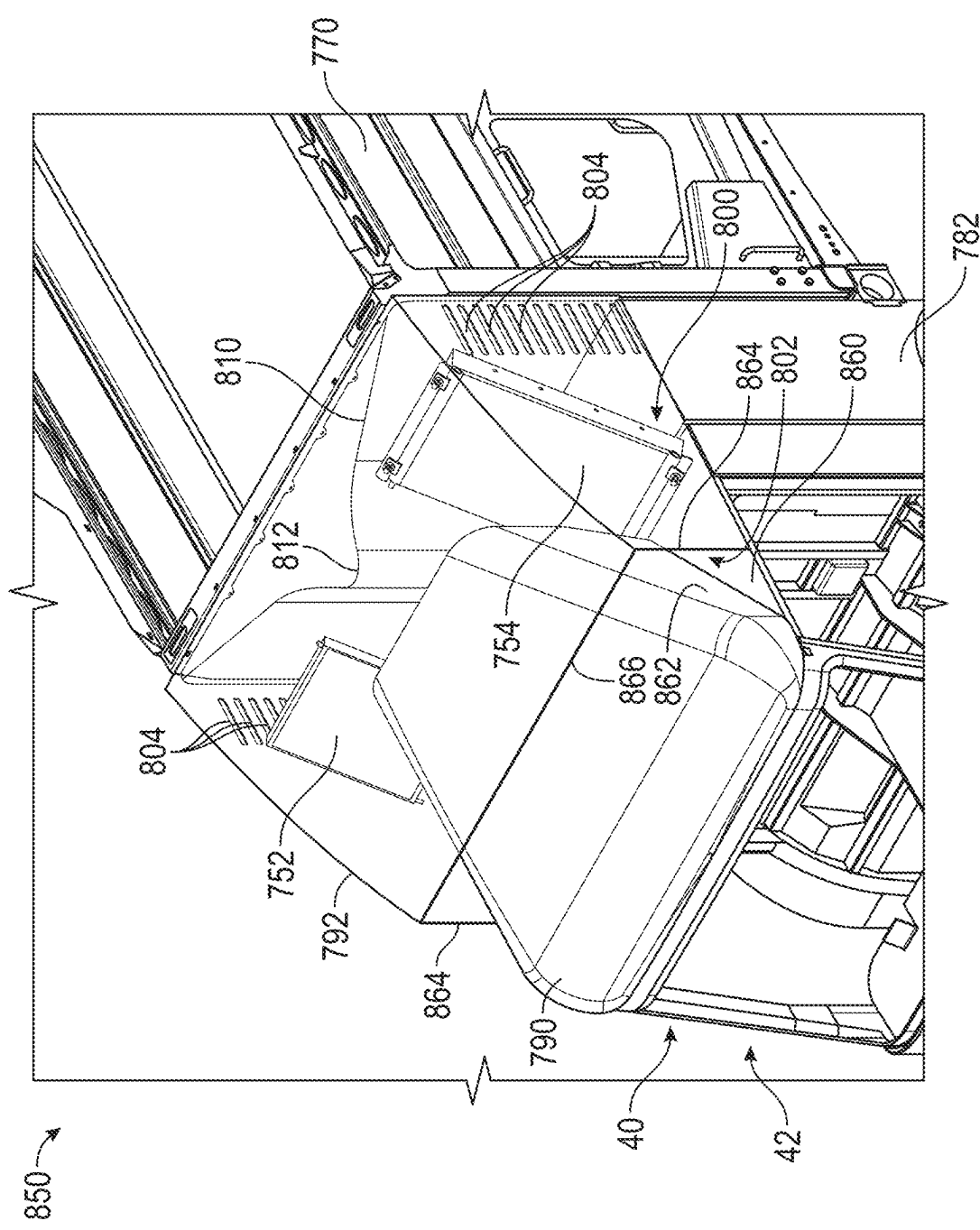


FIG. 20

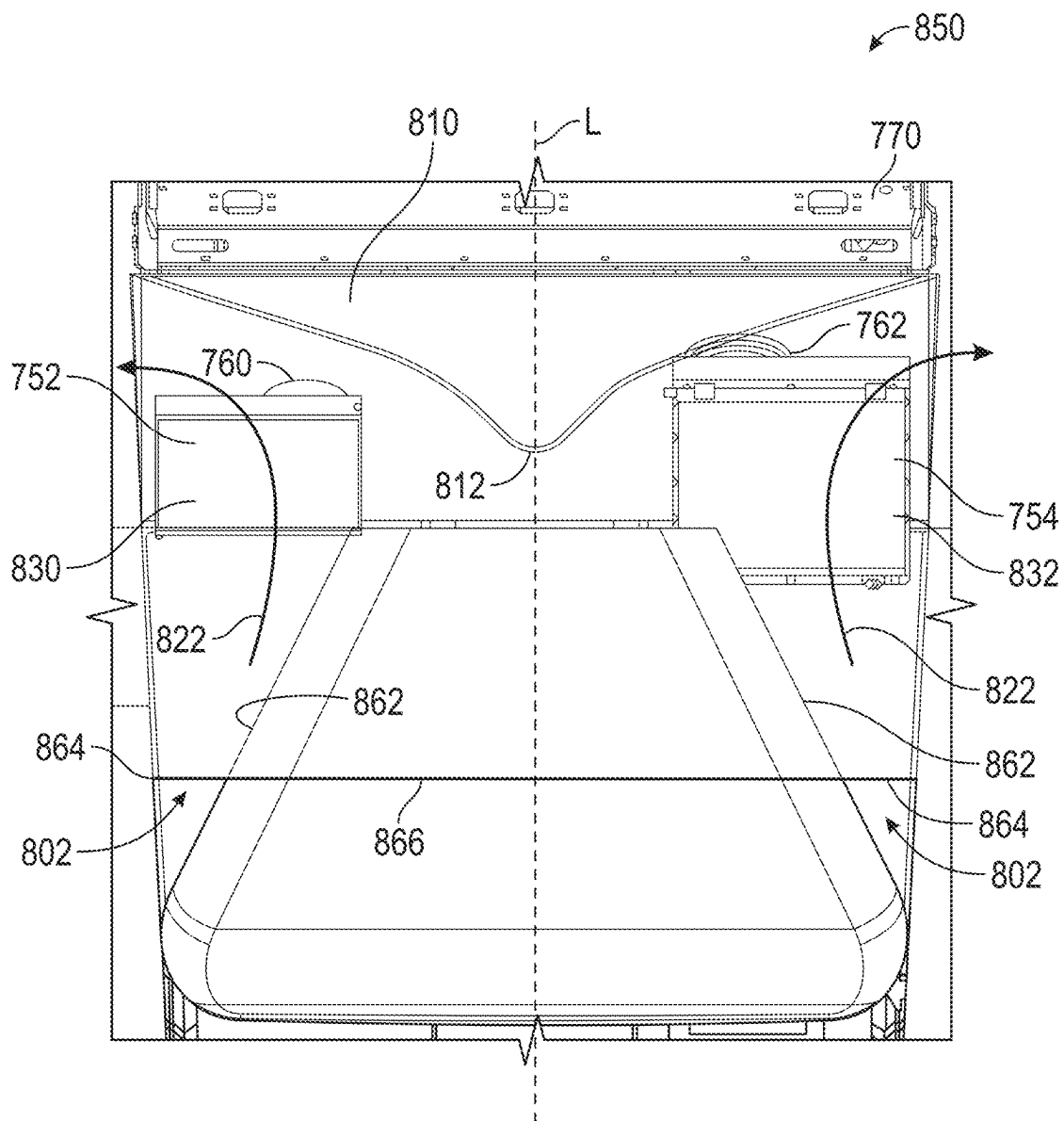
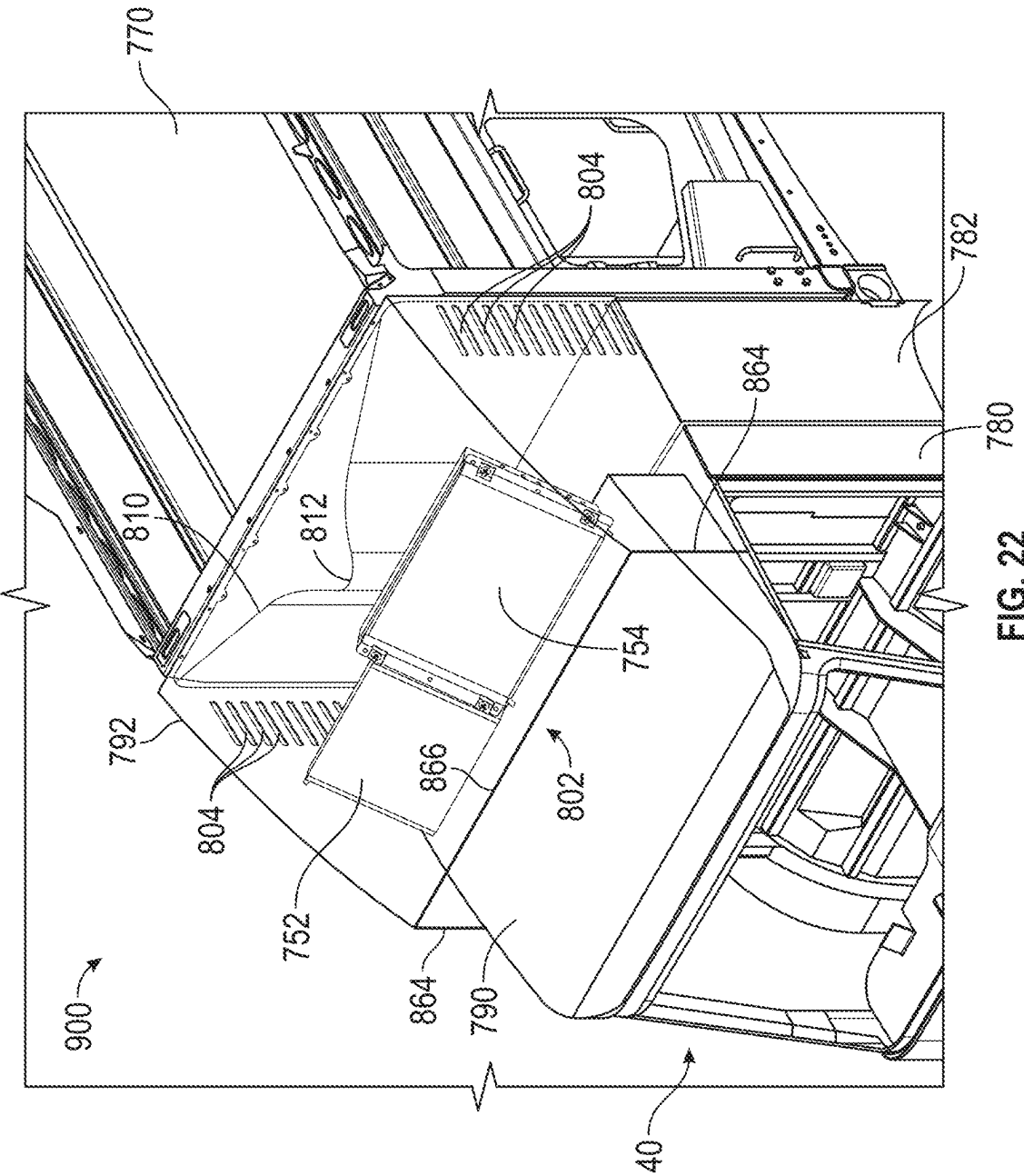


FIG. 21



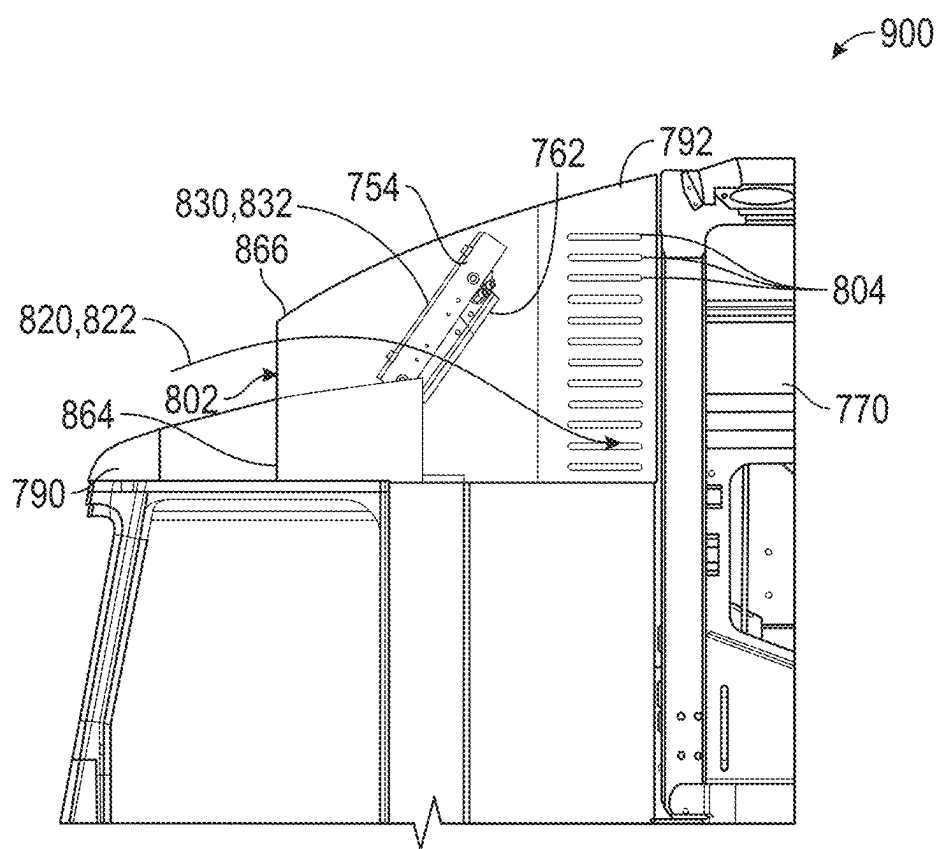


FIG. 23

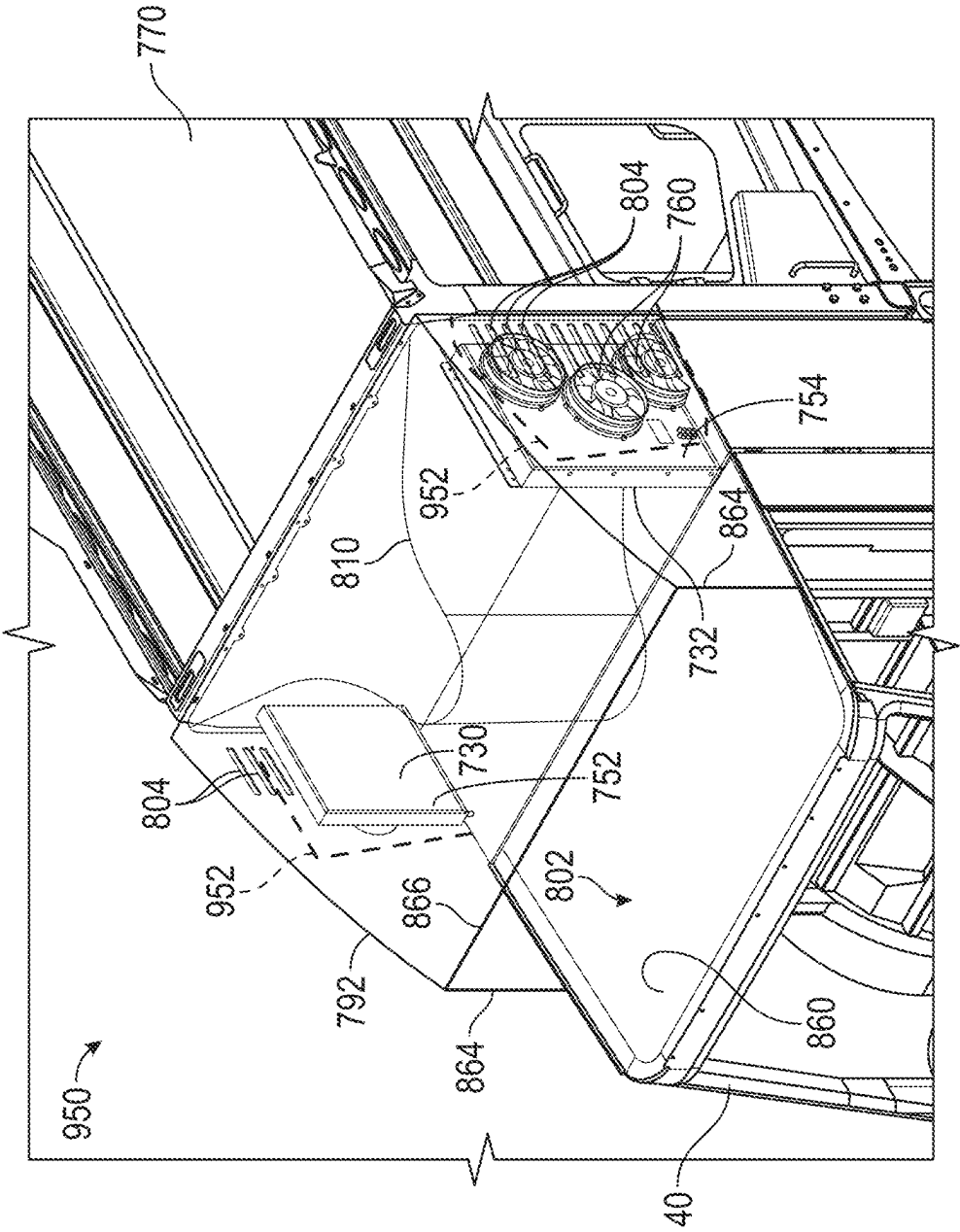
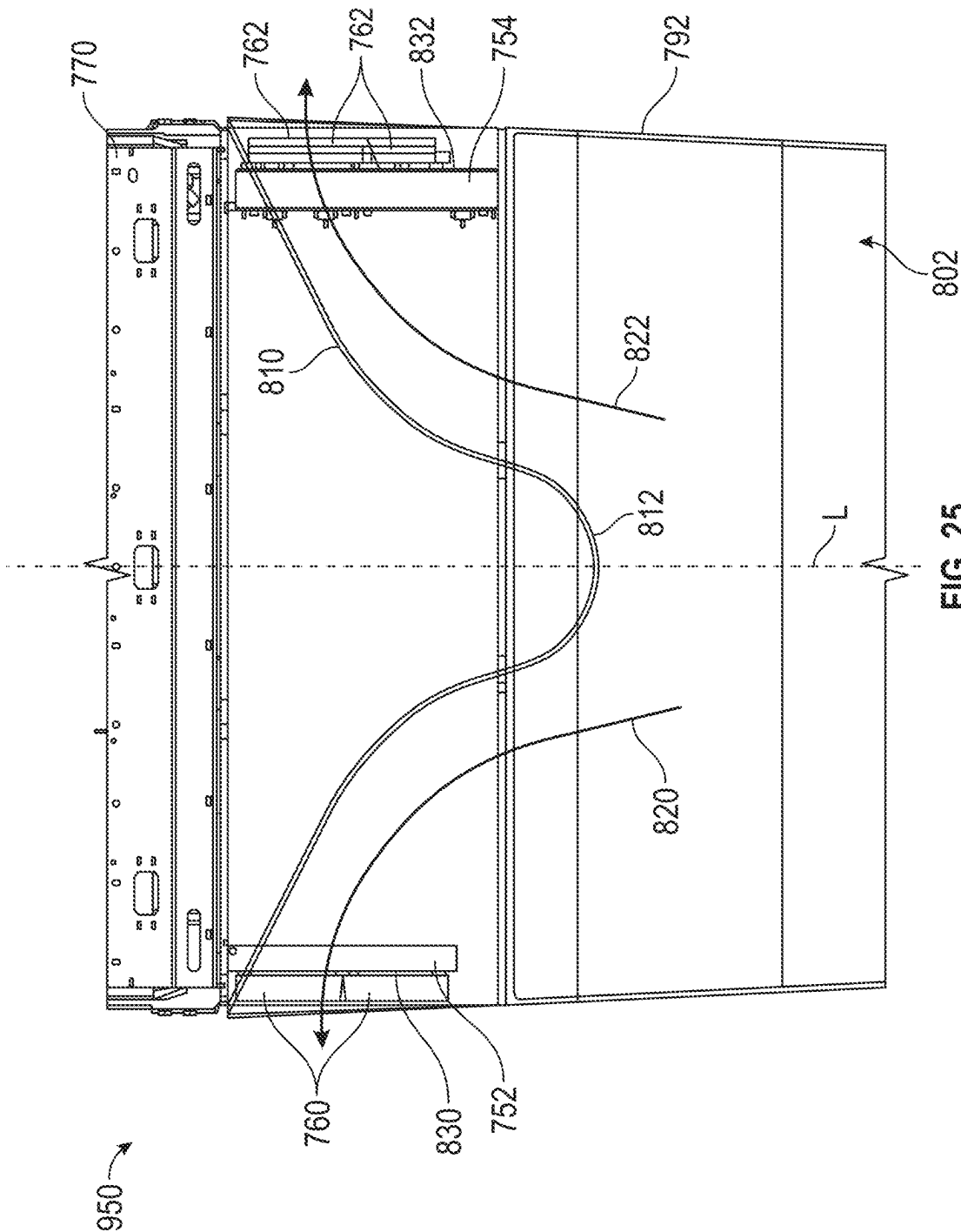
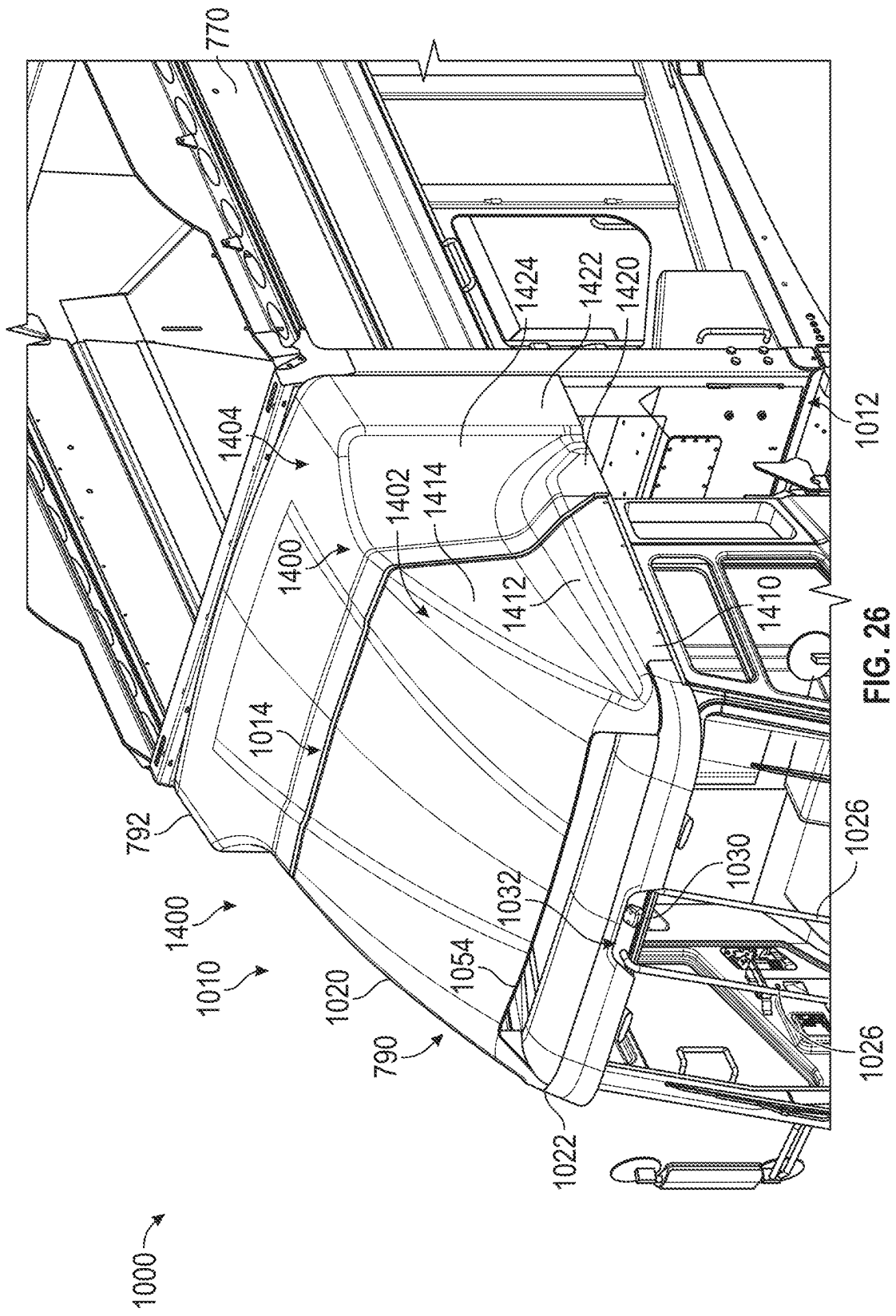


FIG. 24







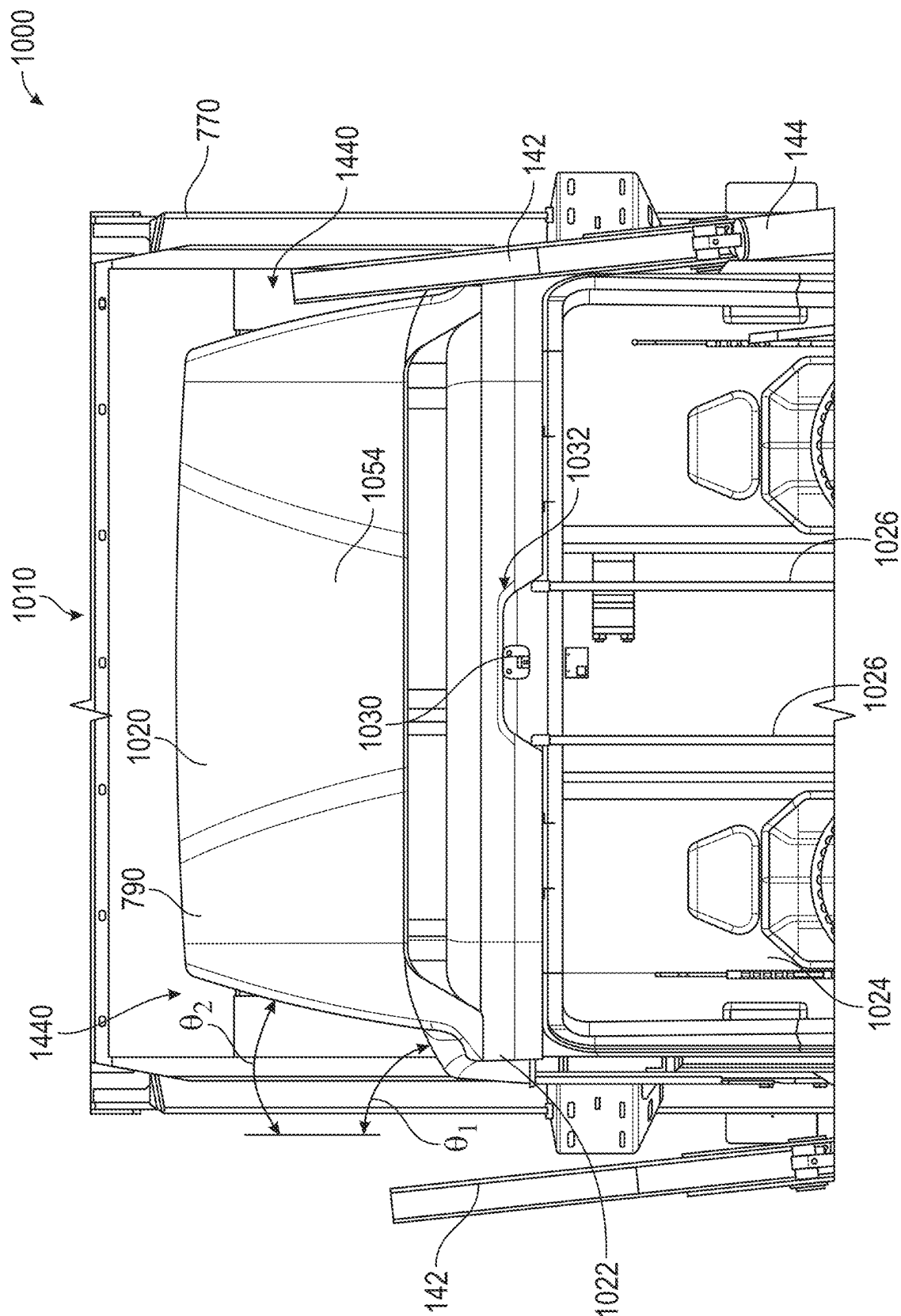


FIG. 27

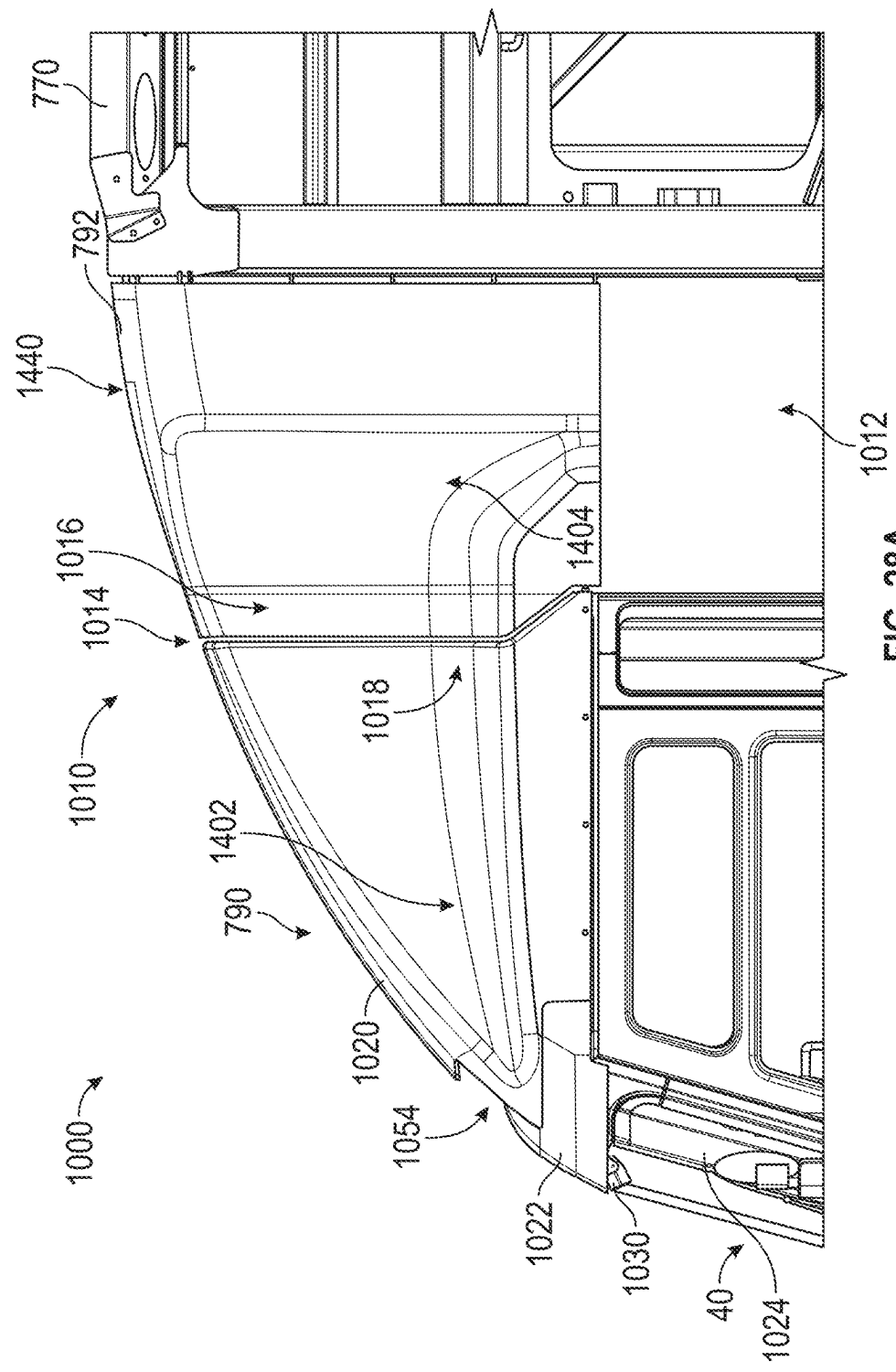
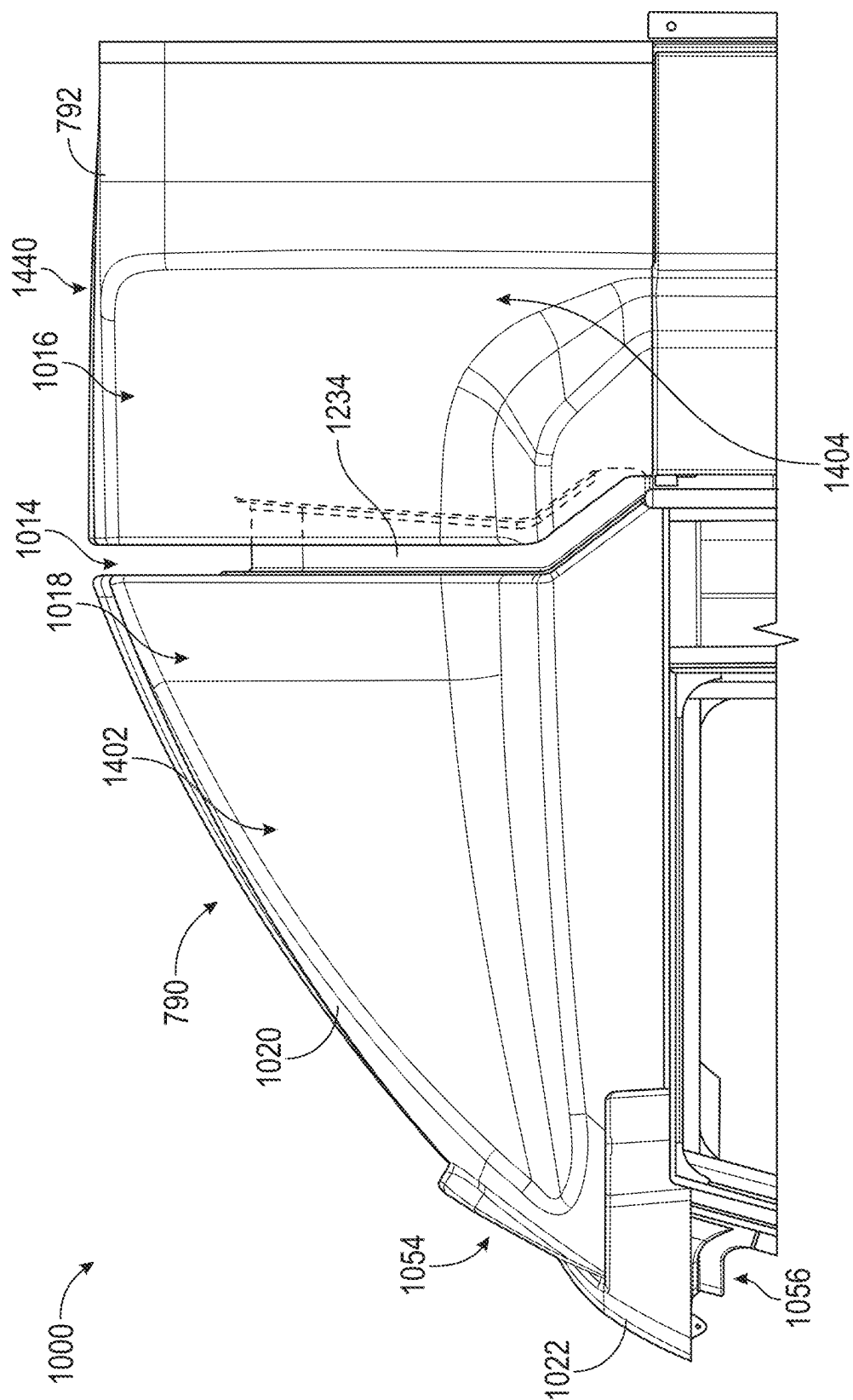


FIG. 28A



**Fig. 28B**

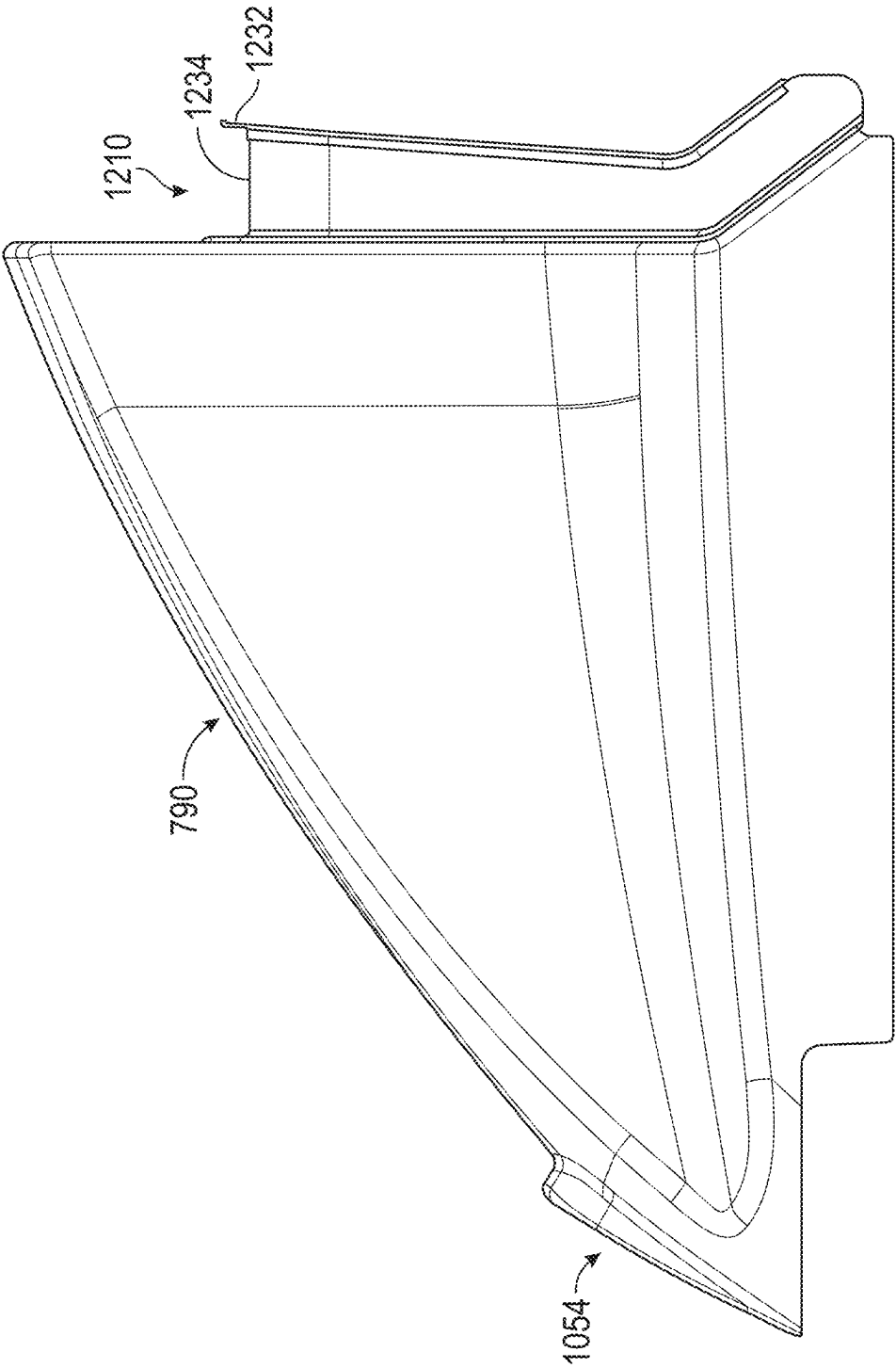


FIG. 28C

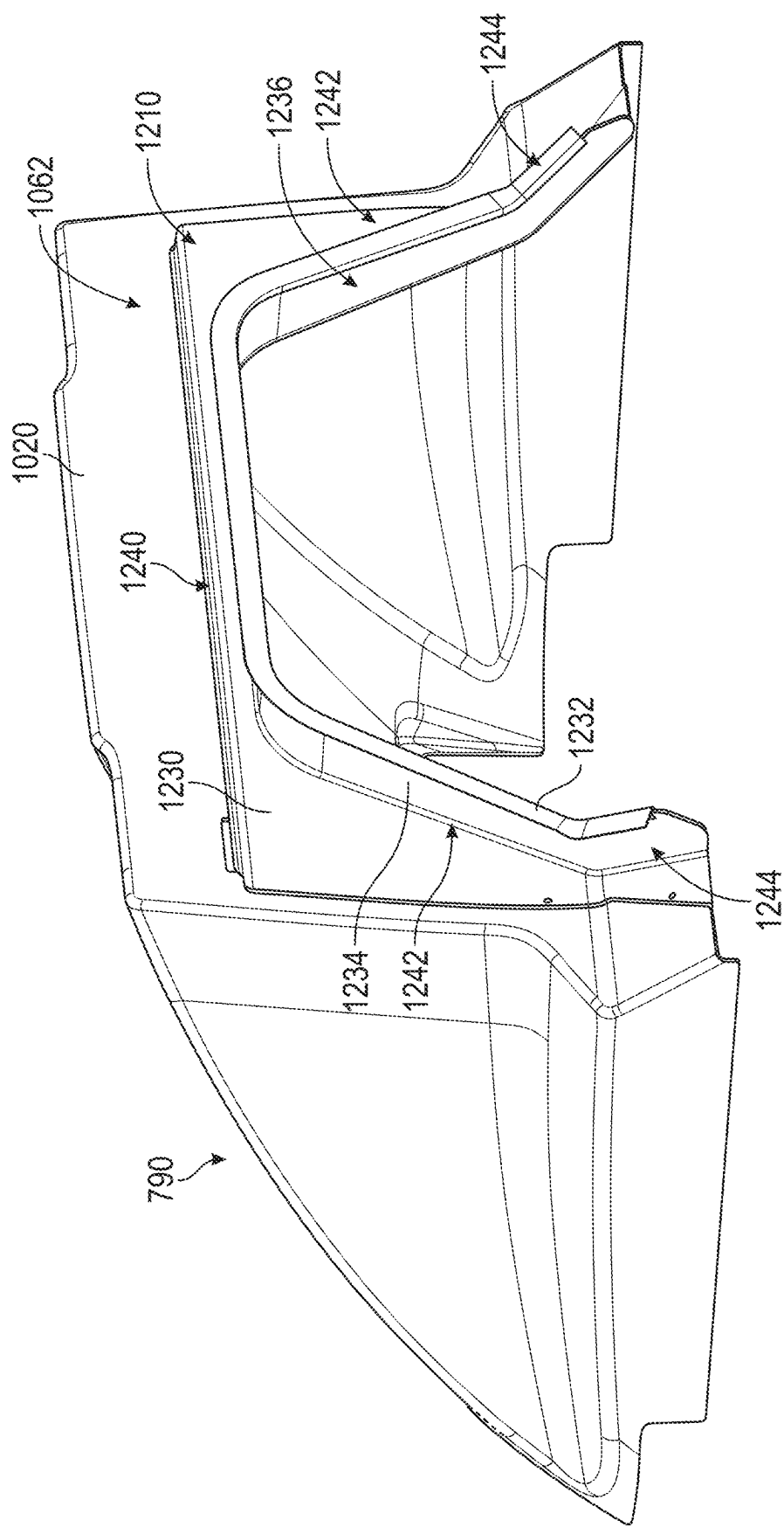


FIG. 28D

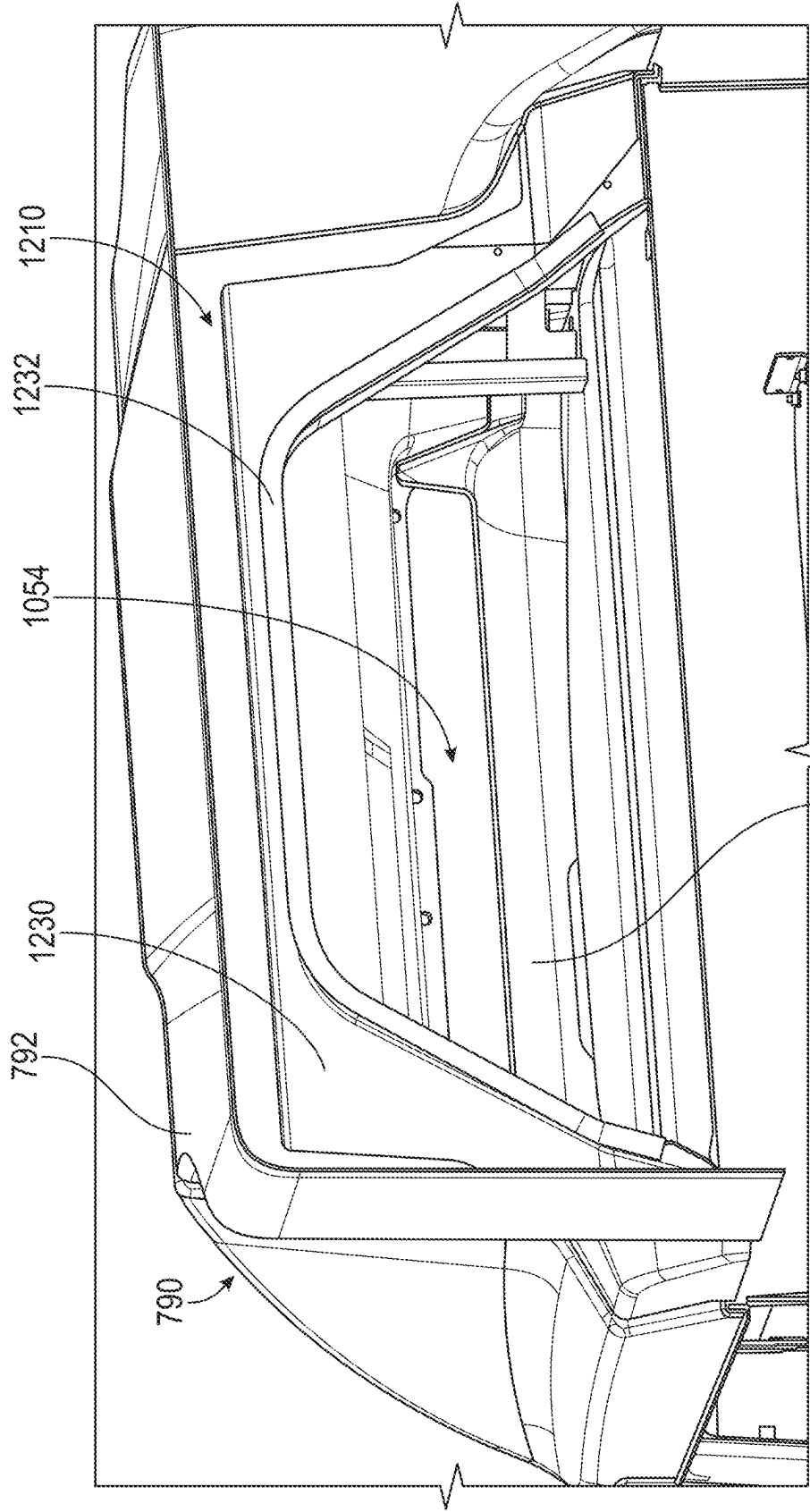


FIG. 28E



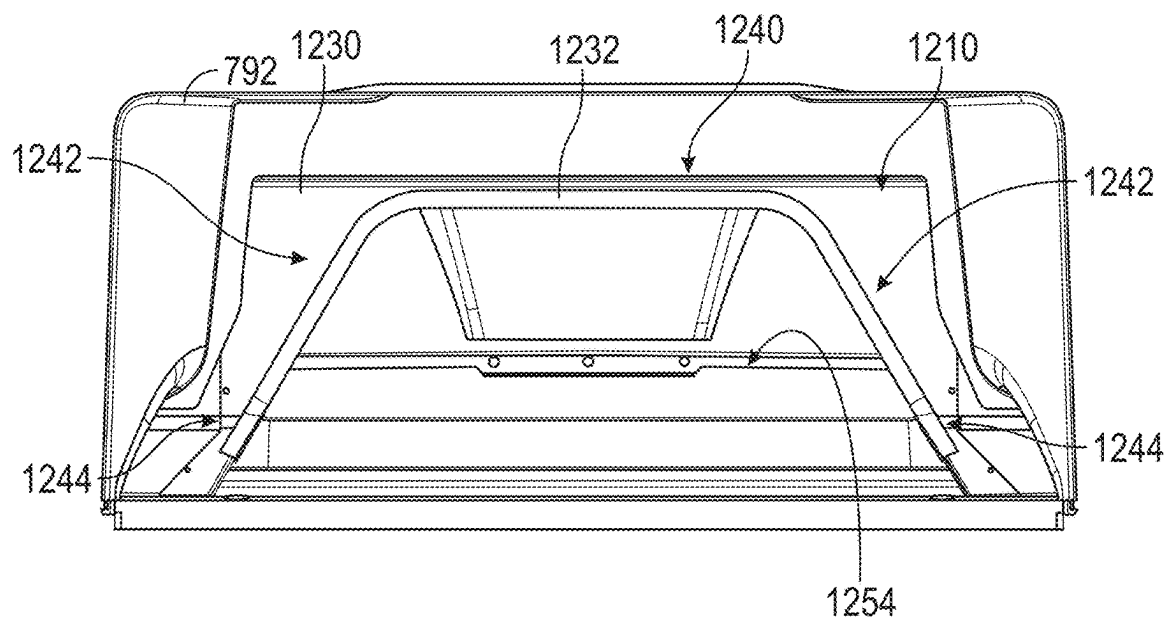


FIG. 28F

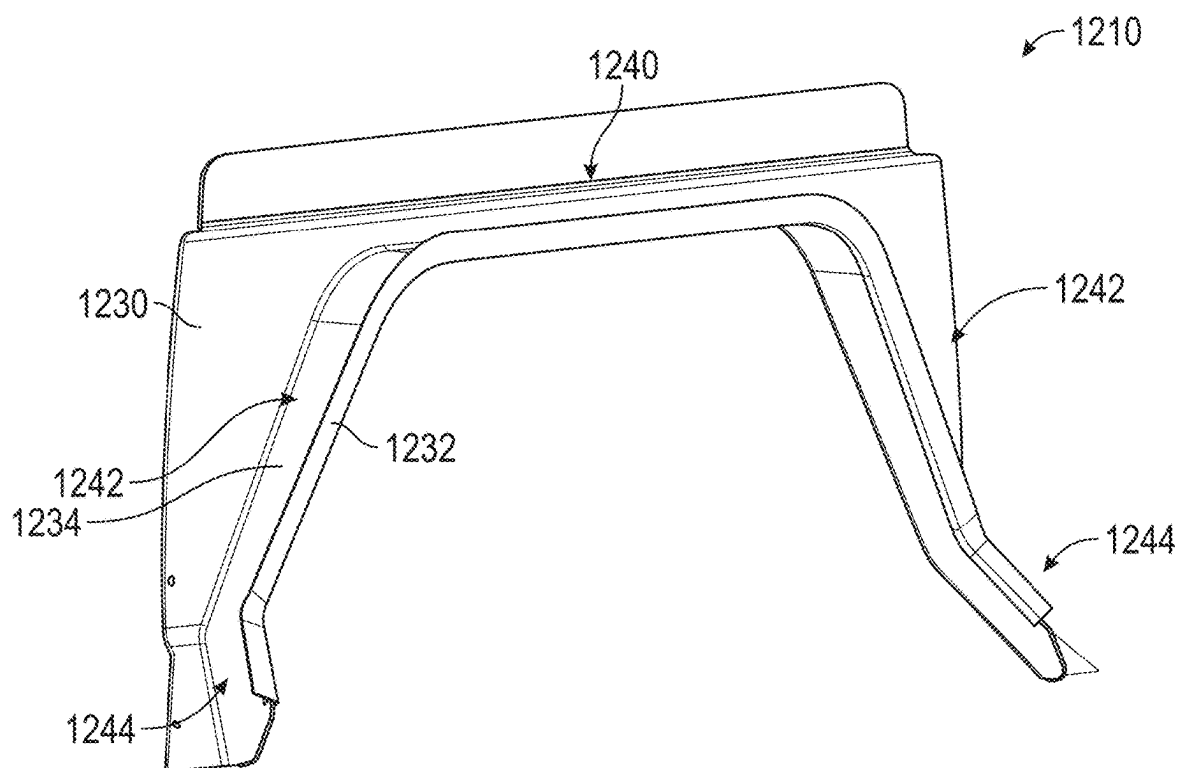


FIG. 28G

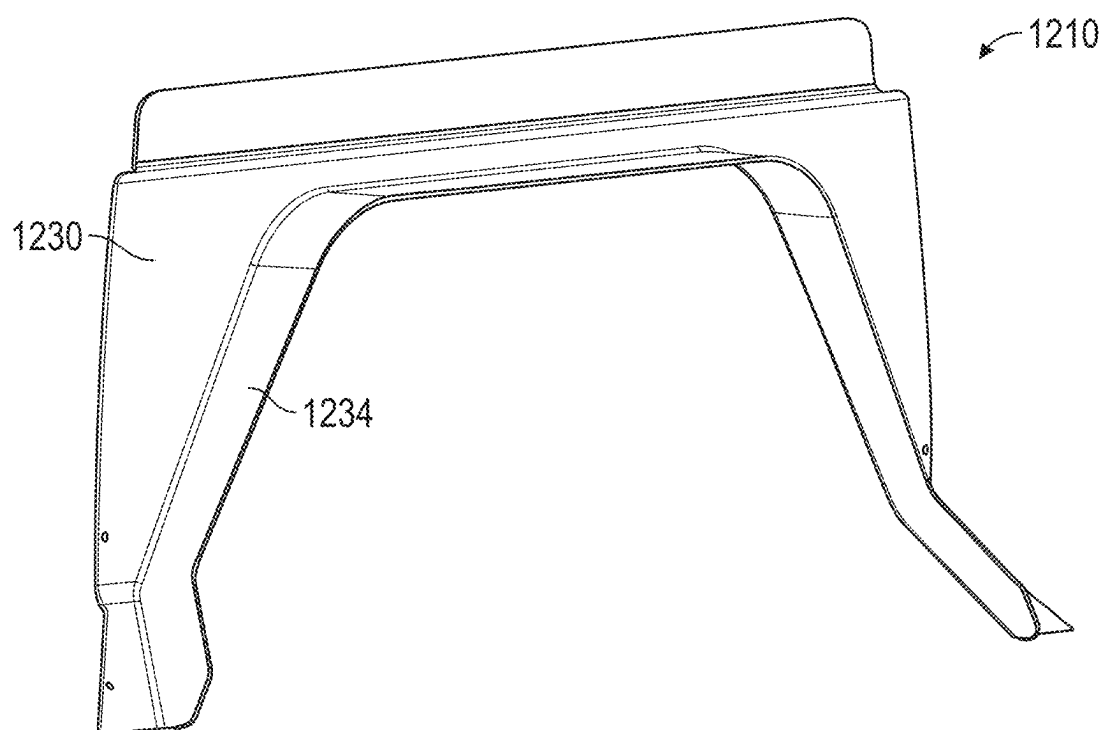
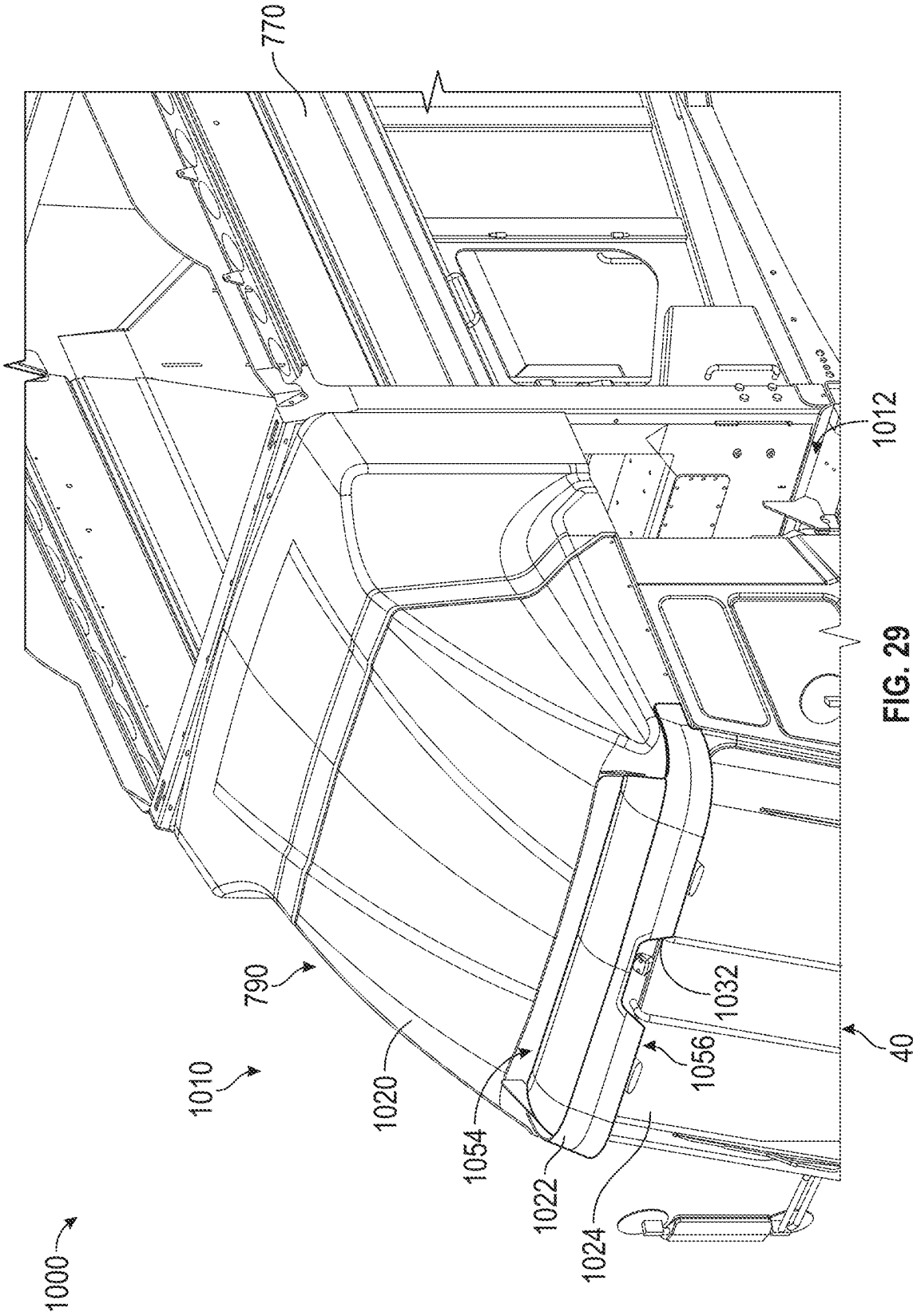


FIG. 28H



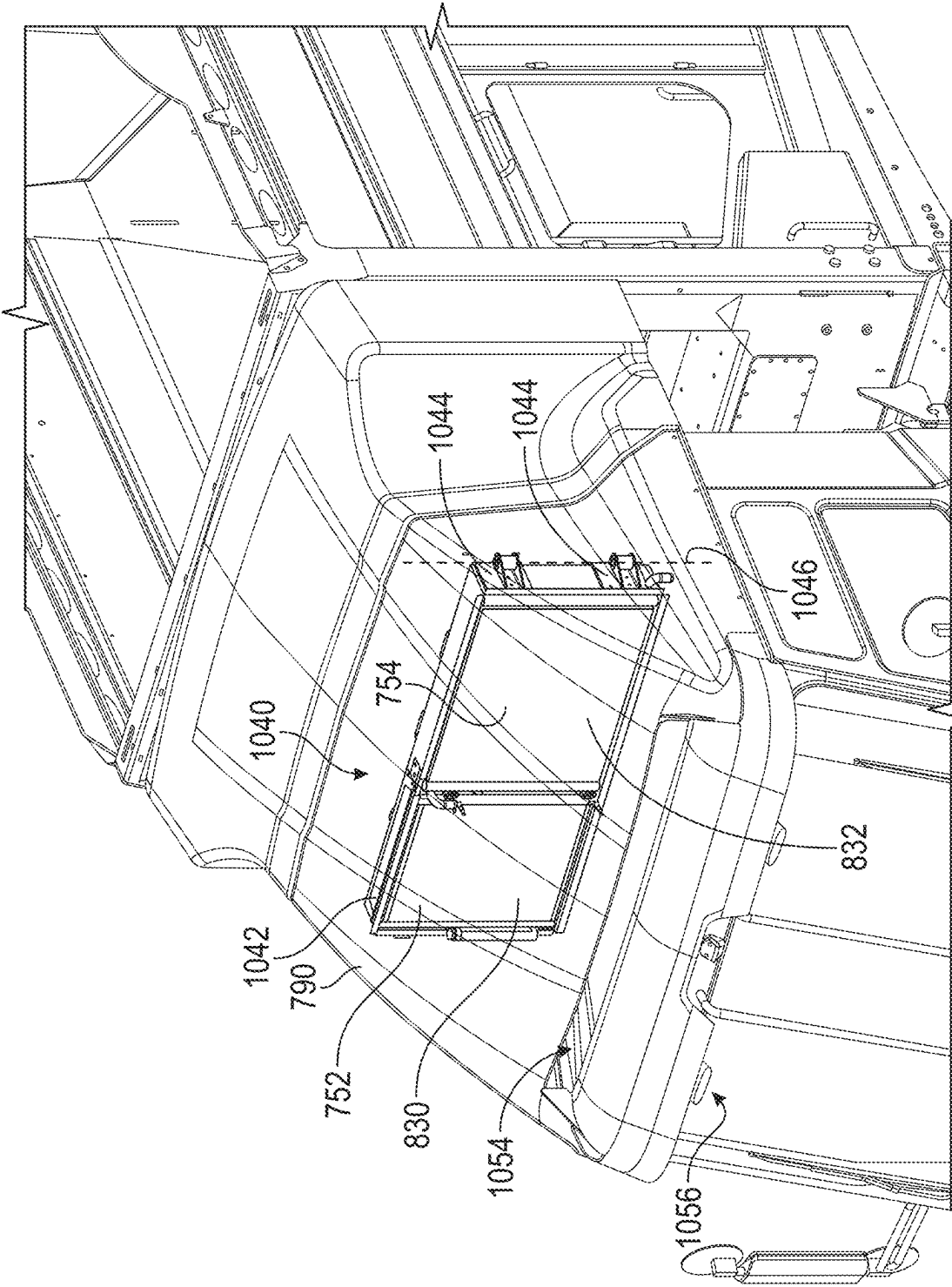


FIG. 30

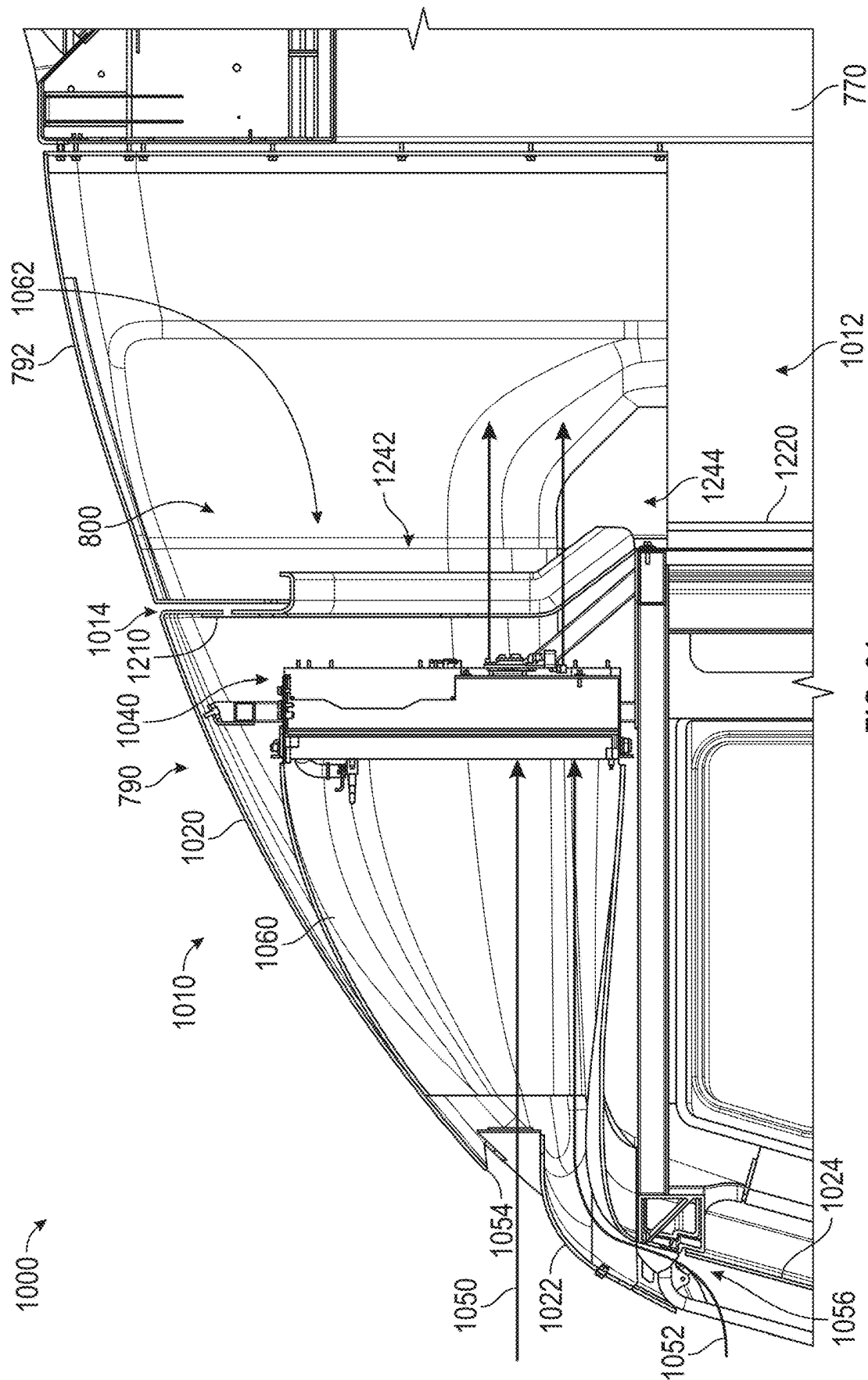


FIG. 31

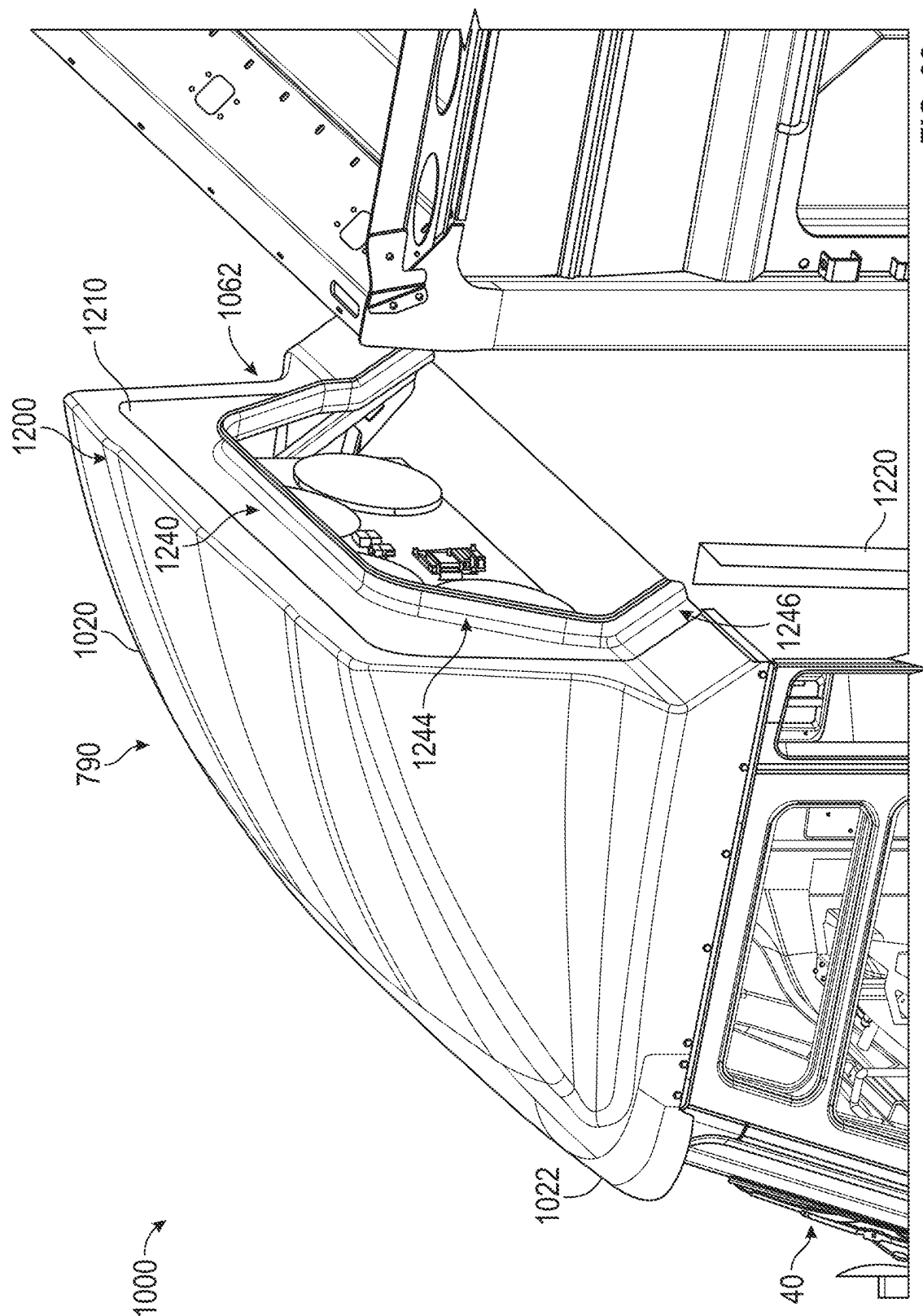


FIG. 32

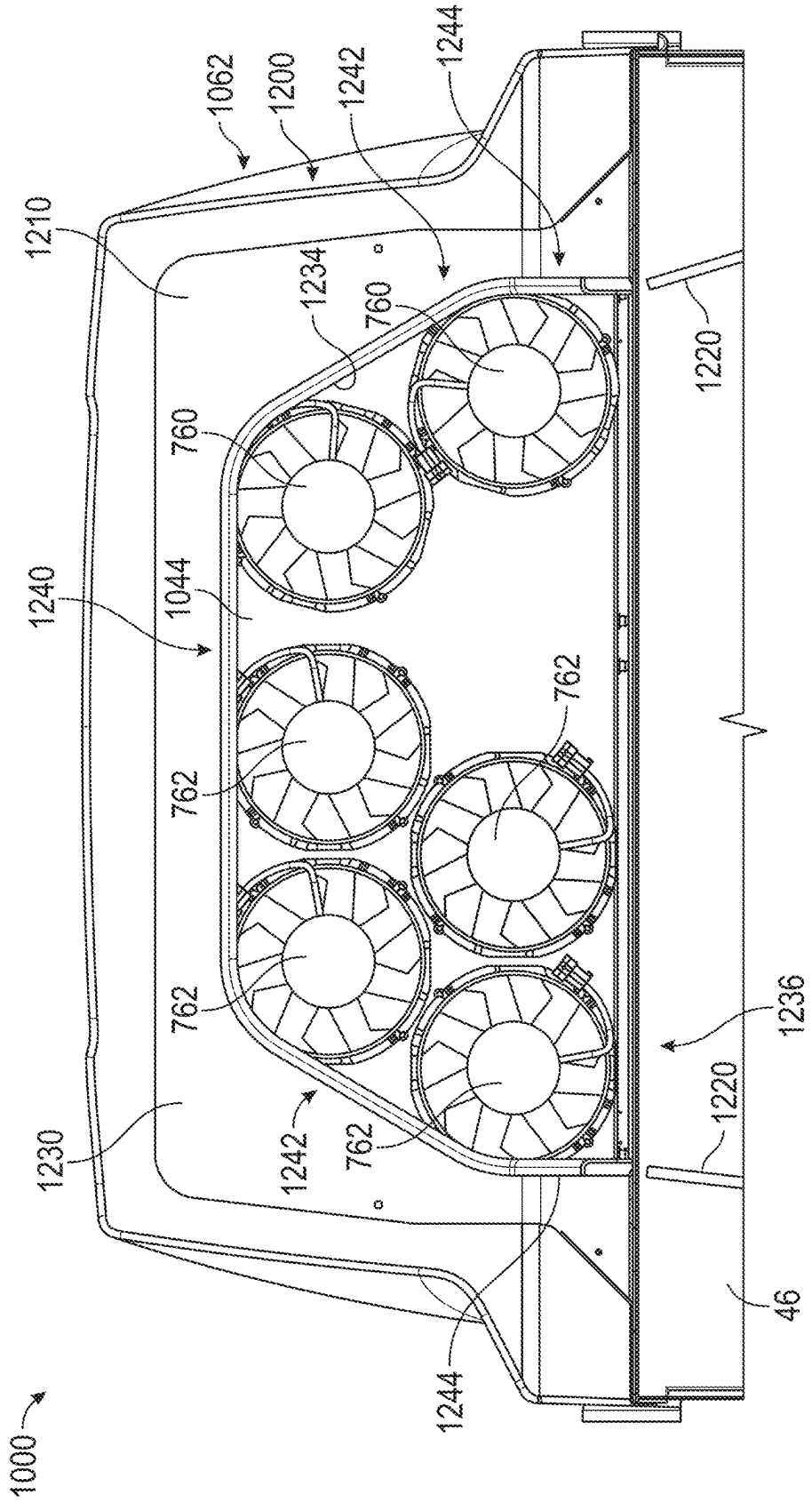


FIG. 33

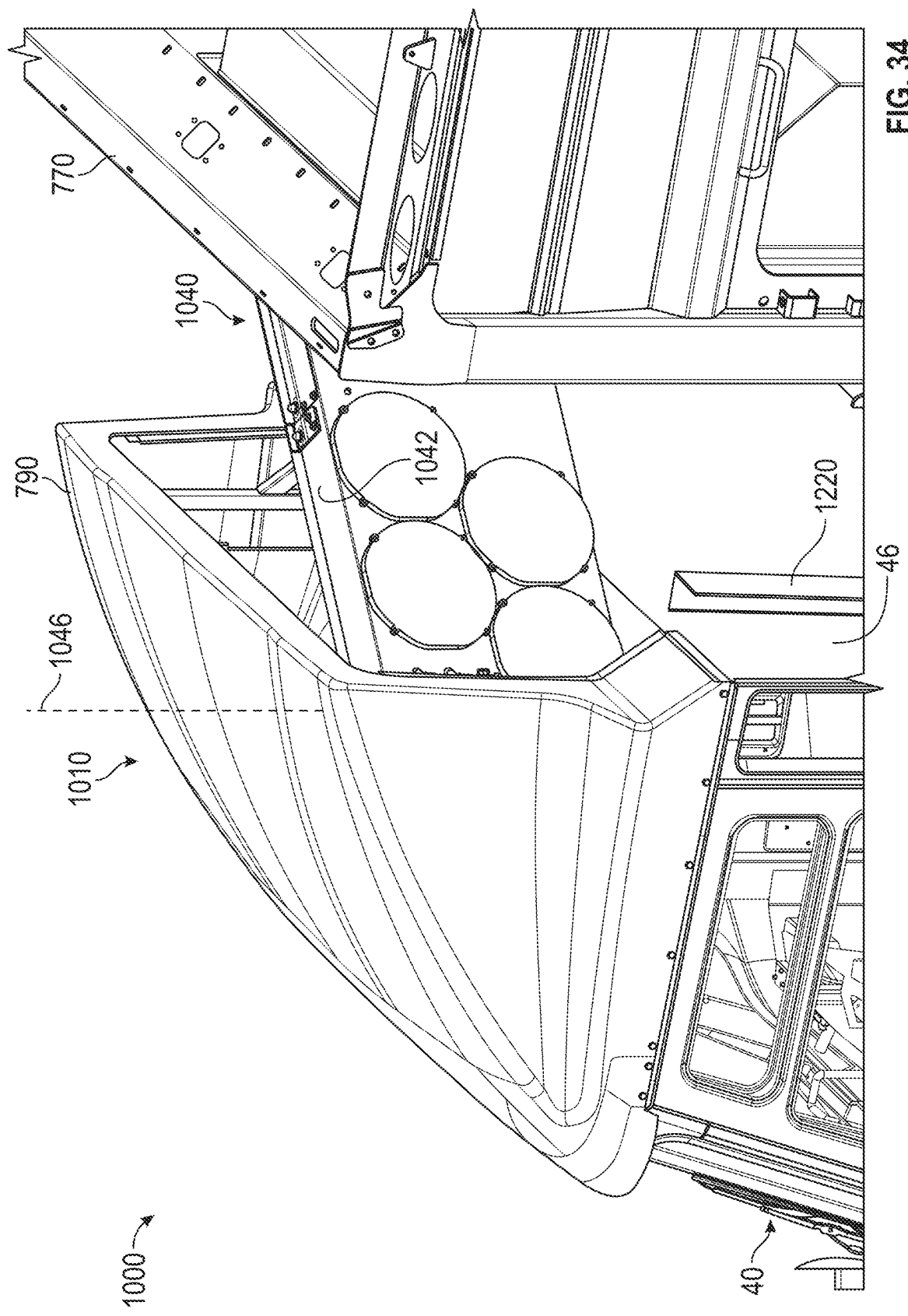


FIG. 34





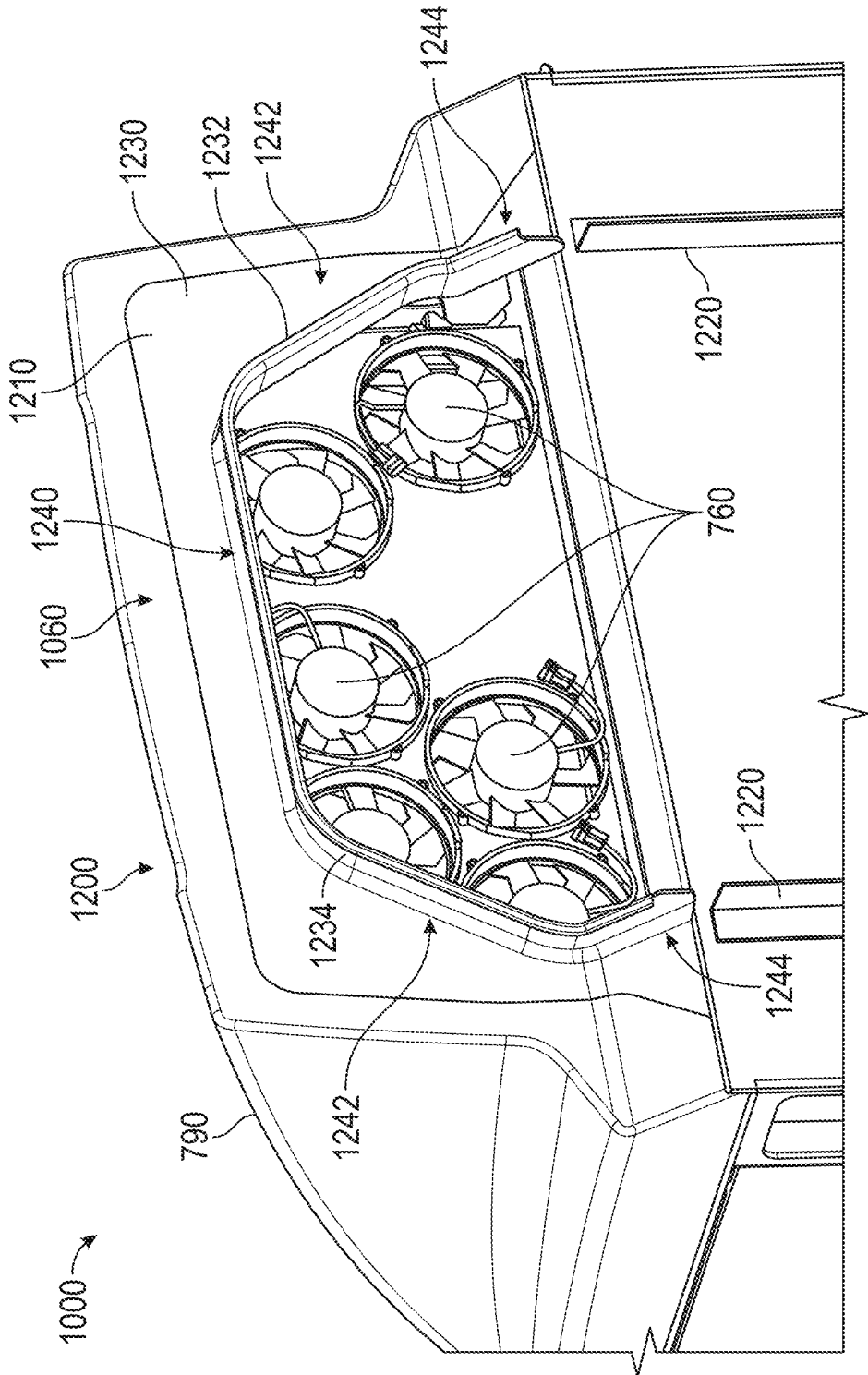


FIG. 36

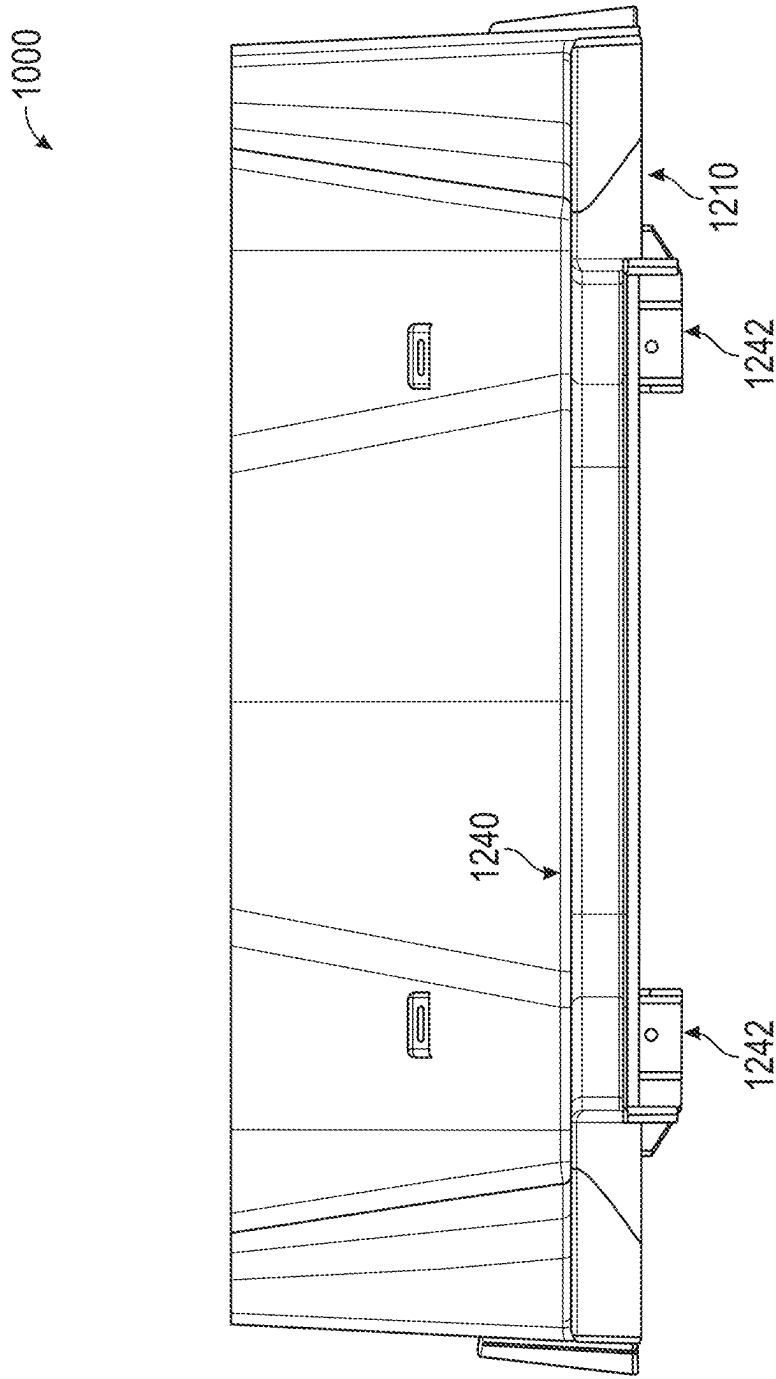


FIG. 37

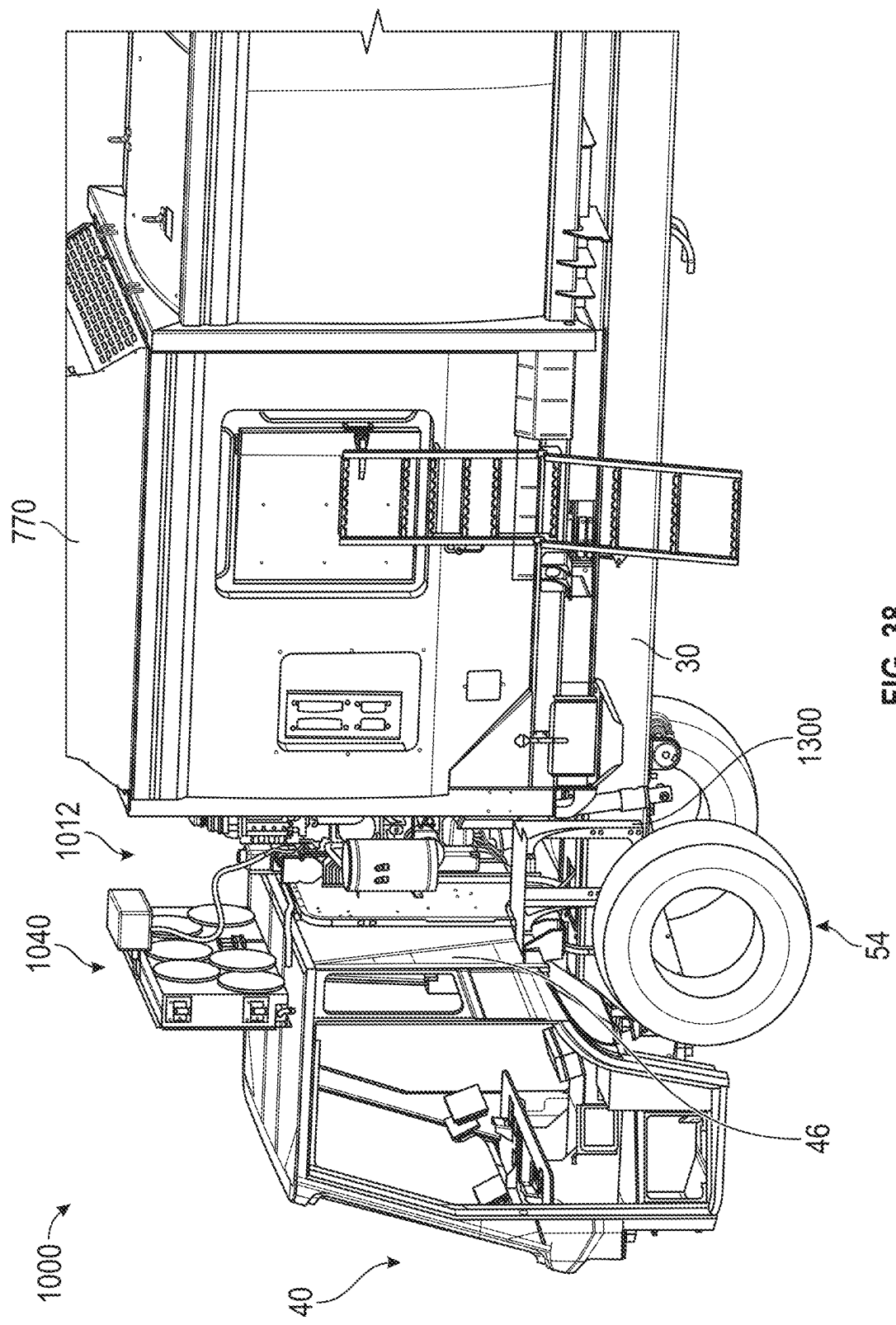


FIG. 38

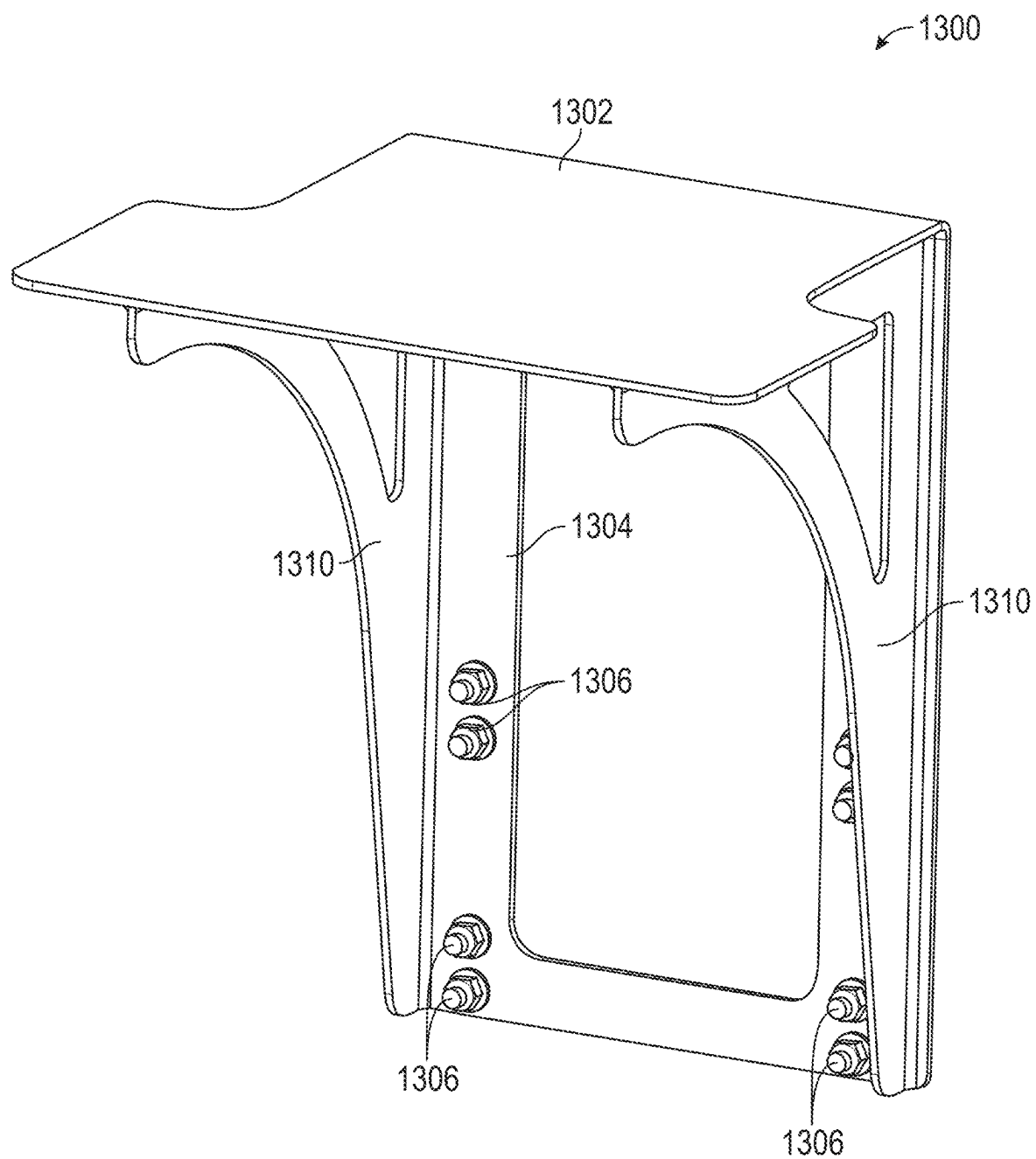


FIG. 39

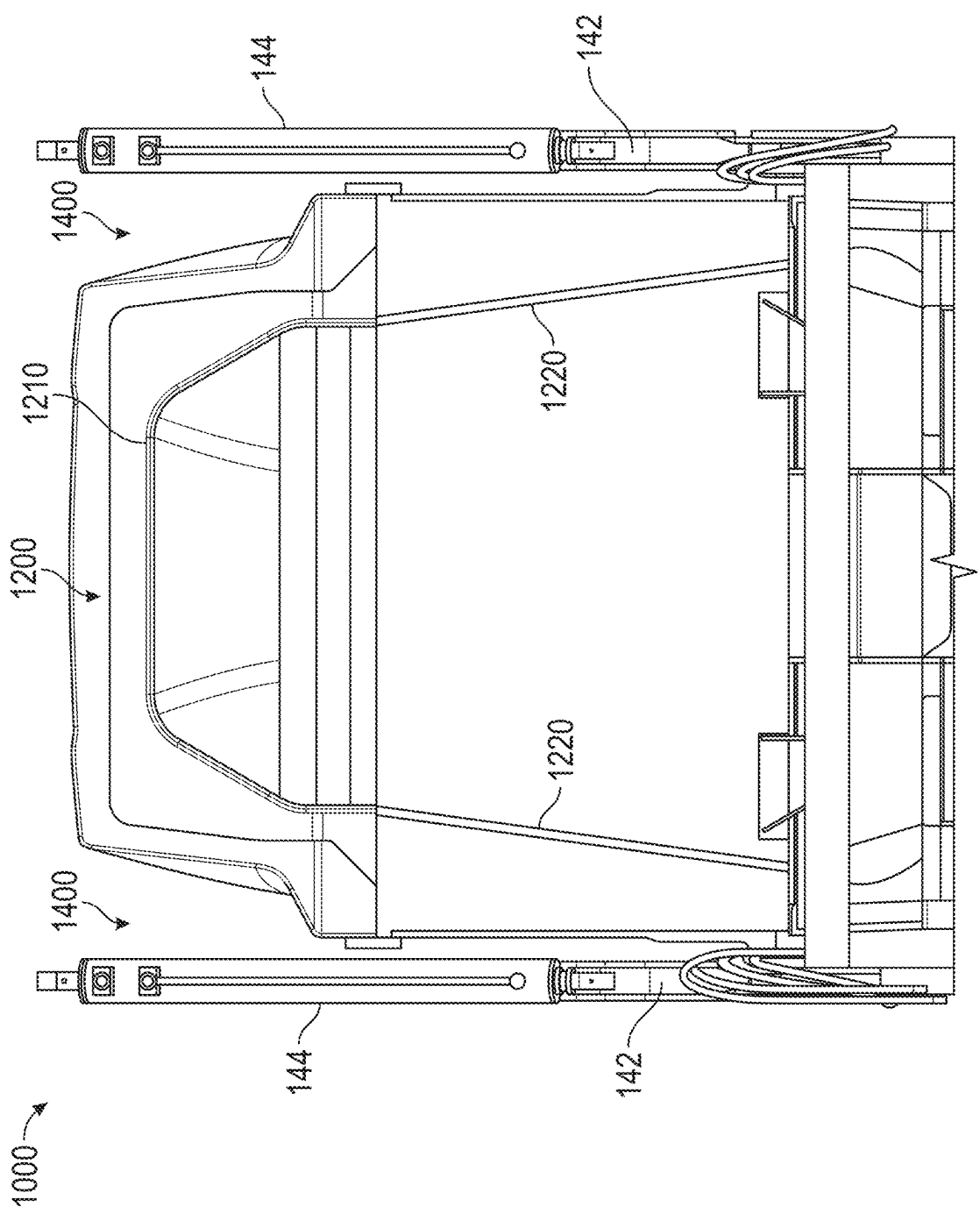
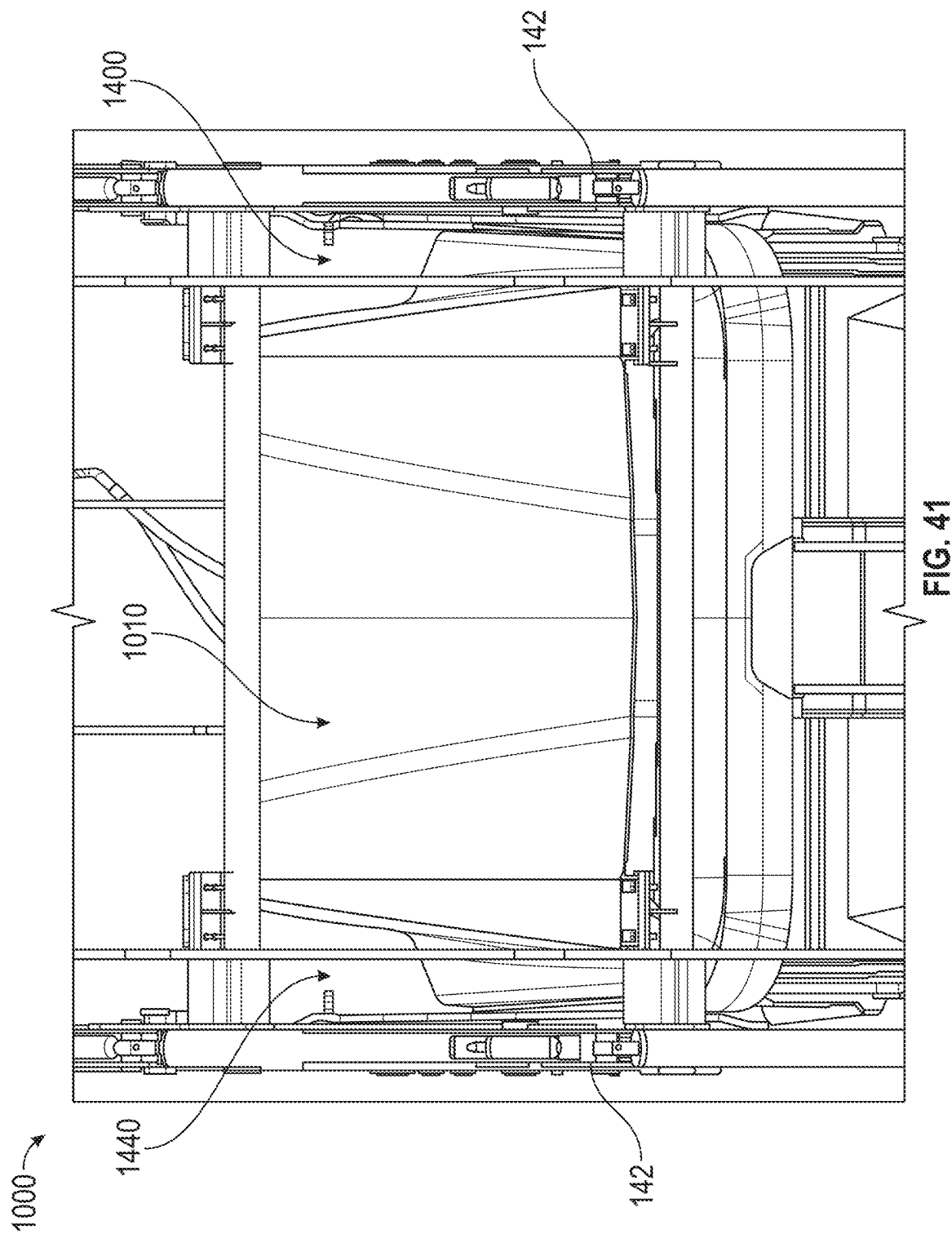
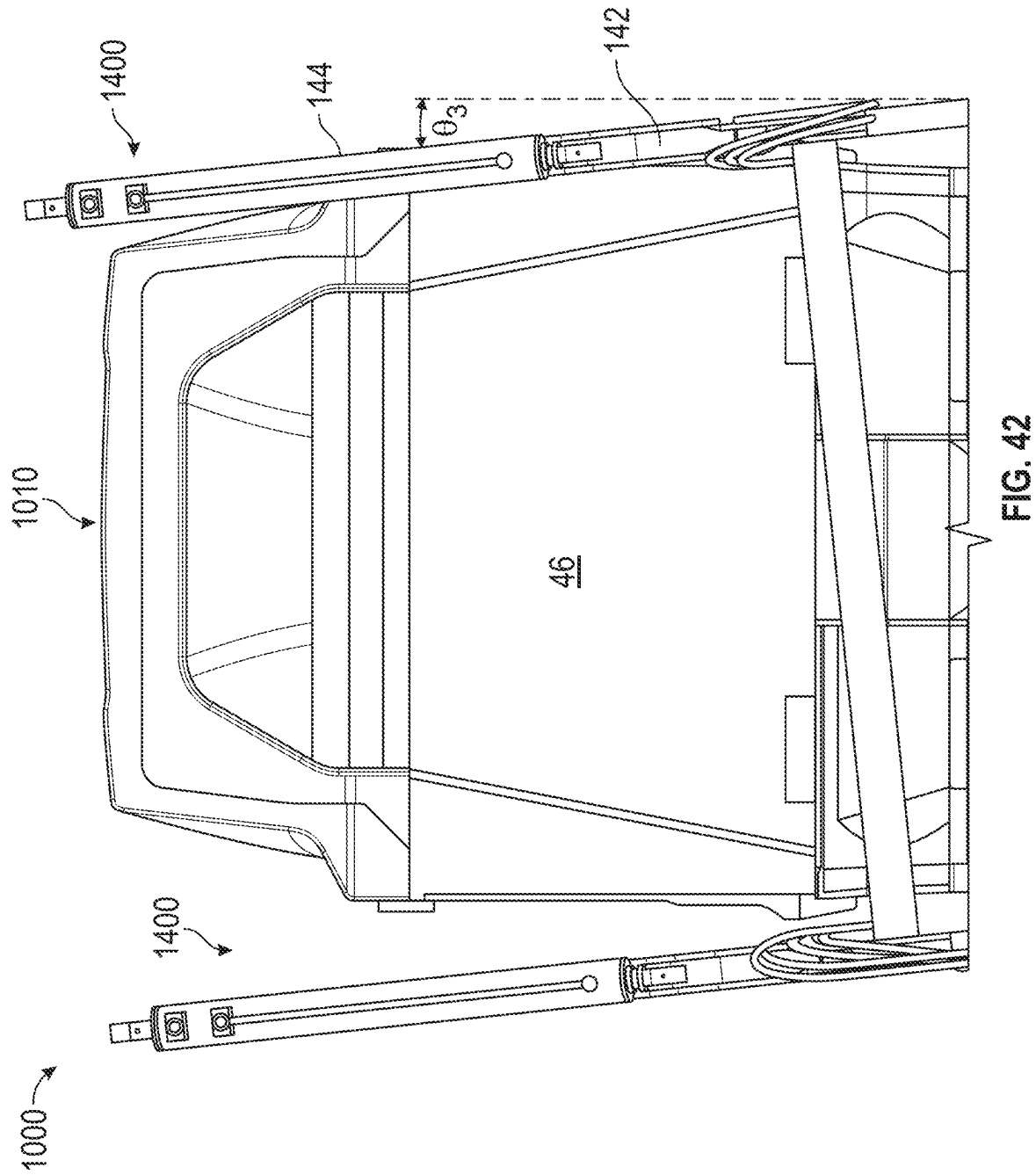
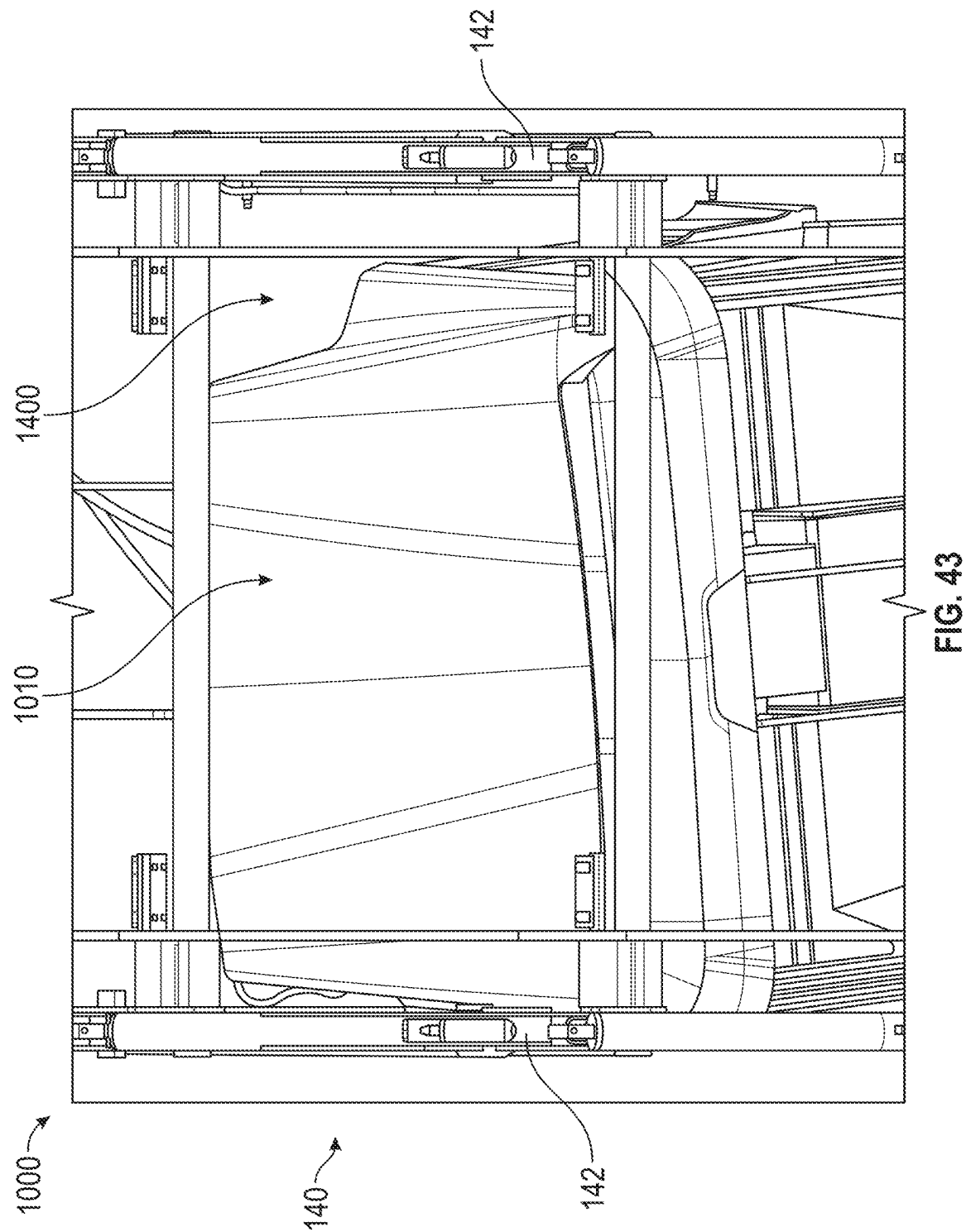


FIG. 40









## VEHICLE WITH THERMAL MANAGEMENT SYSTEM

### CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

[0001] This application is a continuation of Ser. No. 17/852,087, filed on Jun. 28, 2022, which claims the benefit of and priority to U.S. Provisional Application No. 63/216,492, filed on Jun. 29, 2021, U.S. Provisional Application No. 63/325,684, filed on Mar. 31, 2022, and U.S. Provisional Application No. 63/325,810, filed on Mar. 31, 2022, the entire disclosures of which are hereby incorporated by reference herein.

### BACKGROUND

[0002] The present disclosure relates generally to vehicles. More specifically, the present disclosure relates to a vehicle including a thermal management system. Thermal management systems for vehicles generally include one or more radiators that are exposed to airflow to cool one or more fluids of the vehicle.

### SUMMARY

[0003] One embodiment relates to a vehicle including a chassis, a cab coupled to the chassis, a body coupled to the chassis, a cowl coupled to at least one of the body or the cab and extending above the cab, and a heat exchanger coupled to the cab and positioned at least partially within the cowl such that the heat exchanger extends above the cab.

[0004] Another embodiment relates to a vehicle including a chassis, a cab coupled to the chassis, a body coupled to the chassis, a shroud coupled to at least one of the body or the cab, an air volume being defined between the shroud and the cab, and a heat exchanger coupled to the cab and positioned within the air volume. The shroud at least partially defines an inlet and an outlet, and wherein a path for airflow is defined through the inlet, the heat exchanger, and the outlet.

[0005] Another embodiment relates to a refuse vehicle including a chassis, a cab coupled to the chassis, a refuse compartment coupled to the chassis and configured to store a volume of refuse, a lift arm coupled to the chassis and repositionable between a raised position and a lowered position, a front cowl portion coupled to the cab, a front end portion of the front cowl portion defining an inlet extending above the cab, and a rear cowl portion coupled to the refuse compartment. The front cowl portion, the rear cowl portion, and the cab at least partially define an air volume fluidly coupled to the inlet. The refuse vehicle further includes a heat exchanger coupled to the cab and positioned within the air volume, and a fan positioned to direct airflow from the inlet through the heat exchanger. The front cowl portion defines a recess that receives the lift arm when the lift arm is in the lowered position.

[0006] This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

### BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a left side view of a vehicle, according to an exemplary embodiment.

[0008] FIG. 2 is a perspective view of a chassis of the vehicle of FIG. 1.

[0009] FIG. 3 is a perspective view of the vehicle of FIG. 1 configured as a front-loading refuse vehicle, according to an exemplary embodiment.

[0010] FIG. 4 is a left side view of the front-loading refuse vehicle of FIG. 3 configured with a tag axle.

[0011] FIG. 5 is a perspective view of the vehicle of FIG. 1 configured as a side-loading refuse vehicle, according to an exemplary embodiment.

[0012] FIG. 6 is a right side view of the side-loading refuse vehicle of FIG. 5.

[0013] FIG. 7 is a top view of the side-loading refuse vehicle of FIG. 5.

[0014] FIG. 8 is a left side view of the side-loading refuse vehicle of FIG. 5 configured with a tag axle.

[0015] FIG. 9 is a perspective view of the vehicle of FIG. 1 configured as a mixer vehicle, according to an exemplary embodiment.

[0016] FIG. 10 is a perspective view of the vehicle of FIG. 1 configured as a fire fighting vehicle, according to an exemplary embodiment.

[0017] FIG. 11 is a left side view of the vehicle of FIG. 1 configured as an airport fire fighting vehicle, according to an exemplary embodiment.

[0018] FIG. 12 is a perspective view of the vehicle of FIG. 1 configured as a boom lift, according to an exemplary embodiment.

[0019] FIG. 13 is a perspective view of the vehicle of FIG. 1 configured as a scissor lift, according to an exemplary embodiment.

[0020] FIGS. 14 and 15 are block diagrams of a thermal management system for the vehicle of FIG. 1, according to an exemplary embodiment.

[0021] FIGS. 16-19 are various views of a configuration of the vehicle of FIG. 1, according to another exemplary embodiment.

[0022] FIGS. 20 and 21 are various views of a configuration of the vehicle of FIG. 1, according to another exemplary embodiment.

[0023] FIGS. 22 and 23 are various views of a configuration of the vehicle of FIG. 1, according to another exemplary embodiment.

[0024] FIGS. 24 and 25 are various views of a configuration of the vehicle of FIG. 1, according to another exemplary embodiment.

[0025] FIGS. 26-28A are various views of a configuration of the vehicle of FIG. 1 including a cowl assembly, according to another exemplary embodiment.

[0026] FIGS. 28B, 28C, 28D, 28E, and 28F are various views of the cowl assembly of FIG. 26 with an alternative drip channel assembly.

[0027] FIGS. 28G and 28H are various views of the drip channel assembly of FIG. 28B.

[0028] FIG. 29 is a perspective view of the vehicle of FIG. 26 showing a visor of the cowl assembly.

[0029] FIG. 30 is a perspective view of the vehicle of FIG. 26 showing a core assembly positioned within the cowl assembly.

[0030] FIG. 31 is a left section view of the vehicle of FIG. 26.

[0031] FIGS. 32-37 are various views of the vehicle of FIG. 26 with a rear portion of the cowl assembly removed.

[0032] FIG. 38 is a perspective view showing the vehicle of FIG. 26 including a step assembly.

[0033] FIG. 39 is a perspective view of the step assembly of FIG. 38.

[0034] FIGS. 40 and 41 are various views of the vehicle of FIG. 26 in an unstressed state.

[0035] FIGS. 42 and 43 are various views of the vehicle of FIG. 26 in a twisted state.

#### DETAILED DESCRIPTION

[0036] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0037] According to an exemplary embodiment, a vehicle includes a cowl assembly positioned above a cab of the vehicle. The cowl assembly includes a first portion that is coupled to the cab, and a second portion that is coupled to the body. A thermal management system of the vehicle includes a pair of radiators coupled to the cab and positioned between the first portion of the cowl and the roof of the cab. The cowl assembly defines at least one inlet and outlet in fluid communication with the radiators. The radiators may be pivotally coupled to the body to facilitate movement for maintenance purposes. To facilitate this access, a service panel of the cowl assembly may be removed.

#### Overall Vehicle

[0038] Referring to FIGS. 1 and 2, a reconfigurable vehicle (e.g., a vehicle assembly, a truck, a vehicle base, etc.) is shown as vehicle 10, according to an exemplary embodiment. As shown, the vehicle 10 includes a frame assembly or chassis assembly, shown as chassis 20, that supports other components of the vehicle 10. The chassis 20 extends longitudinally along a length of the vehicle 10, substantially parallel to a primary direction of travel of the vehicle 10. As shown, the chassis 20 includes three sections or portions, shown as front section 22, middle section 24, and rear section 26. The middle section 24 of the chassis 20 extends between the front section 22 and the rear section 26. In some embodiments, the middle section 24 of the chassis 20 couples the front section 22 to the rear section 26. In other embodiments, the front section 22 is coupled to the rear section 26 by another component (e.g., the body of the vehicle 10).

[0039] As shown in FIG. 2, the front section 22 includes a pair of frame portions, frame members, or frame rails, shown as front rail portion 30 and front rail portion 32. The rear section 26 includes a pair of frame portions, frame members, or frame rails, shown as rear rail portion 34 and rear rail portion 36. The front rail portion 30 is laterally offset from the front rail portion 32. Similarly, the rear rail portion 34 is laterally offset from the rear rail portion 36. This spacing may provide frame stiffness and space for vehicle components (e.g., batteries, motors, axles, gears, etc.) between the frame rails. In some embodiments, the front rail portions 30 and 32 and the rear rail portions 34 and 36 extend longitudinally and substantially parallel to one

another. The chassis 20 may include additional structural elements (e.g., cross members that extend between and couple the frame rails).

[0040] In some embodiments, the front section 22 and the rear section 26 are configured as separate, discrete subframes (e.g., a front subframe and a rear subframe). In such embodiments, the front rail portion 30, the front rail portion 32, the rear rail portion 34, and the rear rail portion 36 are separate, discrete frame rails that are spaced apart from one another. In some embodiments, the front section 22 and the rear section 26 are each directly coupled to the middle section 24 such that the middle section 24 couples the front section 22 to the rear section 26. Accordingly, the middle section 24 may include a structural housing or frame. In other embodiments, the front section 22, the middle section 24, and the rear section 26 are coupled to one another by another component, such as a body of the vehicle 10.

[0041] In other embodiments, the front section 22, the middle section 24, and the rear section 26 are defined by a pair of frame rails that extend continuously along the entire length of the vehicle 10. In such an embodiment, the front rail portion 30 and the rear rail portion 34 would be front and rear portions of a first frame rail, and the front rail portion 32 and the rear rail portion 36 would be front and rear portions of a second frame rail. In such embodiments, the middle section 24 would include a center portion of each frame rail.

[0042] In some embodiments, the middle section 24 acts as a storage portion that includes one or more vehicle components. The middle section 24 may include an enclosure that contains one or more vehicle components and/or a frame that supports one or more vehicle components. By way of example, the middle section 24 may contain or include one or more electrical energy storage devices (e.g., batteries, capacitors, etc.). By way of another example, the middle section 24 may include fuel tanks. By way of yet another example, the middle section 24 may define a void space or storage volume that can be filled by a user.

[0043] A cabin, operator compartment, or body component, shown as cab 40, is coupled to a front end portion of the chassis 20 (e.g., the front section 22 of the chassis 20). Together, the chassis 20 and the cab 40 define a front end of the vehicle 10. The cab 40 extends above the chassis 20. The cab 40 includes an enclosure or main body that defines an interior volume, shown as cab interior 42, that is sized to contain one or more operators. The cab 40 also includes one or more doors 44 that facilitate selective access to the cab interior 42 from outside of the vehicle 10. The cab interior 42 contains one or more components that facilitate operation of the vehicle 10 by the operator. By way of example, the cab interior 42 may contain components that facilitate operator comfort (e.g., seats, seatbelts, etc.), user interface components that receive inputs from the operators (e.g., steering wheels, pedals, touch screens, switches, buttons, levers, etc.), and/or user interface components that provide information to the operators (e.g., lights, gauges, speakers, etc.). The user interface components within the cab 40 may facilitate operator control over the drive components of the vehicle 10 and/or over any implements of the vehicle 10.

[0044] The vehicle 10 further includes a series of axle assemblies, shown as front axle 50 and rear axles 52. As shown, the vehicle 10 includes one front axle 50 coupled to the front section 22 of the chassis 20 and two rear axles 52 each coupled to the rear section 26 of the chassis 20. In other

embodiments, the vehicle **10** includes more or fewer axles. By way of example, the vehicle **10** may include a tag axle that may be raised or lowered to accommodate variations in weight being carried by the vehicle **10**. The front axle **50** and the rear axles **52** each include a series of tractive elements (e.g., wheels, treads, etc.), shown as wheel and tire assemblies **54**. The wheel and tire assemblies **54** are configured to engage a support surface (e.g., roads, the ground, etc.) to support and propel the vehicle **10**. The front axle **50** and the rear axles **52** may include steering components (e.g., steering arms, steering actuators, etc.), suspension components (e.g., gas springs, dampeners, air springs, etc.), power transmission or drive components (e.g., differentials, drive shafts, etc.), braking components (e.g., brake actuators, brake pads, brake discs, brake drums, etc.), and/or other components that facilitate propulsion or support of the vehicle.

[0045] In some embodiments, the vehicle **10** is configured as an electric vehicle that is propelled by an electric powertrain system. Referring to FIG. 1, the vehicle **10** includes one or more electrical energy storage devices (e.g., batteries, capacitors, etc.), shown as batteries **60**. As shown, the batteries **60** are positioned within the middle section **24** of the chassis **20**. In other embodiments, the batteries **60** are otherwise positioned throughout the vehicle **10**. The vehicle **10** further includes one or more electromagnetic devices or prime movers (e.g., motor/generators), shown as drive motors **62**. The drive motors **62** are electrically coupled to the batteries **60**. The drive motors **62** may be configured to receive electrical energy from the batteries **60** and provide rotational mechanical energy to the wheel and tire assemblies **54** to propel the vehicle **10**. The drive motors **62** may be configured to receive rotational mechanical energy from the wheel and tire assemblies **64** and provide electrical energy to the batteries **60**, providing a braking force to slow the vehicle **10**.

[0046] The batteries **60** may include one or more rechargeable batteries (e.g., lithium-ion batteries, nickel-metal hydride batteries, lithium-ion polymer batteries, lead-acid batteries, nickel-cadmium batteries, etc.). The batteries **60** may be charged by one or more sources of electrical energy onboard the vehicle **10** (e.g., solar panels, etc.) or separate from the vehicle **10** (e.g., connections to an electrical power grid, a wireless charging system, etc.). As shown, the drive motors **62** are positioned within the rear axles **52** (e.g., as part of a combined axle and motor assembly). In other embodiments, the drive motors **62** are otherwise positioned within the vehicle **10**.

[0047] In other embodiments, the vehicle **10** is configured as a hybrid vehicle that is propelled by a hybrid powertrain system (e.g., a diesel/electric hybrid, gasoline/electric hybrid, natural gas/electric hybrid, etc.). According to an exemplary embodiment, the hybrid powertrain system may include a primary driver (e.g., an engine, a motor, etc.), an energy generation device (e.g., a generator, etc.), and/or an energy storage device (e.g., a battery, capacitors, ultra-capacitors, etc.) electrically coupled to the energy generation device. The primary driver may combust fuel (e.g., gasoline, diesel, etc.) to provide mechanical energy, which a transmission may receive and provide to the front axle **50** and/or the rear axles **52** to propel the vehicle **10**. Additionally or alternatively, the primary driver may provide mechanical energy to the generator, which converts the mechanical energy into electrical energy. The electrical energy may be

stored in the energy storage device (e.g., the batteries **60**) in order to later be provided to a motive driver.

[0048] In yet other embodiments, the chassis **20** may further be configured to support non-hybrid powertrains. For example, the powertrain system may include a primary driver that is a compression-ignition internal combustion engine that utilizes diesel fuel.

[0049] Referring to FIG. 1, the vehicle **10** includes a rear assembly, module, implement, body, or cargo area, shown as application kit **80**. The application kit **80** may include one or more implements, vehicle bodies, and/or other components. Although the application kit **80** is shown positioned behind the cab **40**, in other embodiments the application kit **80** extends forward of the cab **40**. The vehicle **10** may be outfitted with a variety of different application kits **80** to configure the vehicle **10** for use in different applications. Accordingly, a common vehicle **10** can be configured for a variety of different uses simply by selecting an appropriate application kit **80**. By way of example, the vehicle **10** may be configured as a refuse vehicle, a concrete mixer, a fire fighting vehicle, an airport fire fighting vehicle, a lift device (e.g., a boom lift, a scissor lift, a telehandler, a vertical lift, etc.), a crane, a tow truck, a military vehicle, a delivery vehicle, a mail vehicle, a boom truck, a plow truck, a farming machine or vehicle, a construction machine or vehicle, a coach bus, a school bus, a semi-truck, a passenger or work vehicle (e.g., a sedan, a SUV, a truck, a van, etc.), and/or still another vehicle. FIGS. 3-13 illustrate various examples of how the vehicle **10** may be configured for specific applications. Although only a certain set of vehicle configurations is shown, it should be understood that the vehicle **10** may be configured for use in other applications that are not shown.

[0050] The application kit **80** may include various actuators to facilitate certain functions of the vehicle **10**. By way of example, the application kit **80** may include hydraulic actuators (e.g., hydraulic cylinders, hydraulic motors, etc.), pneumatic actuators (e.g., pneumatic cylinders, pneumatic motors, etc.), and/or electrical actuators (e.g., electric motors, electric linear actuators, etc.). The application kit **80** may include components that facilitate operation of and/or control of these actuators. By way of example, the application kit **80** may include hydraulic or pneumatic components that form a hydraulic or pneumatic circuit (e.g., conduits, valves, pumps, compressors, gauges, reservoirs, accumulators, etc.). By way of another example, the application kit **80** may include electrical components (e.g., batteries, capacitors, voltage regulators, motor controllers, etc.). The actuators may be powered by components of the vehicle **10**. By way of example, the actuators may be powered by the batteries **60**, the drive motors **62**, or the primary driver (e.g., through a power take off).

[0051] The vehicle **10** generally extends longitudinally from a front side **86** to a rear side **88**. The front side **86** is defined by the cab **40** and/or the chassis. The rear side **88** is defined by the application kit **80** and/or the chassis **20**. The primary, forward direction of travel of the vehicle **10** is longitudinal, with the front side **86** being arranged forward of the rear side **88**.

#### A. Front-Loading Refuse Vehicle

[0052] Referring now to FIGS. 3 and 4, the vehicle **10** is configured as a refuse vehicle **100** (e.g., a refuse truck, a garbage truck, a waste collection truck, a sanitation truck, a

recycling truck, etc.). Specifically, the refuse vehicle **100** is a front-loading refuse vehicle. In other embodiments, the refuse vehicle **100** is configured as a rear-loading refuse vehicle or a front-loading refuse vehicle. The refuse vehicle **100** may be configured to transport refuse from various waste receptacles (e.g., refuse containers) within a municipality to a storage and/or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.).

[0053] FIG. 4 illustrates the refuse vehicle **100** of FIG. 3 configured with a liftable axle, shown as tag axle **90**, including a pair of wheel and tire assemblies **54**. As shown, the tag axle **90** is positioned rearward of the rear axles **52**. The tag axle **90** can be selectively raised and lowered (e.g., by a hydraulic actuator) to selectively engage the wheel and tire assemblies **54** of the tag axle **90** with the ground. The tag axle **90** may be raised to reduce rolling resistance experienced by the refuse vehicle **100**. The tag axle **90** may be lowered to distribute the loaded weight of the vehicle **100** across a greater number of a wheel and tire assemblies **54** (e.g., when the refuse vehicle **100** is loaded with refuse).

[0054] As shown in FIGS. 3 and 4, the application kit **80** of the refuse vehicle **100** includes a series of panels that form a rear body or container, shown as refuse compartment **130**. The refuse compartment **130** may facilitate transporting refuse from various waste receptacles within a municipality to a storage and/or a processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). By way of example, loose refuse may be placed into the refuse compartment **130** where it may be compacted (e.g., by a packer system within the refuse compartment **130**). The refuse compartment **130** may also provide temporary storage for refuse during transport to a waste disposal site and/or a recycling facility. In some embodiments, the refuse compartment **130** may define a hopper volume **132** and storage volume **134**. In this regard, refuse may be initially loaded into the hopper volume **132** and later compacted into the storage volume **134**. As shown, the hopper volume **132** is positioned between the storage volume **134** and the cab **40** (e.g., refuse is loaded into a portion of the refuse compartment **130** behind the cab **40** and stored in a portion further toward the rear of the refuse compartment **130**). In other embodiments, the storage volume may be positioned between the hopper volume and the cab **40** (e.g., in a rear-loading refuse truck, etc.). The application kit **80** of the refuse vehicle **100** further includes a pivotable rear portion, shown as tailgate **136**, that is pivotally coupled to the refuse compartment **130**. The tailgate **136** may be selectively repositionable between a closed position and an open position by an actuator (e.g., a hydraulic cylinder, an electric linear actuator, etc.), shown as tailgate actuator **138** (e.g., to facilitate emptying the storage volume).

[0055] As shown in FIGS. 3 and 4, the refuse vehicle **100** also includes an implement, shown as lift assembly **140**, which is a front-loading lift assembly. According to an exemplary embodiment, the lift assembly **140** includes a pair of lift arms **142** and a pair of actuators (e.g., hydraulic cylinders, electric linear actuators, etc.), shown as lift arm actuators **144**. The lift arms **142** may be rotatably coupled to the chassis **20** and/or the refuse compartment **130** on each side of the refuse vehicle **100** (e.g., through a pivot, a lug, a shaft, etc.), such that the lift assembly **140** may extend forward relative to the cab **40** (e.g., a front-loading refuse truck, etc.). In other embodiments, the lift assembly **140** may extend rearward relative to the application kit **80** (e.g., a

rear-loading refuse truck). As shown in FIGS. 3 and 4, in an exemplary embodiment the lift arm actuators **144** may be positioned such that extension and retraction of the lift arm actuators **144** rotates the lift arms **142** about an axis extending through the pivot. In this regard, the lift arms **142** may be rotated by the lift arm actuators **144** to lift a refuse container over the cab **40**. The lift assembly **140** further includes a pair of interface members, shown as lift forks **146**, each pivotally coupled to a distal end of one of the lift arms **142**. The lift forks **146** may be configured to engage a refuse container (e.g., a dumpster) to selectively couple the refuse container to the lift arms **142**. By way of example, each of the lift forks **146** may be received within a corresponding pocket defined by the refuse container. A pair of actuators (e.g., hydraulic cylinders, electric linear actuators, etc.), shown as articulation actuators **148**, are each coupled to one of the lift arms **142** and one of the lift forks **146**. The articulation actuators **148** may be positioned to rotate the lift forks **146** relative to the lift arms **142** about a horizontal axis. Accordingly, the articulation actuators **148** may assist in tipping refuse out of the refuse container and into the refuse compartment **130**. The lift arm actuators **144** may then rotate the lift arms **142** to return the empty refuse container to the ground.

#### B. Side-Loading Refuse Vehicle

[0056] Referring now to FIGS. 5-8, an alternative configuration of the refuse vehicle **100** is shown according to an exemplary embodiment. Specifically, the refuse vehicle **100** of FIGS. 5-8 is configured as a side-loading refuse vehicle. The refuse vehicle **100** of FIGS. 5-8 may be substantially similar to the front-loading refuse vehicle **100** of FIGS. 3 and 4 except as otherwise specified herein. As shown in FIG. 8, the refuse vehicle **100** of FIGS. 5-7 may be configured with a tag axle **90**.

[0057] Referring still to FIGS. 5-8, the refuse vehicle **100** omits the lift assembly **140** and instead includes a side-loading lift assembly, shown as lift assembly **160**, that extends laterally outward from a side of the refuse vehicle **100**. The lift assembly **160** includes an interface assembly, shown as grabber assembly **162**, that is configured to engage a refuse container (e.g., a residential garbage can) to selectively couple the refuse container to the lift assembly **160**. The grabber assembly **162** includes a main portion, shown as main body **164**, and a pair of fingers or interface members, shown as grabber fingers **166**. The grabber fingers **166** are pivotally coupled to the main body **164** such that the grabber fingers **166** are each rotatable about a vertical axis. A pair of actuators (e.g., hydraulic motors, electric motors, etc.), shown as finger actuators **168**, are configured to control movement of the grabber fingers **166** relative to the main body **164**.

[0058] The grabber assembly **162** is movably coupled to a guide, shown as track **170**, that extends vertically along a side of the refuse vehicle **100**. Specifically, the main body **164** is slidably coupled to the track **170** such that the main body **164** is repositionable along a length of the track **170**. An actuator (e.g., a hydraulic motor, an electric motor, etc.), shown as lift actuator **172**, is configured to control movement of the grabber assembly **162** along the length of the track **170**. In some embodiments, a bottom end portion of the track **170** is straight and substantially vertical such that the grabber assembly **162** raises or lowers a refuse container when moving along the bottom end portion of the track **170**.

In some embodiments, a top end portion of the track **170** is curved such that the grabber assembly **162** inverts a refuse container to dump refuse into the hopper volume **132** when moving along the top end portion of the track **170**.

[0059] The lift assembly **160** further includes an actuator (e.g., a hydraulic cylinder, an electric linear actuator, etc.), shown as track actuator **174**, that is configured to control lateral movement of the grabber assembly **162**. By way of example, the track actuator **174** may be coupled to the chassis **20** and the track **170** such that the track actuator **174** moves the track **170** and the grabber assembly **162** laterally relative to the chassis **20**. The track actuator **174** may facilitate repositioning the grabber assembly **162** to pick up and replace refuse containers that are spaced laterally outward from the refuse vehicle **100**.

### C. Concrete Mixer Truck

[0060] Referring now to FIG. 9, the vehicle **10** is configured as a mixer truck (e.g., a concrete mixer truck, a mixer vehicle, etc.), shown as mixer truck **200**. Specifically, the mixer truck **200** is shown as a rear-discharge concrete mixer truck. In other embodiments, the mixer truck **200** is a front-discharge concrete mixer truck.

[0061] As shown in FIG. 9, the application kit **80** includes a mixing drum assembly (e.g., a concrete mixing drum), shown as drum assembly **230**. The drum assembly **230** may include a mixing drum **232**, a drum drive system **234** (e.g., a rotational actuator or motor, such as an electric motor or hydraulic motor), an inlet portion, shown as hopper **236**, and an outlet portion, shown as chute **238**. The mixing drum **232** may be coupled to the chassis **20** and may be disposed behind the cab **40** (e.g., at the rear and/or middle of the chassis **20**). In an exemplary embodiment, the drum drive system **234** is coupled to the chassis **20** and configured to selectively rotate the mixing drum **232** about a central, longitudinal axis. According to an exemplary embodiment, the central, longitudinal axis of the mixing drum **232** may be elevated from the chassis **20** (e.g., from a horizontal plane extending along the chassis **20**) at an angle in the range of five degrees to twenty degrees. In other embodiments, the central, longitudinal axis may be elevated by less than five degrees (e.g., four degrees, etc.). In yet another embodiment, the mixer truck **200** may include an actuator positioned to facilitate adjusting the central, longitudinal axis to a desired or target angle (e.g., manually in response to an operator input/command, automatically according to a control system, etc.).

[0062] The mixing drum **232** may be configured to receive a mixture, such as a concrete mixture (e.g., cementitious material, aggregate, sand, etc.), through the hopper **236**. In some embodiments, the mixer truck **200** includes an injection system (e.g., a series of nozzles, hoses, and/or valves) including an injection valve that selectively fluidly couples a supply of fluid to the inner volume of the mixing drum **232**. By way of example, the injection system may be used to inject water and/or chemicals (e.g., air entrainers, water reducers, set retarders, set accelerators, superplasticizers, corrosion inhibitors, coloring, calcium chloride, minerals, and/or other concrete additives, etc.) into the mixing drum **232**. The injection valve may facilitate injecting water and/or chemicals from a fluid reservoir (e.g., a water tank, etc.) into the mixing drum **232**, while preventing the mixture in the mixing drum **232** from exiting the mixing drum **232** through the injection system. In some embodiments, one or

more mixing elements (e.g., fins, etc.) may be positioned in the interior of the mixing drum **232**, and may be configured to agitate the contents of the mixture when the mixing drum **232** is rotated in a first direction (e.g., counterclockwise, clockwise, etc.), and drive the mixture out through the chute **238** when the mixing drum **232** is rotated in a second direction (e.g., clockwise, counterclockwise, etc.). In some embodiments, the chute **238** may also include an actuator positioned such that the chute **238** may be selectively pivotable to position the chute **238** (e.g., vertically, laterally, etc.), for example at an angle at which the mixture is expelled from the mixing drum **232**.

### D. Fire Truck

[0063] Referring now to FIG. 10, the vehicle **10** is configured as a fire fighting vehicle, fire truck, or fire apparatus (e.g., a turntable ladder truck, a pumper truck, a quint, etc.), shown as fire fighting vehicle **250**. In the embodiment shown in FIG. 10, the fire fighting vehicle **250** is configured as a rear-mount aerial ladder truck. In other embodiments, the fire fighting vehicle **250** is configured as a mid-mount aerial ladder truck, a quint fire truck (e.g., including an on-board water storage, a hose storage, a water pump, etc.), a tiller fire truck, a pumper truck (e.g., without an aerial ladder), or another type of response vehicle. By way of example, the vehicle **10** may be configured as a police vehicle, an ambulance, a tow truck, or still other vehicles used for responding to a scene (e.g., an accident, a fire, an incident, etc.).

[0064] As shown in FIG. 10, in the fire fighting vehicle **250**, the application kit **80** is positioned mainly rearward from the cab **40**. The application kit **80** includes deployable stabilizers (e.g., outriggers, downriggers, etc.), shown as outriggers **252**, that are coupled to the chassis **20**. The outriggers **252** may be configured to selectively extend from each lateral side and/or the rear of the fire fighting vehicle **250** and engage a support surface (e.g., the ground) in order to provide increased stability while the fire fighting vehicle **250** is stationary. The fire fighting vehicle **250** further includes an extendable or telescoping ladder assembly, shown as ladder assembly **254**. The increased stability provided by the outriggers **252** is desirable when the ladder assembly **254** is in use (e.g., extended from the fire fighting vehicle **250**) to prevent tipping. In some embodiments, the application kit **80** further includes various storage compartments (e.g., cabinets, lockers, etc.) that may be selectively opened and/or accessed for storage and/or component inspection, maintenance, and/or replacement.

[0065] As shown in FIG. 10, the ladder assembly **254** includes a series of ladder sections **260** that are slidably coupled with one another such that the ladder sections **260** may extend and/or retract (e.g., telescope) relative to one another to selectively vary a length of the ladder assembly **254**. A base platform, shown as turntable **262**, is rotatably coupled to the chassis **20** and to a proximal end of a base ladder section **260** (i.e., the most proximal of the ladder sections **260**). The turntable **262** may be configured to rotate about a vertical axis relative to the chassis **20** to rotate the ladder sections **260** about the vertical axis (e.g., up to 360 degrees, etc.). The ladder sections **260** may rotate relative to the turntable **262** about a substantially horizontal axis to selectively raise and lower the ladder sections **260** relative to the chassis **20**. As shown, a water turret or implement, shown as monitor **264**, is coupled to a distal end of a fly

ladder section **260** (i.e., the most distal of the ladder sections **260**). The monitor **264** may be configured to expel water and/or a fire suppressing agent (e.g., foam, etc.) from a water storage tank and/or an agent tank onboard the fire fighting vehicle **250**, and/or from an external source (e.g., a fire hydrant, a separate water/pumper truck, etc.). In some embodiments, the ladder assembly **254** further includes an aerial platform coupled to the distal end of the fly ladder section **260** and configured to support one or more operators.

#### E. ARFF Truck

[0066] Referring now to FIG. 11, the vehicle **10** is configured as a fire fighting vehicle, shown as airport rescue and fire fighting (ARFF) truck **300**. As shown in FIG. 11, the application kit **80** is positioned primarily rearward of the cab **40**. As shown, the application kit **80** includes a series of storage compartments or cabinets, shown as compartments **302**, that are coupled to the chassis **20**. The compartments **302** may store various equipment or components of the ARFF truck **300**.

[0067] The application kit **80** includes a pump system **304** (e.g., an ultra-high-pressure pump system, etc.) positioned within one of the compartments **302** near the center of the ARFF truck **300**. The application kit **80** further includes a water tank **310**, an agent tank **312**, and an implement or water turret, shown as monitor **314**. The pump system **304** may include a high pressure pump and/or a low pressure pump, which may be fluidly coupled to the water tank **310** and/or the agent tank **312**. The pump system **304** may to pump water and/or fire suppressing agent from the water tank **310** and the agent tank **312**, respectively, to the monitor **314**. The monitor **314** may be selectively reoriented by an operator to adjust a direction of a stream of water and/or agent. As shown in FIG. 11, the monitor **314** is coupled to a front end of the cab **40**.

#### F. Boom Lift

[0068] Referring now to FIG. 12, the vehicle **10** is configured as a lift device, shown as boom lift **350**. The boom lift **350** may be configured to support and elevate one or more operators. In other embodiments, the vehicle **10** is configured as another type of lift device that is configured to lift operators and/or material, such as a skid-loader, a telehandler, a scissor lift, a fork lift, a vertical lift, and/or any other type of lift device or machine.

[0069] As shown in FIG. 12, the application kit **80** includes a base assembly, shown as turntable **352**, that is rotatably coupled to the chassis **20**. The turntable **352** may be configured to selectively rotate relative to the chassis **20** about a substantially vertical axis. In some embodiments, the turntable **352** includes a counterweight (e.g., the batteries) positioned near the rear of the turntable **352**. The turntable **352** is rotatably coupled to a lift assembly, shown as boom assembly **354**. The boom assembly **354** includes a first section or telescoping boom section, shown as lower boom **360**. The lower boom **360** includes a series of nested boom sections that extend and retract (e.g., telescope) relative to one another to vary a length of the boom assembly **354**. The boom assembly **354** further includes a second boom section or four bar linkage, shown as upper boom **362**. The upper boom **362** may include structural members that rotate relative to one another to raise and lower a distal end of the boom assembly **354**. In other embodiments, the boom

assembly **354** includes more or fewer boom sections (e.g., one, three, five, etc.) and/or a different arrangement of boom sections.

[0070] As shown in FIG. 12, the boom assembly **354** includes a first actuator, shown as lower lift cylinder **364**. The lower boom **360** is pivotally coupled (e.g., pinned, etc.) to the turntable **352** at a joint or lower boom pivot point. The lower lift cylinder **364** (e.g., a pneumatic cylinder, an electric linear actuator, a hydraulic cylinder, etc.) is coupled to the turntable **352** at a first end and coupled to the lower boom **360** at a second end. The lower lift cylinder **364** may be configured to raise and lower the lower boom **360** relative to the turntable **352** about the lower boom pivot point.

[0071] The boom assembly **354** further includes a second actuator, shown as upper lift cylinder **366**. The upper boom **362** is pivotally coupled (e.g., pinned) to the upper end of the lower boom **360** at a joint or upper boom pivot point. The upper lift cylinder **366** (e.g., a pneumatic cylinder, an electric linear actuator, a hydraulic cylinder, etc.) is coupled to the upper boom **362**. The upper lift cylinder **366** may be configured to extend and retract to actuate (e.g., lift, rotate, elevate, etc.) the upper boom **362**, thereby raising and lowering a distal end of the upper boom **362**.

[0072] Referring still to FIG. 12, the application kit **80** further includes an operator platform, shown as platform assembly **370**, coupled to the distal end of the upper boom **362** by an extension arm, shown as jib arm **372**. The jib arm **372** may be configured to pivot the platform assembly **370** about a lateral axis (e.g., to move the platform assembly **370** up and down, etc.) and/or about a vertical axis (e.g., to move the platform assembly **370** left and right, etc.).

[0073] The platform assembly **370** provides a platform configured to support one or more operators or users. In some embodiments, the platform assembly **370** may include accessories or tools configured for use by the operators. For example, the platform assembly **370** may include pneumatic tools (e.g., an impact wrench, airbrush, nail gun, ratchet, etc.), plasma cutters, welders, spotlights, etc. In some embodiments, the platform assembly **370** includes a control panel (e.g., a user interface, a removable or detachable control panel, etc.) configured to control operation of the boom lift **350** (e.g., the turntable **352**, the boom assembly **354**, etc.) from the platform assembly **370** or remotely. In other embodiments, the platform assembly **370** is omitted, and the boom lift **350** includes an accessory and/or tool (e.g., forklift forks, etc.) coupled to the distal end of the boom assembly **354**.

#### G. Scissor Lift

[0074] Referring now to FIG. 13, the vehicle **10** is configured as a lift device, shown as scissor lift **400**. As shown in FIG. 13, the application kit **80** includes a body, shown as lift base **402**, coupled to the chassis **20**. The lift base **402** is coupled to a scissor assembly, shown as lift assembly **404**, such that the lift base **402** supports the lift assembly **404**. The lift assembly **404** is configured to extend and retract, raising and lowering between a raised position and a lowered position relative to the lift base **402**.

[0075] As shown in FIG. 13, the lift base **402** includes a series of actuators, stabilizers, downriggers, or outriggers, shown as leveling actuators **410**. The leveling actuators **410** may extend and retract vertically between a stored position and a deployed position. In the stored position, the leveling actuators **410** may be raised, such that the leveling actuators

**410** do not contact the ground. Conversely, in the deployed position, the leveling actuators **410** may engage the ground to lift the lift base **402**. The length of each of the leveling actuators **410** in their respective deployed positions may be varied in order to adjust the pitch (e.g., rotational position about a lateral axis) and the roll (e.g., rotational position about a longitudinal axis) of the lift base **402** and/or the chassis **20**. Accordingly, the lengths of the leveling actuators **410** in their respective deployed positions may be adjusted to level the lift base **402** with respect to the direction of gravity (e.g., on uneven, sloped, pitted, etc. terrain). The leveling actuators **410** may lift the wheel and tire assemblies **54** off of the ground to prevent movement of the scissor lift **400** during operation. In other embodiments, the leveling actuators **410** are omitted.

[0076] The lift assembly **404** may include a series of subassemblies, shown as scissor layers **420**, each including a pair of inner members and a pair of outer members pivotally coupled to one another. The scissor layers **420** may be stacked atop one another in order to form the lift assembly **404**, such that movement of one scissor layer **420** causes a similar movement in all of the other scissor layers **420**. The scissor layers **420** extend between and couple the lift base **402** and an operator platform (e.g., the platform assembly **430**). In some embodiments, scissor layers **420** may be added to, or removed from, the lift assembly **404** in order to increase, or decrease, the fully extended height of the lift assembly **404**.

[0077] Referring still to FIG. 13, the lift assembly **404** may also include one or more lift actuators **424** (e.g., hydraulic cylinders, pneumatic cylinders, electric linear actuators such as motor-driven leadscrews, etc.) configured to extend and retract the lift assembly **404**. The lift actuators **424** may be pivotally coupled to inner members of various scissor layers **420**, or otherwise arranged within the lift assembly **404**.

[0078] A distal or upper end of the lift assembly **404** is coupled to an operator platform, shown as platform assembly **430**. The platform assembly **430** may perform similar functions to the platform assembly **370**, such as supporting one or more operators, accessories, and/or tools. The platform assembly **430** may include a control panel to control operation of the scissor lift **400**. The lift actuators **424** may be configured to actuate the lift assembly **404** to selectively reposition the platform assembly **430** between a lowered position (e.g., where the platform assembly **430** is proximate to the lift base **402**) and a raised position (e.g., where the platform assembly **430** is at an elevated height relative to the lift base **402**). Specifically, in some embodiments, extension of the lift actuators **424** moves the platform assembly **430** upward (e.g., extending the lift assembly **404**), and retraction of the lift actuators **424** moves the platform assembly **430** downward (e.g., retracting the lift assembly **404**). In other embodiments, extension of the lift actuators **424** retracts the lift assembly **404**, and retraction of the lift actuators **424** extends the lift assembly **404**.

#### Thermal Management System

[0079] Referring to FIGS. 14 and 15, the vehicle **10** includes a heat dissipation system, a heating, ventilation, and air conditioning (HVAC) system, a temperature control system, or a thermal management system, shown as thermal management system **700**. The thermal management system **700** is configured to vary the temperatures of one or more

components of the vehicle **10** and/or one or more spaces within the vehicle **10**. By way of example, the thermal management system **700** may cool one or more components of the vehicle **10** that generate heat (e.g., thermal energy) during operation. By way of another example, the thermal management system **700** may provide cool air to a space occupied by an operator. By way of another example, the thermal management system **700** may warm the air within a space occupied by the operator (e.g., the cab interior **42**).

[0080] Referring to FIG. 14, the thermal management system **700** includes a heat dissipation circuit, shown as coolant circuit **710**. The coolant circuit **710** is configured to dissipate thermal energy from one or more components of the vehicle **10** to the surrounding atmosphere, decreasing or maintaining a temperature of those components. The coolant circuit **710** includes one or more heat transfer devices (e.g., water jackets, fins, etc.), shown as heat exchangers **712** that place one or more components in thermal communication with the coolant circuit **710**. One heat exchanger **712** may be coupled to multiple components. Alternatively, each component may be coupled to a separate heat exchanger **712**.

[0081] As shown, the heat exchangers **712** are coupled to the batteries **60**, the drive motors **62**, and one or more power conditioners or power converters, shown as inverters **714**. The inverters **714** may be electrically coupled to the batteries **60** and the drive motors **62**. The inverters **714** may convert direct current (DC) electrical energy from the batteries **60** to alternating current (AC) electrical energy and provide the AC electrical energy to the drive motors **62**. Additionally or alternatively, the inverters **714** may convert AC electrical energy from the drive motors **62** to DC electrical energy and provide the DC electrical energy for storage in the batteries **60**. The heat exchangers **712** may thermally couple to the batteries **60**, the drive motors **62** and/or the inverters **714** to remove thermal energy generated during operation (e.g., due to resistance). In other embodiments, the heat exchangers **712** are coupled to other components of the vehicle **10**. By way of example, the heat exchangers **712** may be coupled to pumps, compressors, actuators, or other components of the vehicle **10** that generate thermal energy during operation.

[0082] The heat exchangers **712** are fluidly coupled to a driver or actuator, shown as coolant pump **720**. The coolant pump **720** is configured to receive coolant at a low pressure and provide a flow of fluid at an elevated pressure. As shown, an inlet of the coolant pump **720** is fluidly coupled to the heat exchangers **712**, such that the coolant pump **720** is downstream of the heat exchangers **712**. In other embodiments, the coolant pump **720** is positioned upstream of the heat exchangers **712**.

[0083] The coolant pump **720** is fluidly coupled to a heat dissipater, radiator core, or heat exchanger, shown as radiator **722**. The radiator **722** is fluidly coupled to the heat exchangers **712**, forming a closed loop for coolant flow. In operation, the coolant pump **720** initiates a flow of coolant, which causes the coolant to flow through the heat exchangers **712**. The heat exchangers **712** transfer thermal energy from the batteries **60**, the drive motors **62**, and/or the inverters **714** to the coolant. The coolant is received by the radiator **722**, which transfers thermal energy from the coolant into the surrounding atmosphere.

[0084] In some embodiments, the coolant circuit **710** further includes one or more air movers, coolers, or blowers, shown as fans **724**. The fans **724** are coupled to the radiator



**722.** The fans **724** may be positioned to direct air from the surrounding atmosphere through the radiator **722** (e.g., across fins of the radiator **722**). The fans **724** may improve the cooling performance of the coolant circuit **710** by increasing the heat transfer from the radiator **722** to the surrounding atmosphere.

**[0085]** Referring to FIG. **15**, the thermal management system **700** further includes a refrigeration circuit or air conditioning circuit, shown as air conditioning circuit **730**. The air conditioning circuit **730** is configured to transfer thermal energy from the cab interior **42** to the surrounding atmosphere, decreasing or maintaining a temperature of air within the cab **40** to improve operator comfort. Operation of the air conditioning circuit **730** may be controlled by an operator (e.g., through a user interface that provides commands to a controller that controls the air conditioning circuit **730**).

**[0086]** The air conditioning circuit **730** includes a heat exchanger or radiator core, shown as evaporator **732**. The evaporator **732** is in fluid communication with the cab interior **42** such that air passes through the evaporator **732** and into the cab interior **42**. In some embodiments, the evaporator **732** is positioned within the cab interior **42**.

**[0087]** In some embodiments, the air conditioning circuit **730** includes one or more air movers or blowers, shown as fans **734**. The fans **734** are coupled to the evaporator **732**. The fans **734** may be positioned to direct air from the cab interior **42** and/or from the surrounding atmosphere through the evaporator **732** (e.g., across fins of the evaporator **732**). The fans **734** may improve the cooling performance of the air conditioning circuit **730** by increasing the heat transfer from the air that passes through the evaporator **732** to the evaporator **732**.

**[0088]** An outlet of the evaporator **732** is fluidly coupled to a compressor **736**. The compressor **736** is configured to receive refrigerant at a low pressure and provide a flow of the refrigerant at an elevated pressure. An outlet of the compressor **736** is fluidly coupled to a heat dissipater, radiator core, or heat exchanger, shown as condenser **740**. An outlet of the condenser **740** is fluidly coupled to an expansion valve or throttling valve, shown as expansion valve **742**. The outlet of the expansion valve **742** is fluidly coupled to the inlet of the evaporator **732**, forming a closed loop.

**[0089]** In some embodiments, the air conditioning circuit **730** includes one or more air movers or blowers, shown as fans **744**. The fans **744** are coupled to the condenser **740**. The fans **744** may be positioned to direct air from the surrounding atmosphere through the condenser **740** (e.g., across fins of the condenser **740**). The fans **744** may improve the cooling performance of the air conditioning circuit **730** by increasing the heat transfer from the condenser **740** to the surrounding atmosphere.

**[0090]** In operation, the compressor **736** initiates a flow of compressed refrigerant, which flows into the condenser **740**. The condenser **740** transfers thermal energy from the refrigerant into the surrounding atmosphere. The cooled refrigerant passes through the expansion valve **742**, which expands the refrigerant, further cooling the refrigerant. The cooled refrigerant passes through the evaporator **732**, where thermal energy from the cab interior **42** is transferred to the refrigerant. The heated refrigerant then returns to the compressor **736**.

**[0091]** In some embodiments, the vehicle **10** includes a hydraulic system. By way of example, the vehicle **10** may include a pump that provides hydraulic fluid to one or more hydraulic actuators (e.g., hydraulic motors, hydraulic cylinders, etc.). Hydraulic fluid (e.g., oil) may increase in temperature throughout operation of the hydraulic system. In some embodiments, the thermal management system **700** includes a hydraulic fluid radiator or heat exchanger (e.g., similar to the radiator **722**), through which the hydraulic fluid flows. The hydraulic fluid radiator may transfer heat from the hydraulic fluid to the surrounding atmosphere.

#### A. Configuration 1

**[0092]** Referring to FIGS. **16** and **17**, a vehicle **750** is shown as an exemplary configuration of the vehicle **10**. Accordingly, any description with respect to the vehicle **10** may apply to the vehicle **750**, except as otherwise specified. The vehicle **750** may be configured with any of the application kits **80** described herein.

**[0093]** The thermal management system **700** of the vehicle **750** includes a pair of radiator cores or heat exchangers, shown as core **752** and core **754**. The cores **752** and **754** may transfer thermal energy from any component or system of the vehicle **750** to the surrounding atmosphere. By way of example, the core **752** and/or the core **754** may act as the radiator **722**. By way of another example, the core **752** and/or the core **754** may act as the condenser **740**. By way of another example, the core **752** and/or the core **754** may act as a hydraulic fluid radiator that cools hydraulic fluid.

**[0094]** As shown, the thermal management system **700** of the vehicle **750** includes a series of air movers or blowers, shown as fans **760** and fans **762**. The fans **760** are coupled to the core **752** and configured to direct air through the core **752** (e.g., across fins of the core **752**). The fans **762** are coupled to the core **754** and configured to direct air through the core **754** (e.g., across fins of the core **754**). The fans **760** and the fans **762** may be in a push configuration (e.g., such that air flows first through the fans and then into the corresponding core) and/or in a pull configuration (e.g., such that air flows first through the corresponding core and then into the fans). The fans **760** and/or the fans **762** may include the fans **724** and/or the fans **744**.

**[0095]** As shown in FIGS. **16-19**, the application kit **80** of the vehicle **750** includes a container or structure, shown as body **770**, positioned rearward of the cab **40**. As shown, the body **770** is substantially rectangular (e.g., a rectangular prism). As shown, the body **770** extends above (e.g., is taller than) the cab **40**. The body **770** may be a portion of any of the application kits **80** described herein. By way of example, the body **770** may act as the refuse compartment **130** of the refuse vehicle **100**. By way of another example, the body **770** may act as the compartments **302** of the ARFF truck **300**.

**[0096]** In some embodiments, the body **770** is capable of moving relative to the chassis **20** and the cab **40**. In some such embodiments, the body **770** can be raised and lowered relative to the chassis **20** between a lowered or operating position (e.g., as shown in FIG. **18**) and an elevated position (e.g., as shown in FIG. **19**). In the embodiment shown in FIGS. **18** and **19**, the body **770** is rotatably coupled to the chassis **20** near the rear end of the chassis **20**, and the body **770** rotates about a lateral axis (i.e., a horizontal axis) such that the front end of the body **770** raises and lowers. In other embodiments, the entire body **770** raises and lowers. In

some embodiments, the vehicle 750 includes an actuator (e.g., a hydraulic cylinder, an electric linear actuator, a pneumatic cylinder, etc.), shown as body actuator 772, that is coupled to the chassis 20 and the body 770. The body actuator 772 is configured to selectively move the body 770 between the raised position and the lowered position.

[0097] The cab 40 includes a first portion (e.g., an operator portion), shown as front portion 780, and a second portion (e.g., a storage portion), shown as rear portion 782, each coupled to the chassis 20. The front portion 780 defines the cab interior 42 and is configured to contain one or more operators. The rear portion 782 is positioned between the front portion 780 and the body 770. In some embodiments, the rear portion 782 defines one or more storage spaces that contain components of the vehicle 750 (e.g., pumps, batteries, plumbing, etc.) and/or provide storage for items placed by an operator (e.g., clothing, equipment, etc.). The rear portion 782 may include one or more doors that facilitate accessing the storage spaces. In some embodiments, the front portion 780 and the rear portion 782 are fixedly coupled to the chassis 20 (e.g., the front portion 780 and the rear portion 782 remain stationary when the body 770 is raised).

[0098] The vehicle 750 further includes a pair of covers, guards, diverters, cowls, or flow control members, shown as front shroud 790 (e.g., a front cowl portion, a front shroud portion) and rear shroud 792 (e.g., a rear cowl portion, a rear shroud portion). The front shroud 790 is positioned directly above the front portion 780 of the cab 40 and coupled to the front portion 780. The rear shroud 792 is positioned directly above the rear portion 782 of the cab 40 and coupled to the body 770. The rear shroud 792 is positioned between the front shroud 790 and the body 770. Together, the front shroud 790 and the rear shroud 792 define a top surface of the vehicle 750 that slopes gradually from the front surface of the cab 40 to the top surface of the body 770. This gradual transition may reduce the drag on the vehicle 750 when the vehicle 750 travels in a forward direction.

[0099] Referring again to FIGS. 16 and 17, space or volume (e.g., an air volume, a plenum, a radiator volume, etc.), shown as radiator volume 800, is defined between a top surface of the cab 40, a front surface of the body 770, a bottom surface of the front shroud 790, and a bottom surface of the rear shroud 792. In FIGS. 16 and 17, the front shroud 790 and the rear shroud 792 are illustrated as being transparent for ease of viewing components within the radiator volume 800. However, it should be understood that these components may be opaque. The radiator volume 800 contains the core 752, the fans 760, the core 754, and the fans 760. Accordingly, the cores 752 and 754 and the fans 760 and 762 are positioned above the cab 40 and out of the way of the operators during normal use of the vehicle 750. The cores 752 and 754 and the fans 760 and 762 are protected from rain and falling debris (e.g., acorns, hail, rocks, refuse, etc.) by the front shroud 790 and the rear shroud 792. The cores 752 and 754 may be coupled to the cab 40.

[0100] The cores 752 and 754 are positioned within the radiator volume 800 to facilitate airflow through the cores 752 and 754, maximizing cooling performance of the cores 752 and 754. The front shroud 790 defines a first aperture or front aperture, shown as inlet 802. The inlet 802 is positioned near the front of the front shroud 790 such that air is forced through the inlet 802 when the vehicle 750 travels in a forward direction. The inlet 802 may be approximately

centered about a longitudinal centerline L of the cab 40. The rear shroud 792 defines a series of second apertures or rear apertures, shown as outlets 804. The outlets 804 may be positioned along the lateral sides of the rear shroud 792. Air that flows into the radiator volume 800 through the inlet 802 subsequently exits the radiator volume 800 through the outlets 804.

[0101] To control the airflow through the radiator volume 800, the vehicle 750 includes a baffle, flow separator, or flow divider, shown as flow diverter 810, that facilitates airflow through the cores 752 and 754 and the outlets 804. The flow diverter 810 may be coupled to the cab 40 and/or to the body 770. As viewed from above (e.g., as in FIG. 17), the flow diverter 810 includes a middle portion, protrusion, or thickest portion, shown as frontmost portion 812. The frontmost portion 812 is the closest portion to the front of the vehicle 750, and the flow diverter 810 gradually retreats rearward from the frontmost portion 812 as the flow diverter 810 extends laterally outward. The shape of the flow diverter 810 splits the airflow into a first portion that moves toward the outlets 804 on the left side of the vehicle 750 and a second portion that moves toward the outlets 804 on the right side of the vehicle 750.

[0102] In the embodiment shown in FIG. 17, the core 754 is wider than the core 752. The frontmost portion 812 is offset from the longitudinal centerline L toward the core 752. This positioning of the frontmost portion 812 ensures that a larger portion of the airflow is directed toward the core 754 than to the core 752, taking advantage of the larger size of the core 754. The lateral position of the frontmost portion 812 may vary depending upon the relative sizes of the core 752 and the core 754.

[0103] In operation, airflow through the radiator volume 800 generally follows two paths, illustrated in FIG. 17 as path 820 and path 822. A first portion of the airflow moves along the path 820. Specifically, the first portion enters the inlet 802, is diverted by the flow diverter 810, passes through the core 752 and the fans 760, and exits through the outlets 804 on the passenger or curb side of the vehicle 750. A second portion of the airflow moves along the path 822. Specifically, the second portion enters the inlet 802, is diverted by the flow diverter 810, passes through the fans 760 and the core 754, and exits through the outlets 804 on the driver or street side of the vehicle 750. In some embodiments, the core 752 and the core 754 are arranged to be substantially perpendicular to the path 820 and the path 822, respectively.

[0104] As shown in FIGS. 16 and 17, the front face of the core 752 (e.g., the face through which the airflow enters) is positioned within a plane 830. The plane 830 is substantially vertical. The plane 830 is angularly offset from the longitudinal centerline L by an angle between 0 and 90 degrees. In some embodiments, the angle between the plane 830 and the longitudinal centerline L is approximately 45 degrees. The plane 830 is angularly offset from the longitudinal centerline L by an angle between 0 and 90 degrees. In some embodiments, the angle between the plane 830 and the longitudinal centerline L is approximately 30 degrees. The front face of the core 754 (e.g., the face through which the airflow enters) is positioned within a plane 832. The plane 832 is substantially vertical. In some embodiments, the angle between the plane 832 and the longitudinal centerline L is approximately 45 degrees. The plane 832 is angularly offset from the longitudinal centerline L by an angle between

0 and 90 degrees. In some embodiments, the angle between the plane **832** and the longitudinal centerline **L** is approximately 30 degrees. The core **752** and the core **754** are angularly offset from the longitudinal centerline **L** in opposing directions.

[0105] After operation, it may be advantageous to access the radiator volume **800** to clean and/or otherwise maintain the core **752** and the core **754**. In some embodiments, the front shroud **790** is removably coupled to the front portion **780** of the cab **40**. Accordingly, the front shroud **790** may be removed to facilitate access to the radiator volume **800**. In some embodiments, the rear shroud **792** is coupled to the body **770**. Accordingly, the body **770** may be raised to lift the rear shroud **792** and facilitate access to the radiator volume **800** (e.g., as shown in FIGS. **18** and **19**).

#### B. Configuration 2

[0106] Referring to FIGS. **20** and **21**, a vehicle **850** is shown as an alternative configuration of the vehicle **10**. The vehicle **850** may be substantially similar to the vehicle **750**, except as otherwise specified. The vehicle **850** may be configured with any of the application kits **80** described herein. In FIGS. **20** and **21**, the rear shroud **792** is illustrated as being transparent for ease of viewing components within the radiator volume **800**. However, it should be understood that this component may be opaque.

[0107] In the embodiment shown in FIGS. **20** and **21**, the inlet **802** at the front end of the front shroud **790** is omitted (e.g., the front end of the front shroud **790** is solid). Instead, the front end of the front shroud **790** overlaps the rear end of the rear shroud **792**, defining a pair of inlets **802**. Specifically, each inlet **802** is defined between a top surface **860** of the cab **40**, a lateral side surface **862** of the front shroud **790**, a vertical, side edge **864** of the rear shroud **792**, and a horizontal, top edge **866** of the rear shroud **792**. The lateral side surfaces **862** move toward the longitudinal centerline **L** as the lateral side surfaces **862** extend rearward. Accordingly, the pathway defined for the airflow gradually widens as the pathway moves rearward.

[0108] In operation, airflow through the radiator volume **800** generally follows the paths **820** and **822**. As shown in FIG. **21**, the path **820** enters the inlet **802** on the right side of the vehicle **850**, extends through the core **752** and the fans **760**, is diverted by the flow diverter **810**, and exits through the outlets **804** on the right side of the vehicle **850**. The path **822** enters the inlet **802** on the left side of the vehicle **850**, passes through the core **754** and the fans **762**, is diverted by the flow diverter **810**, and exits through the outlets **804** on the left (e.g., street or driver) side of the vehicle **850**. In the vehicle **850**, the frontmost portion **812** of the flow diverter **810** is located at or near the longitudinal centerline **L**.

[0109] As shown in FIGS. **20** and **21**, the core **752** and the core **754** each extend from the top surface **860** of the cab **40** to a bottom surface of the rear shroud **792**. The plane **830** and the plane **832** are each substantially parallel to a lateral axis. The plane **830** and the plane **832** are each angled relative to a horizontal plane. Specifically, the plane **830** and the plane **832** each extend upward as the plane **830** and the plane **832** extend rearward. The plane **830** and the plane **832** are each angularly offset from the horizontal plane by an angle between 0 and 90 degrees. In some embodiments, the angles between the planes **830** and **832** and the horizontal plane are each approximately 45 degrees.

[0110] In some embodiments, the front shroud **790** is removably coupled to the front portion **780** of the cab **40**. Accordingly, the front shroud **790** may be removed to facilitate access to the radiator volume **800**. In some embodiments, the angled orientations of the lateral side surfaces **862** of the front shroud **790** facilitate access to the cores **752** and **754** without removing the front shroud **790**. In some embodiments, the rear shroud **792** is coupled to the body **770**. Accordingly, the body **770** may be raised to lift the rear shroud **792** and facilitate access to the radiator volume **800**.

#### C. Configuration 3

[0111] Referring to FIGS. **22** and **23**, a vehicle **900** is shown as an alternative configuration of the vehicle **10**. The vehicle **900** may be substantially similar to the vehicle **850**, except as otherwise specified. The vehicle **900** may be configured with any of the application kits **80** described herein. In FIGS. **22** and **23**, the rear shroud **792** is illustrated as being transparent for ease of viewing components within the radiator volume **800**. However, it should be understood that this component may be opaque.

[0112] In the vehicle **900**, the front end of the front shroud **790** overlaps the rear end of the rear shroud **792**, defining a single inlet **802**. Specifically, the inlet **802** is defined by the exterior surfaces of the front shroud **790**, the side edges **864** of the rear shroud **792**, and the top edge **866** of the rear shroud **792**.

[0113] In operation, airflow through the radiator volume **800** generally follows the paths **820** and **822**. The path **820** enters the inlet **802**, extends through the core **752** and the fans **760**, is diverted by the flow diverter **810**, and exits through the outlets **804** on the passenger side of the vehicle **900**. The path **822** enters the inlet **802**, passes through the core **754** and the fans **762**, is diverted by the flow diverter **810**, and exits through the outlets **804** on the left (e.g., street or driver) side of the vehicle **900**. In the vehicle **900**, the frontmost portion **812** of the flow diverter **810** is located at or near the longitudinal centerline **L**.

[0114] As shown in FIGS. **22** and **23**, the core **752** and the core **754** each extend from a top surface of the front shroud **790** to a bottom surface of the rear shroud **792**. The plane **830** and the plane **832** are each substantially parallel to a lateral axis. The plane **830** and the plane **832** are each angled relative to a horizontal plane. Specifically, the plane **830** and the plane **832** each extend upward as the plane **830** and the plane **832** extend rearward. The plane **830** and the plane **832** are each angularly offset from the horizontal plane by an angle between 0 and 90 degrees. In some embodiments, the angles between the planes **830** and **832** and the horizontal plane are each approximately 60 degrees.

[0115] In some embodiments, the rear shroud **792** is coupled to the body **770**. Accordingly, the body **770** may be raised to lift the rear shroud **792** and facilitate access to the radiator volume **800**. Because the cores **752** and **754** are positioned above the front shroud **790**, the cores **752** and **754** may be accessed without removing the front shroud **790**.

#### D. Configuration 4

[0116] Referring to FIGS. **24** and **25**, a vehicle **950** is shown as an alternative configuration of the vehicle **10**. The vehicle **950** may be substantially similar to the vehicle **900**, except as otherwise specified. The vehicle **950** may be configured with any of the application kits **80** described

herein. In FIGS. 24 and 25, the rear shroud 792 is illustrated as being transparent for ease of viewing components within the radiator volume 800. However, it should be understood that this component may be opaque.

[0117] In the vehicle 950, the front shroud 790 is omitted. Instead, the inlet 802 is defined by the top surface 860 of the cab 40, the side edges 864 of the rear shroud 792, and the top edge 866 of the rear shroud 792. The radiator volume 800 is defined between the cab 40, the body 770, and the rear shroud 792.

[0118] In operation, airflow through the radiator volume 800 generally follows the paths 820 and 822. The path 820 enters the inlet 802, is diverted by the flow diverter 810, extends through the core 752 and the fans 760, and exits through the outlets 804 on the passenger side of the vehicle 950. The path 822 enters the inlet 802, is diverted by the flow diverter 810, passes through the core 754 and the fans 762, and exits through the outlets 804 on the driver side of the vehicle 950. In the vehicle 900, the frontmost portion 812 of the flow diverter 810 is located at or near the longitudinal centerline L.

[0119] As shown in FIGS. 24 and 25, the rear shroud 792 defines a pair of access panels, shown as doors 952. Each of the doors 952 is selectively repositionable between an open position, in which the door 952 permits access to the radiator volume 800 through an aperture defined by the rear shroud 792, and a closed position, in which the door 952 extends across the aperture to prevent access to the radiator volume 800. By way of example, the doors 952 may be hingedly coupled to the rear shroud 792 such that the doors 952 are rotatable about an axis. The doors 952 are positioned along the lateral sides of the rear shroud 792, near the body 770. The doors 952 define at least a portion of the outlets 804. The core 752 and the fans 760 are coupled to a first door 952. The core 754 and the fans 762 are coupled to a second door 952. Accordingly, a user may simply open the doors 952 to access the cores 752 and 754 for cleaning.

[0120] The plane 830 and the plane 832 are each substantially parallel to a vertical axis and to a longitudinal axis. The plane 830 and the plane 832 are laterally offset from one another. In some embodiments, the plane 830 and the plane 832 are equidistant from the longitudinal center line L.

#### E. Configuration 5—Cowl Position and Thermal Management

[0121] Referring to FIGS. 26-28A and 29-43, a vehicle 1000 is shown as an alternative configuration of the vehicle 10. The vehicle 1000 may be substantially similar to the vehicle 750, except as otherwise specified. The vehicle 1000 may be configured with any of the application kits 80 described herein.

[0122] As shown, the vehicle 1000 includes a cowl assembly 1010 including a first cowl portion, shown as front shroud 790, and a second cowl portion, shown as rear shroud 792. The front shroud 790 is coupled to the cab 40 and positioned above the cab 40. The rear shroud 792 is coupled to the body 770 of the application kit (e.g., a refuse compartment 130 when the vehicle 1000 is configured as a refuse vehicle) and extends forward, toward the front shroud 790. A space, volume, or compartment, shown as storage compartment 1012, is defined between the body 770, the cab 40, the rear shroud 792, and the chassis 20. Specifically, the storage compartment 1012 is positioned below the rear shroud 792, above the chassis 20, forward of the body 770,

and behind the cab 40. In some embodiments, the storage compartment 1012 occupies the same space as the rear portion 782 of the cab 40 shown in FIGS. 18 and 19.

[0123] Similar to the arrangement of FIGS. 18 and 19, the front shroud 790 and the rear shroud 792 of the cowl assembly 1010 are movable relative to one another. By way of example, when the body 770 is raised (e.g., by the body actuator 772), the rear shroud 792 is raised relative to the front shroud 790. Raising the rear shroud 792 in this way may facilitate access to components within the front shroud 790 (e.g., for cleaning or maintenance).

[0124] As best shown in FIG. 28A, a space or gap, shown as body gap 1014, is formed between the front shroud 790 and the rear shroud 792. The body gap 1014 extends laterally through the entirety of the cowl assembly 1010, from a left side of the cowl assembly 1010 to a right side of the cowl assembly 1010. The body gap 1014 extends vertically through the entirety of the cowl assembly 1010, from a top side of the cowl assembly 1010 to a bottom side of the cowl assembly 1010. The body gap 1014 extends from immediately behind the cab 40, upward and forward for a first distance, and vertically for a second distance to the top surface of the cowl assembly 1010. Accordingly, the front shroud 790 and the rear shroud 792 form corresponding overlapping sections. Specifically, rear shroud 792 forms an overhanging portion 1016 that extends forward from the rear shroud 792, and the front shroud 790 forms a recess 1018 that receives the overhanging portion 1016. The overhanging portion 1016 extends over the front shroud 790 such that upward movement of the overhanging portion 1016 is permitted.

[0125] Referring to FIGS. 26-28A and 29, the front shroud 790 is an assembly including a main portion or fixed portion, shown as hood 1020, and a replaceable portion or removable portion, shown as visor 1022. The hood 1020 is fixedly coupled to the roof of the cab 40, such that the hood 1020 remains during normal maintenance procedures. The hood 1020 is positioned directly above the cab 40 and extends along the body gap 1014. The visor 1022 is removably coupled to the hood 1020 and the cab 40, such that the visor 1022 can be removed and replaced (e.g., if the visor 1022 becomes damaged, to exchange the visor 1022 with a different aesthetic appearance or made from a different material, etc.). The visor 1022 is positioned at the front of the cab 40, directly above a front windscreen or windshield 1024 of the cab 40. The visor 1022 extends between the hood 1020 and the windshield 1024. This position at the front of the vehicle 1000 may make the visor 1022 more likely (e.g., than the hood 1020) to come into contact with debris, obstacles, or other objects that may contact the visor 1022, causing damage. By making this front portion of the cowl assembly 1010 easily replaceable, any damage caused to the front portion of the cowl assembly 1010 can be easily and quickly repaired. The visor 1022 also extends above and forward of the windshield 1024, protecting the windshield 1024 from debris falling downward toward the windshield 1024.

[0126] In some embodiments, the cab 40 includes a pair of guards or rails, shown as bars 1026. The bars 1026 are arranged substantially vertically, extending across the windshield 1024. The bars 1026 are laterally offset from one another. The bars 1026 may protect the windshield 1024 from contact with other objects. By way of example, the bars

**1026** may prevent a refuse container from coming into contact with the windshield **1024**.

[0127] The vehicle **1000** further includes a sensor, imaging device, or viewer, shown as front camera **1030**, coupled to the cab **40**. The front camera **1030** may provide image data regarding (e.g., showing) an area forward of the vehicle **1000**. The image data may be utilized by a control system of the vehicle **10** (e.g., to provide information that is used by a controller to control operation of the vehicle **1000**, to provide a real-time video output to an operator, etc.). As shown, the front camera **1030** is positioned along a front side of the cab **40**, above the windshield **1024**, and below the visor **1022**. The position of the visor **1022** above the front camera **1030** causes the visor **1022** to protect the front camera **1030** from falling rain and debris, preventing damage to the front camera **1030** and keeping the viewing portion (e.g., the lens) of the front camera **1030** clean. The visor **1022** includes a recessed portion, cutout, or notch, shown as camera cutout **1032**, that is positioned around the front camera **1030**. The camera cutout **1032** may provide clearance around the front camera **1030** to prevent the visor **1022** obstructing the view of the front camera **1030**. In some embodiments, the front camera **1030** and the camera cutout **1032** are both approximately laterally centered relative to the cab **40**. In some embodiments, the front camera **1030** is positioned between the bars **1026** such that the bars **1026** protect the front camera **1030** from contact with other objects.

[0128] Referring to FIGS. 30-33, the vehicle **1000** includes a cooling assembly, shown as core assembly **1040**, positioned within a radiator volume **800** defined by the cowl assembly **1010**. Specifically, the core assembly **1040** is positioned between the roof of the cab **40** and the front shroud **790**. The core assembly **1040** includes a core **752** and a core **754** each coupled to a subframe, shown as core frame **1042**. The core frame **1042** is pivotally coupled to the cab **40** by a pair of couplers, shown as hinges **1044**, such that the core assembly **1040** is rotatable relative to the cab **40** about a substantially vertical axis, shown as axis of rotation **1046**. The core **752** and the core **754** may be fixedly coupled to the core frame **1042**, such that the core assembly **1040** moves together as one assembly about the axis of rotation **1046**. The axis of rotation **1046** is laterally offset from the center of the vehicle **1000** to be positioned near the end of the core assembly **1040**. As shown, the axis of rotation **1046** is offset to the left. In other embodiments, the arrangement is mirrored about a longitudinal center plane, such that the axis of rotation **1046** is offset to the right.

[0129] The core assembly **1040** is selectively repositionable about the axis of rotation **1046** between a default position, stored position, or use position, shown in FIG. 30, and an extended position or maintenance position, shown in FIG. 34. The core assembly **1040** may normally remain in the use position (e.g., unless the vehicle **1000** is undergoing maintenance). By way of example, the core assembly **1040** may include a latch, fastener, or another type of coupler that selectively limits (e.g., prevents) movement of the core assembly **1040** out of the use position. The core assembly may be released by the coupler and rotated backward toward the maintenance position. The maintenance position may facilitate access by a user (e.g., through the storage compartment **1012**) to both the front and rear sides of the core assembly **1040** to facilitate cleaning and/or maintenance.

The maintenance position may also facilitate access to the portion of the radiator volume **800** defined between the front shroud **790** and the cab **40**.

[0130] Referring to FIG. 30, in the use position, the core assembly **1040** extends in a substantially vertical and lateral plane that is substantially perpendicular to a longitudinal axis. Specifically, the plane **830** and the plane **832** extend vertically and laterally and perpendicular to a longitudinal axis. In FIG. 30, the front shroud **790** is shown as being transparent for ease of viewing the core assembly **1040**. Referring to FIG. 34, in the maintenance position, the core assembly **1040** rotationally offset about the axis of rotation **1046** (e.g., clockwise as viewed from above) relative to the use position. In the maintenance position, the core assembly **1040** extends rearward, beyond the front shroud **790**, such that a space or gap is formed between the core assembly **1040** and the front shroud **790**. The core assembly **1040** extends in a substantially vertical plane that is skewed relative to a longitudinal axis. Specifically, the plane **830** and the plane **832** extend vertically and are skewed relative to a longitudinal axis.

[0131] Referring to FIGS. 26-28A and 29-33, airflow through the core assembly **1040** generally flows along one of two airflow paths: a path **1050**, and a path **1052**. The path **1050** and the path **1052** may be consistent between multiple lateral positions, such that the path **1050** and the path **1052** can each reach the core **752** and the core **754**. Although the paths **1050** and **1052** generally indicate the direction of airflow through the radiator volume **800**, it should be understood that the airflow may deviate from the exact paths shown (e.g., in order to fill the radiator volume **800** with pressurized air, due to turbulent flow, etc.).

[0132] The path **1050** enters the radiator volume **800** of the cowl assembly **1010** through a first opening, shown as inlet **1054**, that is positioned along a front of the cowl assembly **1010**. The inlet **1054** is defined by and between the hood **1020** and the visor **1022**. Specifically, the top side of the inlet **1054** is defined by the hood **1020**. The left, right, and bottom sides of the inlet **1054** are defined by the visor **1022**. In other embodiments, the left side and/or the right side of the inlet **1054** are defined by the hood **1020**.

[0133] As shown, the inlet **1054** extends laterally and vertically, the width of the inlet **1054** measured laterally being substantially larger (e.g., 5 times larger, 10 times larger, etc.) than the height of the inlet **1054** measured vertically. As shown, the inlet **1054** is generally oriented perpendicular to a longitudinal axis, such that airflow along the path **1050** enters the inlet **1054** longitudinally. In some embodiments, the vertical position of the inlet **1054** is selected such that airflow entering longitudinally through the inlet **1054** is permitted to pass directly to the core assembly **1040** (e.g., without encountering any obstructions). Such an arrangement may facilitate supplying unobstructed airflow to the core assembly **1040** through the inlet **1054** when the vehicle **1000** is traveling forward.

[0134] The path **1052** enters the radiator volume **800** of the cowl assembly **1010** through a second opening, shown as inlet **1056**, that is positioned along the bottom of the cowl assembly **1010**. The inlet **1056** is defined by and between the visor **1022** and the cab **40**. Specifically, the top, left, and right sides of the inlet **1056** are defined by the visor **1022**. The bottom side of the inlet **1056** is defined by the cab **40**. Specifically, the bottom side of the inlet **1056** may be defined by an upper portion of the windshield **1024**. As

shown, the inlet **1056** extends laterally and longitudinally, the width of the inlet **1056** measured laterally being substantially larger (e.g., 5 times larger, 10 times larger, etc.) than the depth of the inlet **1056** measured longitudinally.

[0135] As shown, the inlet **1056** generally extends perpendicular to the front surface of the cab **40**, such that the inlet **1056** generally faces downward. The front face of the cab **40** (e.g., the windshield **1024**) is generally inclined, such that the front face extends upward as the cab **40** extends rearward. By way of example, the front face of the cab **40** may be offset from vertical by 5 degrees, 10 degrees, 15 degrees, 30 degrees, 45 degrees, etc.). Accordingly, longitudinal airflow (e.g., from the vehicle **1000** traveling forward) strikes the front face of the cab **40** and is directed upward, along the front face, toward the inlet **1056**. The visor **1022** is curved to redirect the airflow in a longitudinal direction toward the core assembly **1040**. Accordingly, due to the position of the visor **1022**, the vehicle **1000** is able to direct airflow from the front face of the cab **40** toward the core assembly **1040**, increasing the total airflow through the core assembly **1040** (e.g., the total of the airflow along the path **1050** and the path **1052**) and more effectively cooling the core assembly **1040**.

[0136] Referring to FIG. 31, the front shroud **790** further includes a flow collector or plenum, shown as sleeve **1060**. The sleeve **1060** is positioned within the hood **1020** and coupled to the cab **40**. The sleeve **1060** has a first end or opening that is in fluid communication with the inlet **1054** and the inlet **1056**. Specifically, the sleeve **1060** is sealed against the hood **1020** and the cab **40** such that substantially all airflow through the inlet **1054** and the inlet **1056** passes into the sleeve **1060** through the first end. The sleeve **1060** has a second end or opening that is in fluid communication with the core assembly **1040**. Specifically, the sleeve **1060** is sealed against the core assembly **1040** such that substantially all airflow through the second end of the sleeve **1060** passes into the core **752** or the core **754**. Accordingly, the sleeve **1060** may facilitate forcing all of the airflow through the inlet **1054** and the inlet **1056** through the core **752** and/or the core **754**. In some embodiments, the cross-sectional area of the sleeve **1060** generally increases as the sleeve **1060** passes rearward, from the inlet **1054** and the inlet **1056** toward the core assembly **1040**.

[0137] Referring to FIGS. 31-33, an aperture or outlet, shown as outlet **1062**, is defined between the hood **1020** and the top of the cab **40**. The outlet **1062** is generally oriented parallel to the core assembly **1040**. The outlet **1062** is fluidly coupled to the storage compartment **1012**. As shown, the fans **760** and the fans **762** are positioned longitudinally between the cores **752** and **754** and the outlet **1062**. In operation, the path **1050** extends from outside the vehicle **1000**, through the inlet **1054**, through the sleeve **1060**, through the core **752** and/or the core **754**, through the fans **760** and/or the fans **762**, and out through the outlet **1062**. The path **1052** extends from outside the vehicle **1000**, through the inlet **1056**, through the sleeve **1060**, through the core **752** and/or the core **754**, through the fans **760** and/or the fans **762**, and out through the outlet **1062**. The airflow that exits the outlet **1062** may have an elevated temperature after removing thermal energy from the core assembly **1040**. The airflow that exits the outlet **1062** may exit the vehicle **1000** through a variety of paths. By way of example, the airflow may exit through a gap between two or more components (e.g., the body gap **1014**, between the rear shroud **792** and

the body **770**, laterally outward from the side of the storage compartment **1012**, downward through the chassis **20**, etc.).

[0138] Referring to FIGS. 31-37, the vehicle further includes a drainage assembly, rain channel, or fluid diversion system, shown as drip channel assembly **1200**. The drip channel assembly **1200** is configured to receive fluid that passes into the vehicle **1000** through the body gap **1014** and direct the received fluid away from certain components of the vehicle **1000**. By way of example, the drip channel assembly **1200** may receive and direct water (e.g., from rain, when washing the vehicle **1000**, etc.). By way of another example, in an embodiment where the vehicle **1000** is configured as a front-loading refuse vehicle (e.g., the refuse vehicle **100**), the lift assembly **140** may raise a refuse container above the body gap **1014** in order to deposit refuse from the container into the hopper volume **132**. During such a motion, fluid from the refuse container may drip through the body gap **1014** and into the drip channel assembly **1200**. Accordingly, the drip channel assembly **1200** may improve the overall cleanliness of the vehicle **1000** and prevent fluid from reaching components that may be sensitive to contaminants or difficult to clean. By way of example, the drip channel assembly **1200** may prevent fluid from passing through the outlet **1062** and reaching the core assembly **1040**, which may be difficult to clean, and which may be adversely affected by contamination from the fluid.

[0139] The drip channel assembly **1200** includes a first section, portion, or service panel, shown as upper channel portion **1210**, and a pair of second sections or portions, shown as lower channel portions **1220**. The upper channel portion **1210** is positioned above the cab **40** and is approximately laterally centered relative to the cab **40**. The lower channel portions **1220** are each positioned below an end of the upper channel portion **1210**. The upper channel portion **1210** directs fluid downward and laterally outward from a longitudinal centerline of the vehicle **1000**. The lower channel portions **1220** are each positioned to be in fluid communication with an end of the upper channel portion **1210**, such that fluid exiting the upper channel portion **1210** is received by one of the lower channel portions **1220**. The lower channel portions **1220** each then direct the fluid further downward and laterally outward.

[0140] The upper channel portion **1210** includes a first flange or inner flange, shown as backing plate **1230**, a second flange, shown as outer flange **1232**, and a connecting portion, shown as channel bottom **1234**. Together, the backing plate **1230**, the outer flange **1232**, and the channel bottom **1234** form a channel, gutter, trough, or drain of the upper channel portion **1210** having a U-shaped cross section. In some embodiments, the backing plate **1230**, the outer flange **1232**, and the channel bottom **1234** are integrally formed as a single, continuous piece. In other embodiments, the backing plate **1230**, the outer flange **1232**, and/or the channel bottom **1234** are formed separately and coupled to one another (e.g., using fasteners, using adhesive, etc.). The backing plate **1230** is removably coupled (e.g., by one or more fasteners) to the hood **1020** of the front shroud **790**. The backing plate **1230** extends in a substantially vertical and lateral plane. The backing plate **1230** extends along the periphery of the outlet **1062** defined by the hood **1020**, sealing the outlet **1062**. The channel bottom **1234** extends rearward from a bottom edge of the backing plate **1230**. The channel bottom **1234** defines an aperture **1236**, through which the airflow through the outlet **1062** is forced by the

backing plate **1230**. The aperture **1236** has a reduced cross-sectional area relative to the outlet **1062**, but generally surrounds the fans **760** and **762**. The outer flange **1232** is positioned at a distal end of the channel bottom **1234** and extends upward from the channel bottom **1234**. In some embodiments, the backing plate **1230**, the channel bottom **1234**, and the outer flange **1232** are formed as a single, continuous piece.

[0141] The backing plate **1230**, the channel bottom **1234**, and the outer flange **1232** form the channel of the upper channel portion **1210**, which includes a center portion **1240**, a pair of intermediate portions or middle portions **1242**, and a pair of drip portions or end portions **1244**. The center portion **1240** is approximately laterally centered and extends laterally outward. The center portion **1240** is generally horizontal. In some embodiments, the center portion **1240** has an arc shape or a pair of pitched portions such that the center portion **1240** slopes downward as the center portion **1240** extends laterally outward from the center of the vehicle **1000**.

[0142] Each end of the center portion **1240** meets one of the middle portions **1242**. The middle portions **1242** are generally sloped downward and laterally outward. In some embodiments, the slopes of the middle portions **1242** are significantly steeper than the slope of the center portion **1240** (e.g., 30 to 60 degrees versus 0 to 10 degrees). In some embodiments, the center portion **1240** and the middle portions **1242** lie in a common lateral and vertical plane such that the center portion **1240** and the middle portions **1242** do not extend longitudinally forward or rearward (e.g., have a substantially constant longitudinal position).

[0143] The end of each middle portion **1242** meets one of the end portions **1244**. The end portions **1244** are generally sloped downward and longitudinally rearward. In some embodiments, the lateral slopes of the end portions **1244** are significantly steeper than the slope of the center portion **1240** (e.g., 30 to 60 degrees versus 0 to 10 degrees). In some embodiments, each end portion **1244** lies in a longitudinal and vertical plane such that the end portions **1244** do not extend laterally inward or outward (e.g., have a substantially constant lateral position). The shapes and sizes of the center portion **1240**, the middle portions **1242**, and the end portions **1244** may be selected to match the shape and size of the body gap **1014**, such that fluid entering the body gap **1014** at any point along the length of the body gap **1014** is captured by the channel of the upper channel portion **1210**.

[0144] FIGS. 28B-28H illustrate an alternative configuration of the upper channel portion **1210**. This alternative upper channel portion **1210** may be substantially similar to the upper channel portion **1210** shown in FIG. 35, except as otherwise specified herein, and may be utilized in place of the upper channel portion **1210** of FIG. 35. In the upper channel portion **1210** of FIGS. 28B-28H, the channel bottom **1234** and the outer flange **1232** are formed as separate pieces that are removably coupled to one another (e.g., by one or more fasteners). Such a configuration may facilitate manufacturing of the upper channel portion **1210**. In the configuration of FIGS. 28B-28H, the center portion **1240** is wider (e.g., as measured perpendicular to the outer flange **1232**) than the end portions **1244**. The middle portions **1242** taper from the relatively large width of the center portion **1240** to the relatively small width of the end portions **1244**. The end portions **1244** are generally sloped downward, longitudinally rearward, and laterally outward. In some embodi-

ments, the lateral outward slopes of the end portions **1244** match (e.g., are substantially equal to) the lateral outward slopes of the middle portions **1242** (e.g., as shown in FIG. 28F).

[0145] The lower channel portions **1220** are positioned directly beneath the end portions **1244**, such that fluid that drips off of the end portions **1244** is received by the lower channel portions **1220**. The lower channel portions **1220** are each coupled (e.g., fixedly, removably, etc.) to a rear wall **46** of the cab **40**. By way of example, the lower channel portions **1220** may be fastened or welded to the rear wall **46** of the cab **40**. The lower channel portions **1220** extend rearward from the rear wall **46** of the cab **40**. In some embodiments, the lower channel portions **1220** each have a U-shaped cross section, similar to the channel of the upper channel portion **1210**. In such an embodiment, the U-shaped cross section opens laterally outward to facilitate capturing and retaining fluid drippings. The lower channel portions **1220** each slope downward and laterally outward. In some embodiments, the lower channel portions **1220** terminate near the bottom of the cab **40** and/or near the outer walls (e.g., left and right walls) of the cab **40**.

[0146] In operation, fluid may pass through the body gap **1014**. If the fluid is near the longitudinal center of the vehicle **10**, the fluid is captured within the center portion **1240**. If the fluid is to the left of center upon contact with the center portion **1240**, gravity will force the fluid to flow laterally outward to the left. If the fluid is to the right of center upon contact with the center portion **1240**, gravity will force the fluid to flow laterally outward to the right. The fluid may pass through the middle portion **1242** and the end portion **1244** and subsequently fall onto one of the lower channel portions **1220**. The lower channel portions **1220** then direct the fluid even further laterally outward, away from any components that might be sensitive to contaminants. If the fluid initially enters the body gap **1014** farther from the longitudinal center of the vehicle **10**, the fluid may fall onto a middle portion **1242**, an end portion **1244**, or a lower channel portion **1220** and be directed accordingly.

[0147] Referring to FIGS. 32 and 34, when installed, the upper channel portion **1210** may obstruct access to the core assembly **1040**. Additionally, the upper channel portion **1210** may obstruct movement of the core assembly **1040**, preventing the core assembly **1040** from reaching the maintenance position. In order to facilitate maintenance and cleaning of the core assembly **1040**, the upper channel portion **1210** may be removably coupled to the front shroud **790** and/or the cab **40**. By way of example, the front shroud **790** may be removably coupled to the front shroud **790** by a series of fasteners. By removing the fasteners, the upper channel portion **1210** may be disconnected from the front shroud **790** and removed. This removed state may facilitate movement of the core assembly **1040** and free access to the core assembly **1040** by a user. Additionally, removing the upper channel portion **1210** may facilitate cleaning the upper channel portion **1210**. Once the cleaning and maintenance have been completed, the upper channel portion **1210** may be reattached to the front shroud **790**. In other embodiments, the upper channel portion **1210** is fixedly and permanently coupled to the front shroud **790**. By way of example, the upper channel portion **1210** may be formed as a single, continuous piece with the front shroud **790**.

[0148] Referring to FIGS. 38 and 39, to further facilitate user access to the core assembly **1040**, the vehicle **1000**

includes a maintenance step or service step, shown as step assembly 1300. The step assembly 1300 is coupled to the front rail portion 30 (e.g., fixedly coupled, coupled by one or more fasteners, etc.). As shown, the step assembly 1300 is positioned between the rear wall 46 of the cab 40 and the body 770. The step assembly 1300 extends above the front rail portion 30 and below the roof of the cab 40. The step assembly 1300 extends directly above a wheel and tire assembly 54 of the front axle 50. Accordingly, the step assembly 1300 is generally positioned within the storage compartment 1012. Although the step assembly 1300 is shown on one side of the vehicle 1000 (e.g., a driver side), the step assembly 1300 may additionally or alternatively be positioned on the opposite side of the vehicle 1000 (e.g., a passenger side).

[0149] The step assembly 1300 includes a step surface or top member, shown as top plate 1302. The top plate 1302 extends in a generally horizontal plane and provides a surface onto which a user can step. Extending downward from the top plate 1302 along the back of the step assembly 1300 is an attachment plate or back member, shown as back plate 1304. The back plate 1304 is configured to be directly coupled to the front rail portion 30 to couple the step assembly 1300 to the chassis 20. In some embodiments, the back plate 1304 defines a series of apertures that each receive a fastener, shown as bolt 1306, to couple the step assembly 1300 to the chassis 20. The bolts 1306 may be received by corresponding threaded apertures defined by the chassis 20. Additionally or alternatively, the step assembly 1300 may be welded to the chassis 20.

[0150] A pair of support members, shown as gusset plates 1310, extend between the back plate 1304 and the top plate 1302. The gusset plates 1310 may be substantially perpendicular to the top plate 1302 and the back plate 1304. The gusset plates 1310 transfer forces from the top plate 1302 to the back plate 1304, strengthening the step assembly 1300. In some embodiments, the gusset plates 1310, the top plate 1302, and the back plate 1304 are fixedly coupled to one another. In some embodiments, the gusset plates 1310, the top plate 1302, and the back plate 1304 form a single, continuous piece.

[0151] A user may stand on the step assembly 1300 when accessing the core assembly 1040. By way of example, the user may step directly onto the step assembly 1300 from the ground. By way of another example, the user may first step onto the wheel and tire assembly 54 and then onto the step assembly 1300. By placing the step assembly 1300 directly above a wheel and tire assembly 54, the wheel and tire assembly 54 may facilitate the user climbing the vehicle 1000 without requiring an additional step.

#### F. Configuration 5—Cowl Clearance for Lift Assembly of Front-Loading Refuse Vehicle

[0152] During operation, the vehicle 1000 may experience various loadings that cause the body 770 to move relative to the cab 40. By way of example, if the vehicle 1000 drives over a bump or depression in a road surface, the body 770 may rotate relative to the cab 40 about a longitudinal axis. Because the front shroud 790 is coupled to the cab 40 and the rear shroud 792 is coupled to the body 770, this relative movement of the cab 40 and the body 770 causes a corresponding relative movement of the front shroud 790 and the rear shroud 792. The body gap 1014 may be sized to prevent

or minimize contact between the front shroud 790 and the rear shroud 792 when such relative movement occurs.

[0153] When the vehicle 1000 is configured as a front-loading refuse vehicle (e.g., the refuse vehicle 100), the lift arms 142 are coupled to the body 770 (e.g., the refuse compartment 130). At various points of operation (e.g., when lifting or lowering a refuse container, when the lift arms 142 are between a fully raised position and a fully lowered position), the lift arms 142 may be positioned adjacent the cab 40. Accordingly, when the chassis 20 twists about the longitudinal axis, the lift arms 142 may move relative to the cab 40, bringing the lift arms 142 closer to the cab 40. In other vehicles, such a twisting movement could cause the lift arms of a refuse vehicle to come into contact with the cab 40, causing damage to the cab and/or the arms.

[0154] Instead, the cowl assembly 1010 is specifically configured to avoid contact between the lift arms 142 in the event that the body 770 twists relative to the cab 40. Specifically, the cowl assembly 1010 defines cowl recesses 1400 positioned at the top left and top right corners of the cowl assembly 1010. These cowl recesses 1400 reduce the size of the cowl assembly 1010, moving the extremities of the cowl assembly 1010 away from the path of the lift arms 142, even when the body 770 twists relative to the cab 40.

[0155] As shown in FIGS. 26 and 27, each cowl recess 1400 includes a front recess portion 1402 defined by the front shroud 790 and a rear recess portion 1404 defined by the rear shroud 792. The front recess portion 1402 is positioned forward of the rear recess portion 1404. The front recess portion 1402 is continuous with the rear recess portion 1404, and together the front recess portion 1402 and the rear recess portion 1404 define the cowl recess 1400.

[0156] The front shroud 790 includes a first surface, shown as cab interface surface 1410, a second surface, shown as transition surface 1412, and a third surface, shown as side surface 1414. The cab interface surface 1410 is positioned along a top edge of the cab 40. The cab interface surface 1410 may meet the top of the cab 40 to form a smooth transition from the cab 40 to the front shroud 790. In some embodiments, a gap is formed between the interface surface 1410 to permit fluid to drain between the interface surface 1410 and the cab 40. In such embodiments, the interface surface 1410 may be the closest surface of the front shroud 790 to the cab 40. The cab interface surface 1410 is substantially vertical and extends longitudinally along the front shroud 790. The cab interface surface 1410 may taper laterally outward as the cab interface surface 1410 extends rearward.

[0157] The transition surface 1412 is positioned above the cab interface surface 1410 and is continuous with the cab interface surface 1410. The transition surface 1412 is oriented at an angle  $\theta_1$  relative to a vertical axis. The transition surface 1412 extends upward and laterally inward from the cab interface surface 1410. In some embodiments, the angle  $\theta_1$  is between 45 degrees and 90 degrees (i.e., horizontal). The transition surface 1412 may extend gradually upward as the transition surface 1412 extends rearward.

[0158] The side surface 1414 is positioned above the transition surface 1412 and is connected with the transition surface 1412 through a concave filleted surface. Together, the side surface 1414, the transition surface 1412, and the concave filleted surface define the front recess portion 1402. The side surface 1414 is oriented at an angle  $\theta_2$  relative to a vertical axis. The side surface 1414 extends upward and



laterally inward from the transition surface 1412. In some embodiments, the angle  $\theta_2$  is less than the angle  $\theta_1$ , such that the side surface 1414 is more steeply inclined than the transition surface 1412. In some embodiments, the angle  $\theta_2$  is between 0 degrees (i.e., vertical) and 45 degrees.

[0159] The rear shroud 792 includes a first surface, shown as cab interface surface 1420, a second surface, shown as body interface surface 1422, and a third surface, shown as transition surface 1424. The cab interface surface 1420 is aligned with the cab interface surface 1410. The cab interface surface 1420 generally continues the same shape, size, and orientation as the cab interface surface 1410, such that the cab interface surface 1420 generally acts as a continuation of the cab interface surface 1410 on the opposite side of the body gap 1014. The cab interface surface 1420 is substantially vertical and extends longitudinally along the rear shroud 792. The cab interface surface 1420 may taper laterally outward as the cab interface surface 1420 extends rearward.

[0160] The body interface surface 1422 is positioned along a front edge of the body 770. The body interface surface 1422 may meet the left or right side of the cab 40 to form a smooth transition from the cab 40 to the rear shroud 792. The body interface surface 1422 is substantially vertical and extends longitudinally along the front shroud 790. The body interface surface 1422 may taper laterally outward as the body interface surface 1422 extends rearward.

[0161] The transition surface 1424 generally provides a smooth transition between the cab interface surface 1420, the body interface surface 1422, the transition surface 1412, and the side surface 1414. The cab interface surface 1420, the body interface surface 1422, the transition surface 1412, and the side surface 1414 have different lateral positions, and the transition surface 1424 provides a curved (e.g., filleted) surface to smoothly transition between each of the surfaces.

[0162] FIGS. 27 and 40-43 illustrate how the cowl recesses 1400 facilitate relative movement of the body 770 and the cab 40 without the lift assembly 140 contacting the cowl assembly 1010. In each of these figures, the lift assembly 140 is in a lowered position, bringing the lift arms 142 and the corresponding actuators in close proximity to the cowl assembly 1010.

[0163] FIGS. 40 and 41 illustrate the vehicle 1000 in a neutral or unstressed state in which the body 770 is minimally twisted relative to the cab 40 (e.g., in which the frame rails of the chassis 20 are straight). In this state, the cowl assembly 1010 is approximately centered between the lift arms 142, and the lift arms 142 are not received by the cowl recesses 1400. Accordingly, in the neutral state, clearance between the lift arms 142 and the cowl assembly 1010 can be achieved, even without the cowl recesses 1400.

[0164] FIGS. 27, 42, and 43 illustrate the vehicle 1000 in a stressed or twisted state in which the chassis 20 twists, causing relative movement between (a) the body 770 and the lift assembly 140 and (b) the cab 40. Specifically, (a) the body 770 and the lift assembly 140 rotates relative to (b) the cab 40 about a longitudinal axis (e.g., laterally centered on the vehicle 10, positioned below the cab 40, positioned within the chassis 20, positioned below the body 770, etc.). Specifically, as shown in FIG. 42, the relative angle of twist between the body 770 and the cab 40 is shown as angle  $\theta_3$ . The angle  $\theta_3$  shown in FIG. 42 may represent the maximum twist angle that the vehicle 1000 would experience in normal

operation. By way of example, the angle  $\theta_3$  may be between 1 degree and 30 degrees, between 5 degrees and 20 degrees, between 10 degrees and 15 degrees, etc.

[0165] As shown in FIGS. 27, 42, and 43, in the twisted state, the lift assembly 140 (e.g., one of the lift arms 142) is received within one of the cowl recesses 1400. Specifically, one of the lift arms 142 is received within the cowl recess 1400 on the driver side of the vehicle 1000. It should be understood that the cowl recesses 1400 may be symmetrical with one another, such that the vehicle 1000 is capable of addressing an opposite twisting state (e.g., one of the lift arms 142 is received within a cowl recess 1400 on the passenger side of the vehicle 1000). The cowl recess 1400 is sufficiently sized that the lift assembly 140 does not contact the cowl assembly 1010 in the twisted state. By way of example, in some embodiments, the angle  $\theta_3$  is greater than or equal to the angle  $\theta_2$  to provide angular clearance for the lift arms 142.

[0166] Because the rear shroud 792 is coupled directly to the body 770, the rear shroud 792 moves with the body 770 in the event of a twisting of the chassis 20. Accordingly, even if the cab 40 twists about the longitudinal axis relative to the body 770, the rear shroud 792 remains stationary relative to the body 770. As such, the lateral positions of the lift arms 142 relative to the rear shroud 792 remain substantially constant throughout operation of the vehicle 1000. Because of this consistent positioning, the rear recess portion 1404 may be shallower than the front recess portion 1402 while still avoiding contact between the lift arms 142 and the rear shroud 792. In other words, a lateral width of the rear shroud 792 adjacent the lift arms 142 (e.g., a distance between the body interface surfaces 1422) may be larger than a lateral width of the front shroud 790 adjacent the lift arms 142 (e.g., a distance between the side surfaces 1414) for a given position of the lift assembly 140.

[0167] However, it may be advantageous (e.g., for aesthetics, for improved aerodynamics, etc.) for the surfaces of the front recess portion 1402 to be continuous with the surfaces of the rear recess portion 1404. The transition surface 1424 may facilitate this continuous transition between the surfaces. By way of example, the transition surface 1424 may align with the corresponding side surface 1414, transition surface 1412, and cab interface surface 1410.

[0168] As utilized herein, the terms “approximately,” “about,” “substantially,” and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0169] It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to

connote that such embodiments are necessarily extraordinary or superlative examples).

**[0170]** The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

**[0171]** References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and that such variations are intended to be encompassed by the present disclosure.

**[0172]** The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

**[0173]** The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

**[0174]** Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

**[0175]** It is important to note that the construction and arrangement of the vehicle **10** and the systems and components thereof as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with any other embodiment disclosed herein. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

**[0176]** **1-20.** (canceled)

**21.** A vehicle, comprising:

- a chassis;
- a cab coupled to the chassis;
- a body coupled to the chassis;
- a cowl including a first cowl portion coupled to the cab and a second cowl portion coupled to the body, wherein the second cowl portion is movable relative to the first cowl portion; and
- a heat exchanger positioned at least partially within the cowl.

**22.** The vehicle of claim **21**, wherein the cowl at least partially defines an inlet positioned forward of the heat

exchanger such that airflow is permitted to pass from a front end of the vehicle toward a rear end of the vehicle through the inlet and the heat exchanger.

**23.** The vehicle of claim **22**, wherein the first cowl portion at least partially defines the inlet, and wherein the inlet extends above the cab.

**24.** The vehicle of claim **21**, wherein the heat exchanger is a first heat exchanger, further comprising a second heat exchanger positioned at least partially within the cowl.

**25.** The vehicle of claim **24**, further comprising a first fan coupled to the first heat exchanger and a second fan coupled to the second heat exchanger.

**26.** The vehicle of claim **21**, wherein the vehicle further comprises:

an electric motor;

a battery; and

an inverter configured to supply electrical energy from the battery to the electric motor,

wherein the heat exchanger is a radiator that is configured to dissipate thermal energy from at least one of the electric motor, the battery, or the inverter.

**27.** The vehicle of claim **21**, wherein the heat exchanger is a condenser of an air conditioning circuit.

**28.** A vehicle, comprising:

a chassis;

a cab coupled to the chassis;

a body coupled to the chassis;

a shroud coupled to at least one of the body or the cab and at least partially defining an air volume, an inlet, and an outlet; and

a heat exchanger positioned within the air volume, wherein a path for airflow is defined through the inlet, the heat exchanger, and the outlet; and  
wherein the heat exchanger is selectively repositionable about a substantially vertical axis between a use position and a maintenance position.

**28.** The vehicle of claim **28**, further comprising a service panel removably coupled to the shroud and positioned to obstruct movement of the heat exchanger from the use position to the maintenance position when the service panel is coupled to the shroud.

**29.** The vehicle of claim **29**, wherein the shroud defines a body gap, and wherein the service panel defines a channel positioned to receive fluid passing through the body gap.

**31.** The vehicle of claim **30**, wherein the channel is sloped to direct the fluid laterally outward from a longitudinal centerline of the vehicle.

**32.** The vehicle of claim **31**, wherein the channel is a first channel, further comprising a second channel coupled to the cab and configured to remain coupled to the cab when the service panel is removed, and wherein the first channel is configured to direct the fluid onto the second channel.

**33.** The vehicle of claim **28**, further comprising a step coupled to the chassis and positioned at least partially within a space between the cab and the body.

**34.** The vehicle of claim **33**, wherein the path for the airflow is defined through the inlet, the outlet, the heat exchanger, and the space defined between the cab and the body.

**35.** The vehicle of claim **34**, further comprising a flow collector positioned within the shroud and fluidly coupling the inlet to the inlet to the heat exchanger.

**36.** A vehicle, comprising:

a chassis;

a cab coupled to the chassis;

a body coupled to the chassis;

a cowl including a first cowl portion coupled to the cab and a second cowl portion coupled to the body, wherein the cowl at least partially defines an inlet; and

a heat exchanger in fluid communication with the inlet such that airflow is permitted to pass through the inlet to the heat exchanger.

**37.** The vehicle of claim **36**, wherein the second cowl portion is movable relative to the first cowl portion.

**38.** The vehicle of claim **37**, wherein the first cowl portion at least partially defines the inlet, and wherein the inlet extends above the cab.

**39.** The vehicle of claim **37**, wherein the heat exchanger is a first heat exchanger, further comprising a second heat exchanger positioned at least partially within the cowl.

**40.** The vehicle of claim **39**, further comprising a first fan coupled to the first heat exchanger and a second fan coupled to the second heat exchanger.

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